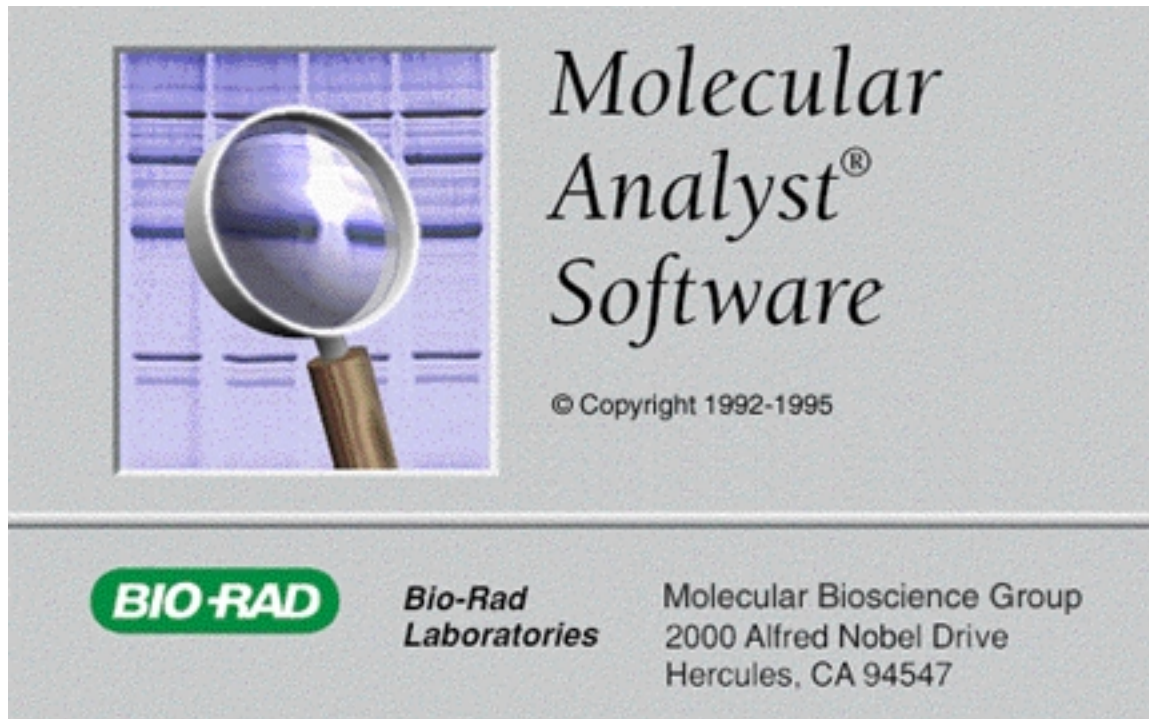




# **Molecular Analyst<sup>®</sup>/Macintosh**

## **Macintosh Software for Bio-Rad's Image Analysis Systems**

**Version 2.1**

### **Instruction Manual**

For Technical Service  
Call Your Local Bio-Rad Office or  
in the U.S. Call **1-800-4BIORAD**  
(1-800-424-6723)

**Part Number 4000000 Rev D**

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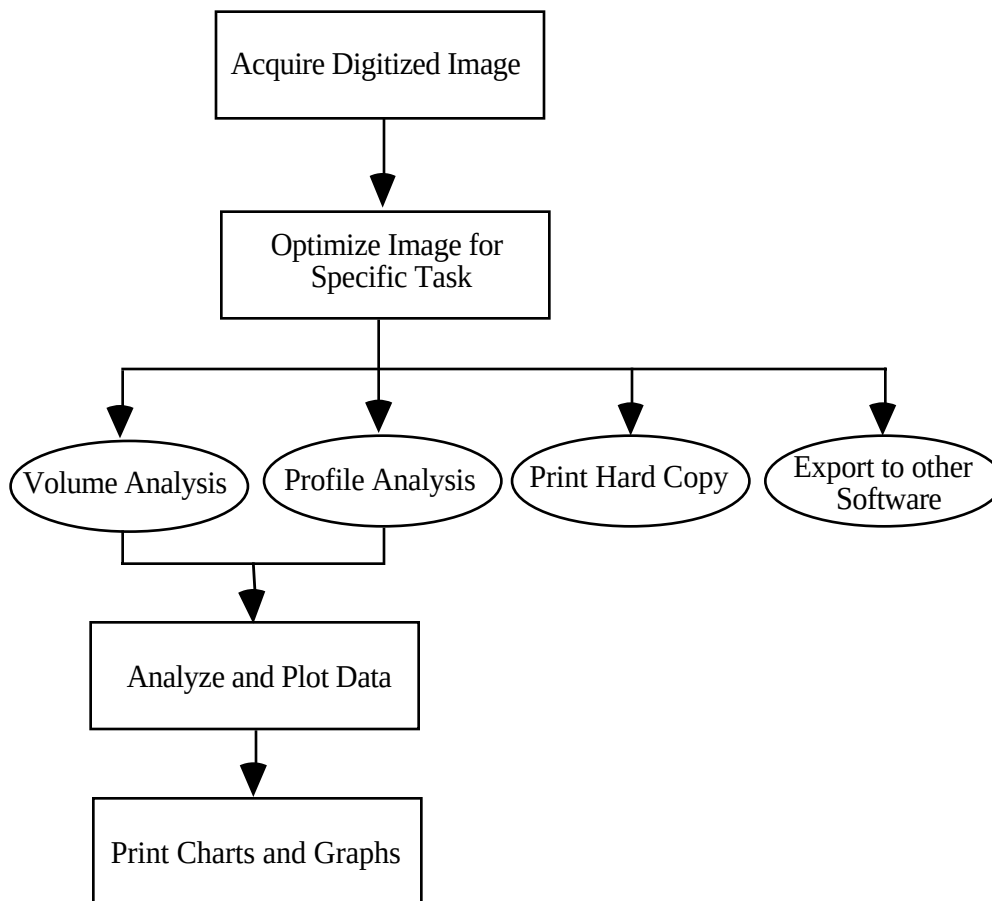
## Section 1 Molecular Analyst/Macintosh Software

### 1.1 Introduction

The Molecular Analyst<sup>®</sup>/Macintosh data analysis software is a powerful analytical tool for controlling and analyzing images and data from Bio-Rad's image analysis systems. The Molecular Analyst/Macintosh software uses a series of pull down menus and tool palettes to allow you to easily scan, display, analyze, and output data from many types of samples. The determination of valuable information such as sample concentration or amount of macromolecule, relative mobility, and molecular weight, is easily performed with the software's graphical interface. It is recommended that the first-time user follow the tutorial section of this manual to become acquainted with some of the Molecular Analyst/Macintosh functions and features. More detailed information about each function is contained in the section for that function.

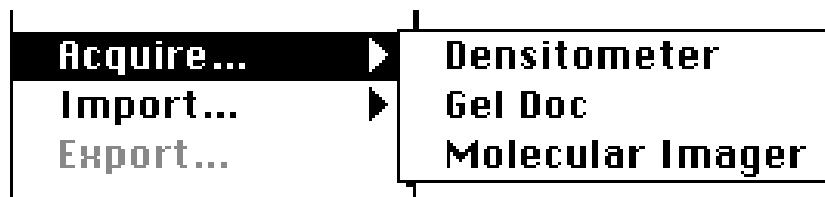
### 1.2 Image Analysis Process: Overview

An overview of the image analysis process is diagrammed below. After an image is acquired (either from the Densitometer, Molecular Imager, or Gel Doc system) and saved in a digitized form, the image can be adjusted or optimized for (1) printing, (2) exporting to other software, (3) performing volume analysis, or (4) performing profile analysis. The quantified data can then be easily exported to other spreadsheet software and the data plotted as a chart or graph. Regression analysis, molecular weight determination, and other functions can also be carried out once the data has been obtained by volume and profile analysis. Each step of image analysis process will be described in detail in the tutorial and the sections following.



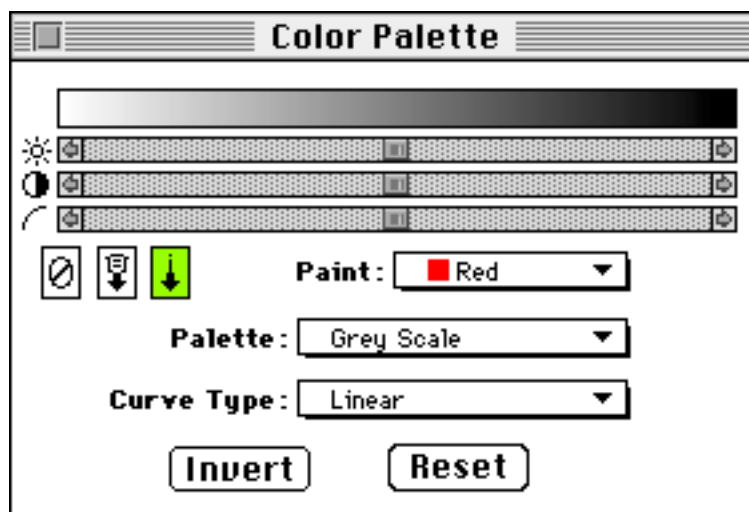
## Acquiring Images

Molecular Analyst/Mac software allows the acquisition of images from Bio-Rad's Densitometer, Gel Documentation, and Molecular Imager systems. This integrated image analysis software provides a fast method to acquire images from radioactive, chemiluminescent, white-light, and fluorescent samples. This software will also open images acquired from Molecular Analyst/PC and TIFF files.



## Optimizing Images

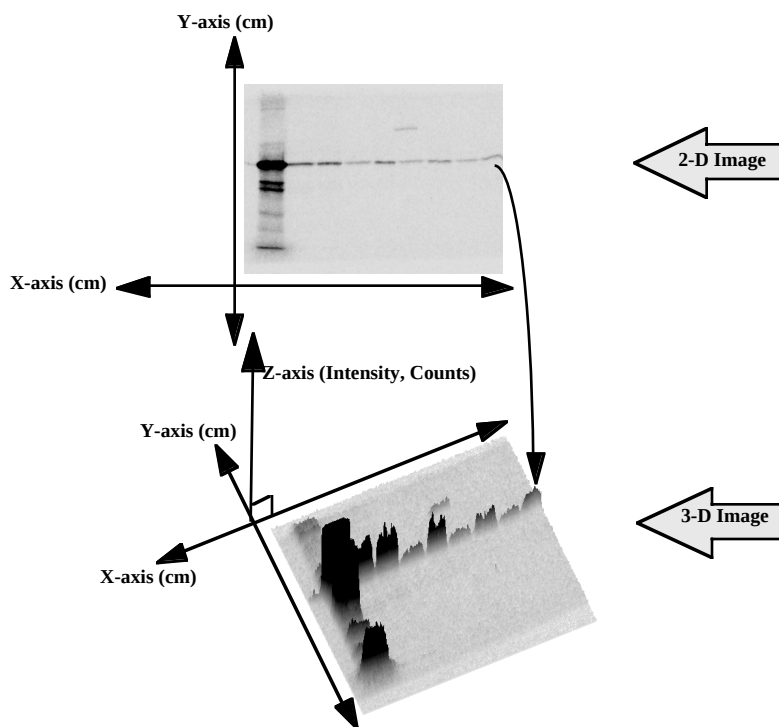
Images from Molecular Analyst/Mac can be optimized using a variety of tools. The color palette, transformation tool, and a variety of filters can visually adjust faint or saturated images to aid in quantitation and printing.



## Volume Analysis

The Gel Doc 1000, Densitometer, and Molecular Imager systems are light or radiation detectors that are capable of converting biological signals into digital data. The digital data is then displayed on the computer in a 2-dimensional format using the Molecular Analyst software. The intensity of the various signals (i.e. bands or spots) is differentiated by the intensity of the gray levels displayed on the computer monitor. The stronger the signal, the darker the image on the monitor and vice versa.

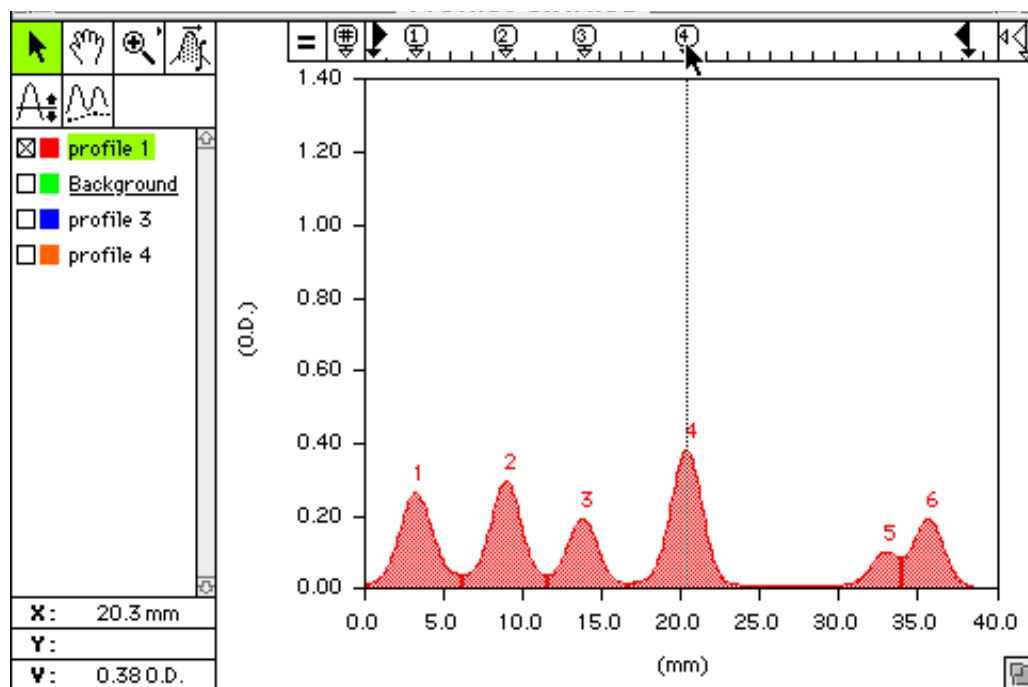
Currently, most image analysis software displays data in a 2-dimensional format. However, the 2-dimensional display does not reflect all the information stored in the digital data. A 3-dimensional representation of the data gives a more realistic and clear picture. (Figure 1). In a 3-dimensional view, signals are displayed as mountains or hills in a landscape surrounding. Therefore, the strength or intensity of the signal is represented by the volumetric capacity of the mountains or hills. It is essential to be aware that the 2-dimension image display represents 3-dimensional data. Understanding this concept is essential for accurate quantitative and qualitative analysis of the digital data.



**Figure 1.** Two and 3-dimensional representations of a digitized image.

### Profile Analysis

Profile analysis determines relative differences in intensities and positions of bands or spots in an image. Using the profile tools, a 2-dimensional graph or profile is generated and values such as “Percent of Total” and “Molecular Weight” can be calculated.



## **Printing**

Molecular Analyst/Mac will print images and reports to a variety of printers. The Gel Doc 1000 will print live and saved images to a Mitsubishi P68 Thermal Printer. Please see Appendix E for configuring the Mitsubishi P68 printer.

## **Data Export**

The results obtained from Molecular Analyst/Mac software can be exported to a variety of other software applications. For example, using simple Copy and Paste commands, the data can be transferred to Excel for further analysis, and graph and chart generation.

## **1.3 System Requirements**

Apple Macintosh Quadra 700 or better computer with NuBus slot, 256 gray level display, 20 mb RAM(20+ mb recommend to allow processing of large images), 230 mb hard disk (230+mb recommended), and Apple Macintosh System 7.1 system software or better.

Compatibility and performance of all other computers not purchased from Bio-Rad are the sole responsibility of the user. Please check with your local Bio-Rad representative regarding compatibility of your specific Macintosh computer.

## **1.4 System Setup**

### **Molecular Imager**

Installation of Bio-Rad's Molecular Imager Systems is completed by a field service technician. When connecting the Molecular Imager to a Power Macintosh's built-in SCSI port, you must turn on the Molecular Imager before turning on the Power Macintosh.

### **Gel Doc 1000**

The host computer must be connected to the Gel Doc 1000 via Bio-Rad's Frame Grabber board interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION).

- 1) Insert the frame grabber card into an available NuBus slot.
- 2) Next, plug the 9 pin DB video input connector of the integrating cable into the frame grabber adapter in your computer.
- 3) Connect the BNC video output connector (labeled Camera-Video) and 8 pin "MINI DIN" integration control connector (labeled Camera-AUX) into the camera "Video" and "AUX" ports respectively.
- 4) If using the P68 video printer, connect the 1/8" mono male jack labeled "Printer Remote" into the remote outlet on the back of the printer and connect the BNC connector labeled "Video In" into the printers video-in port. SEE APPENDIX E TO SETUP THE PRINTER FOR THE MACINTOSH.

### **GS-700 Densitometer/ GS-690 Densitometer**

The host computer must be connected to the Model GS-700 Densitometer or GS-690 Densitometer via the Macintosh's built-in SCSI interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION).

- 1) Simply attach the SCSI-Macintosh cable to the SCSI port in the Macintosh.
- 2) Next connect the SCSI cable to the SCSI port on the back of the densitometer.

### GS-670 Densitometer

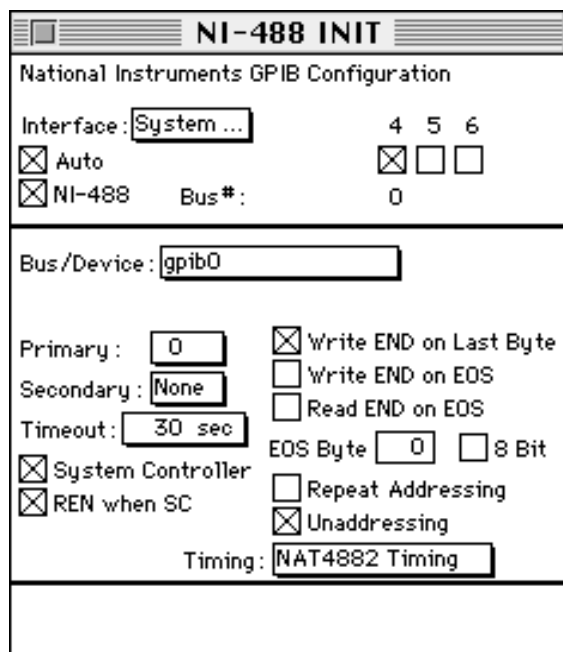
The host computer must be connected to the Model GS-670 Densitometer via a NB-GPIB interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION).

- 1) Insert the GPIB card into an available NuBus slot.
- 2) Attach the GPIB cable to the NB-GPIB card port.
- 3) Next connect the GPIB cable to the GPIB port on the back of the densitometer.

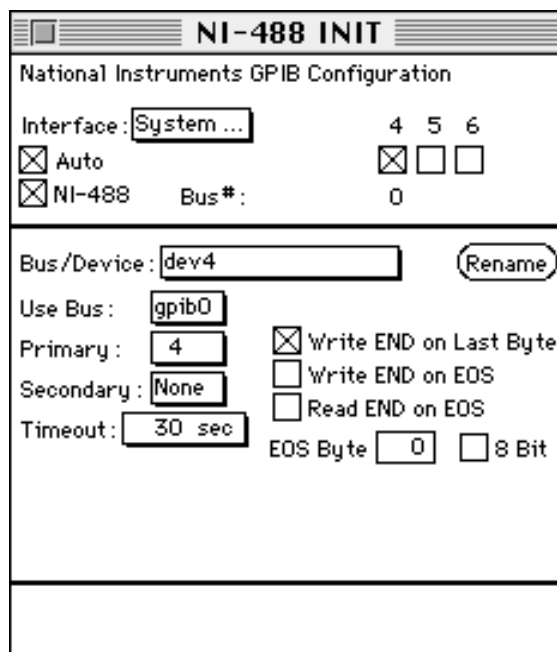
### Installing the Densitometer's GPIB NB Handler Software

Refer to the instructions which accompany the NB-GPIB card for proper installation. The NB Handler Init should be automatically set up so that it will load when the computer starts up. To verify whether your NB Handler Init is setup properly, double click on the system folder, double click on the control panels folder, then double click on the NB Handler Init icon. The settings should be as displayed below for the Bus/Device popup menu **gpib0** and **0 Dev4**:

**Note: Molecular Analyst/Macintosh is not compatible with Symantec's Directory Assistance (part of Norton Utilities). Please disable Directory Assistance before using Molecular Analyst.**

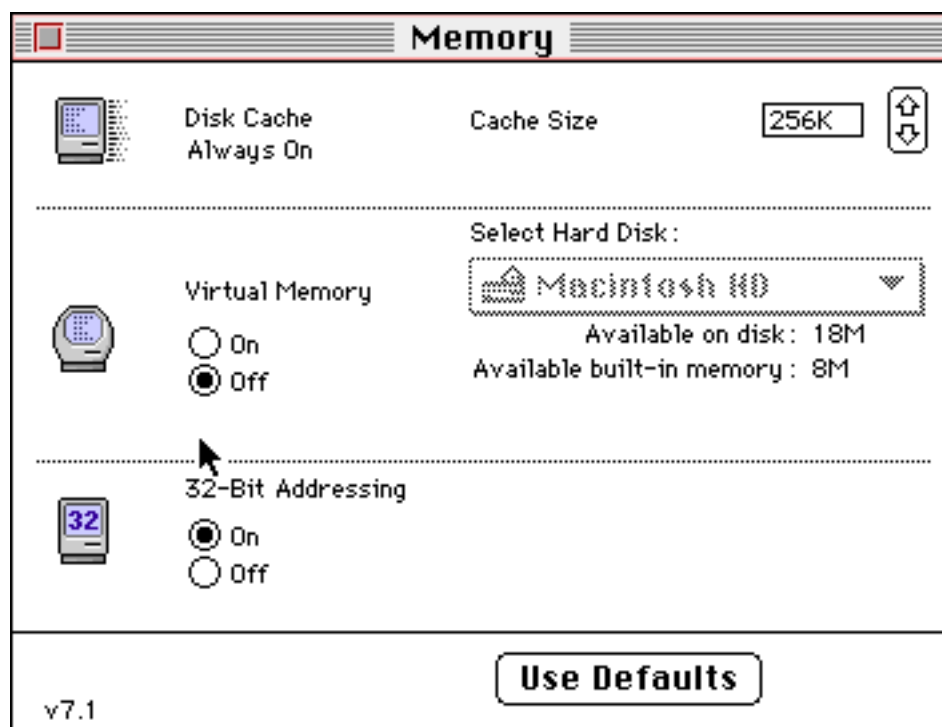


GPIB settings for Bus/Device gpib0



GPIB settings for Bus/Device Dev4

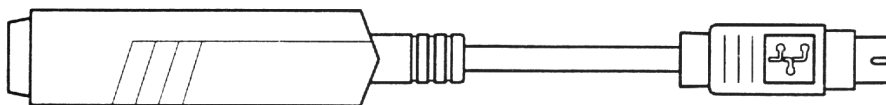
With the *Auto* box checked, the software should automatically configure the bus to the proper hardware interface association after the computer is restarted. Bus slots may be specified as numbers (4,5,6) or as letters. Next, from the Control Panels folder in the Apple menu, select 'Memory'. When this panel runs, it should look like the illustration below.



*Memory control panel dialog*

### 1.5 Molecular Analyst Hardware Protection Key

A hardware protection key (HPK) is required before using Molecular Analyst/Macintosh software. A single user HPK is included with each copy of Molecular Analyst software. The Macintosh HPK is first connected to any free Apple Desktop Bus connector at the rear of the Macintosh. Next, the keyboard cable is connected into the matching connector on the HPK. For an additional cost, multi-user network hardware protection keys (170-7623) are also available through Bio-Rad. This network key allows multiple users to use the software over the network at the same time. To install network hardware protection keys, first connect the hardware key as described above. Next, install or reinstall Molecular Analyst software and select "Custom Installation" from the "Easy Install" pull-down menu. Click on the check box labeled "Network Hardware Key Extension 2.1" and click "Install". Once installed reboot the computer. (ALL POWER MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION TO HPK).



## 1.6 Installing the Molecular Analyst\Macintosh Software

The Molecular Analyst\Macintosh system software comes with different software utilities and its own installation program. Before running the installer, make sure all extensions are disabled. To do this, restart the computer while holding down the <Shift> key until the computer has restarted. To load the program:

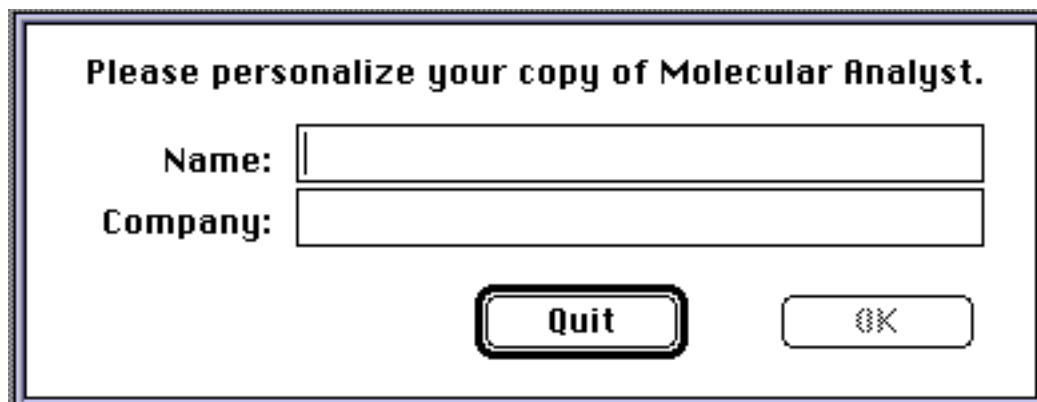
1. Insert the Molecular Analyst\Macintosh, "DISK 1" disk into a floppy disk drive. Double click on the "INSTALLER" icon and click OK. Next, select the installation location for the software and click on "Install". The Molecular Analyst\Macintosh installation program will then prompt you whether you want to run both Macintosh and Power Macintosh systems. Click on "YES". The program will then automatically prompt you for the proper disks.
2. The screen will display a dialog box indicating the status of the installation procedure.
3. The Molecular Analyst installation program will have created a new folder titled Molecular Analyst *f*. Double click on this folder to view the contents. The installation program installs the application Molecular Analyst, and NIH Image.

## 1.7 Startup

Start the Molecular Analyst\Macintosh program by double clicking on the Molecular Analyst icon. The program will load. After displaying the title screen, the menu bar will appear.

When the application starts up it will first check the system configuration. If the configuration does not meet the minimum requirements to run, an appropriate dialog will be displayed to the user then the application will quit.

If the system configuration meets the specified requirements then application will check itself to see if it has been personalized. If not then the user will be prompt for the user's name and company. The OK button will remain disabled until the user enters at least 3 characters in the Name field.



**Please personalize your copy of Molecular Analyst.**

**Name:**

**Company:**

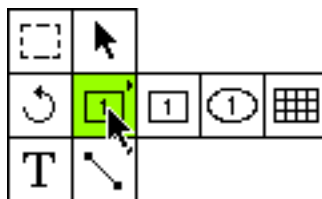
**Quit** **OK**

## Section 2 Image Tool Palette

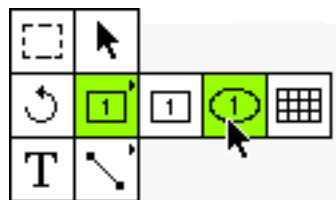
Throughout this manual you will refer to the tool palette. Familiarity with this palette is necessary for the Tutorial in Section 3. The tool palette, located on the left side of display windows, provide the user with tools for fast and simple image display and data optimization. These tools contain the most commonly used functions as listed below. Users select a tool by clicking on it at which point that tool will remain active until another tool is selected. When a tool has a triangle in the upper right corner, it means that there are different versions of that tool available. Simply clicking on that tool will make that tool active.



But if the user clicks and holds on the tool, the other versions of the tool will pop-up.



The user can then move the mouse and select the version of the tool to use.



When the user releases the mouse button that version will become selected and the tool icon will change accordingly.





## Object selection and positioning

Allows the user to select one or more objects. Also allow the user to reposition the currently selected objects.

Click selection - When the user clicks on an object it will become selected, and all previously selected objects will be de-selected.

Drag selection - The user can select multiple objects at once by dragging a selection rectangle around the objects. The user clicks in an area where there are no objects and drags a rectangle around the objects to be selected then releases the button.

Shift key extension - The user can make objects active or inactive by holding down the shift key while clicking on an object. If an object is not currently active, clicking on it while holding down the shift key will make it active. This function also applies to Drag Selection.

Positioning - The user can move the currently selected objects by moving the cursor within one of the selected objects and dragging the objects. This will also work for image selection objects.



## Image Selection

Provides the user with the capability of selecting an area to be used with the Macintosh Edit functions (copy and select all only). All image operations are based upon the current selection (the entire image if nothing is selected).

Rectangular selection - To select an area of the image with the rectangle selection tool, the user simply has to click and drag to mark an area.



## Rotate

Rotates the contents of the current rectangular selection or object at an arbitrary number of degrees. Rotation can be clockwise or counter-clockwise.

Using the cursor, the user clicks and grabs the corner of an object and rotates an outline of that object. When the user releases the button the object is then actually rotated at which point any needed recalculations will be performed.



## Text

Provides a text tool which allows the user to type text anywhere within the image displayed in the active window.

## Volume Integration Tools



### Rectangle



Allows the user to manually identify an image area to be integrated. The user will drag a rectangle to define the spot at which point the program can quickly integrate the complete area within the rectangle.

### Ellipse



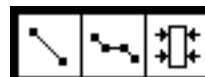
Allows the user to manually identify an image area to be integrated. The user will drag an ellipse to define the spot at which point the program can quickly integrate the complete area within the ellipse.

### Two-Dimensional Grid



Provides a pre-defined as well as user defined grid templates which when selected will display a cell grid on top of the image in the active image window. The grid's dimensions are defined by the grid parameters (in the OBJECT menu). The program will then integrate the area within the grid and display the information.

## Profile Integration Tools



### Single Line



Allows the user to select a single pixel "slice" of the image for exportation to the Profile window for additional analysis. To use this tool the user simply clicks and holds the mouse button to start the line. Releasing the mouse button creates the line. Holding the Shift Key will restrict movement in the X or Y direction.

### Multi-Segment Line



Allows the user to select a single pixel "slice" of the image for exportation to the Profile window for additional analysis. To use this tool the user simply clicks to start the segments. Move to the next location (a line will follow the cursor), and click again to mark another anchor. To mark the last anchor, the user double-clicks to define that anchor and end the line drawing. Holding the Shift Key will restrict movement in the X or Y direction.

### Rectangle



Allows the user to select an area of the image (pixel averaging) for analysis with the profile analysis module of the program. The profile will always be compressed along the

narrow side of the rectangle. Holding the Shift Key will restrict movement in the X or Y direction.

### Zoom Levels



Located at the bottom left of a saved image, the zoom feature provides the user a means of zooming in or out of the image. When the user clicks on the left or small mountains the entire image will be zoomed out by one level. When the user clicks on the right or large mountains the image will be zoomed in by one level. The middle zoom level display serves two purposes. One, it displays the current zoom level, and two it provides the user a way to toggle between the current zoom level and 100 percent (not zoomed). When the user clicks on the zoom level display the image will toggle between the current zoom level and 100 percent (not zoomed).



### Normal Molecular Imager Image Processing

Normal provides simple processing, a median filter and auto-transformation adjustment, for acquired images (using the *Info* button in the Acquisition window) and saved images. This only effects display data and does not effect raw data or calculations. To process a previously acquired Molecular Imager image with the “Normal” image processing

option, click on the  that is displayed under the tool palette in the display window.

The icon will change to  and the image will be processed. To undo the processing, simply click on the  and the icon will return to .

## Section 3 Molecular Analyst/Macintosh Tutorial

Familiarity with your Macintosh Operating System is assumed. If you are unfamiliar with the operation of the densitometer, Molecular Imager, or Gel Doc, please refer to your user manual for more information.

In the following tutorial, you will learn to display and optimize the example image provided with the software. You will then learn how to use the different tools to perform volume and area integration, and send that data to the results table. Depending upon the image analysis instrument used, data values can be displayed as O.D. (Densitometer) or PDU (Gel Doc 1000 and Molecular Imager).

### 3.1 Display and Optimization of an Image

In this section, you will work with an example image provided with the Molecular Analyst/Macintosh software.

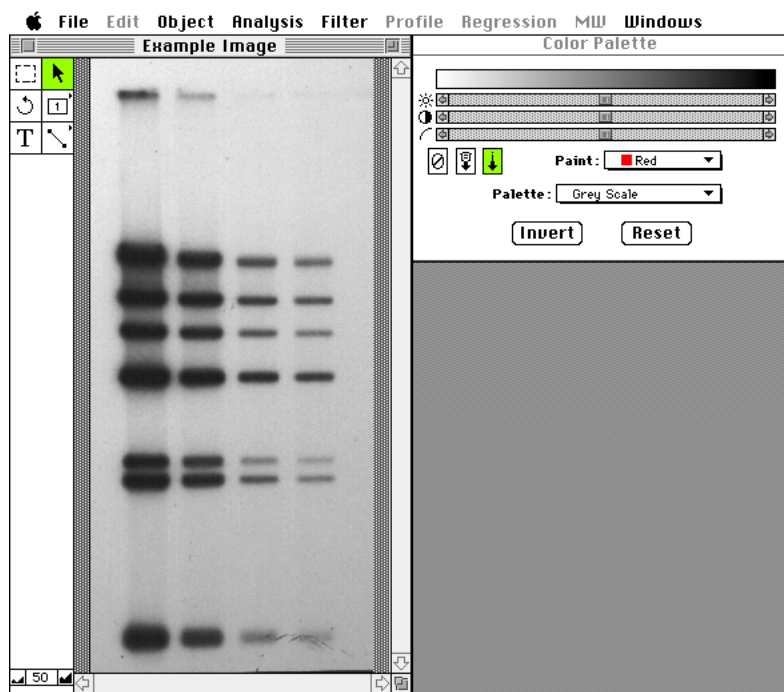
- 1) You will first start the program
- 2) You will then display the example image included with the software
- 3) You will prepare it for analysis by adjusting gray levels using the histogram palette.
- 4) You will then adjust image brightness / contrast of the image.
- 5) You will next magnify the example image.
- 6) You will scroll throughout a zoomed image.
- 7) Next, you will reduce high frequency noise in the image.
- 8) You will print the image to a laser printer.
- 9) Finally, you will save the image to file under a new name.

#### **1) Starting the program**

Open the folder that contains the Molecular Analyst/Macintosh program. Start the program by double clicking on the Molecular Analyst icon. The program will load. After the title screen, the menu bar for the program and color palette will remain.

#### **2) Displaying an image**

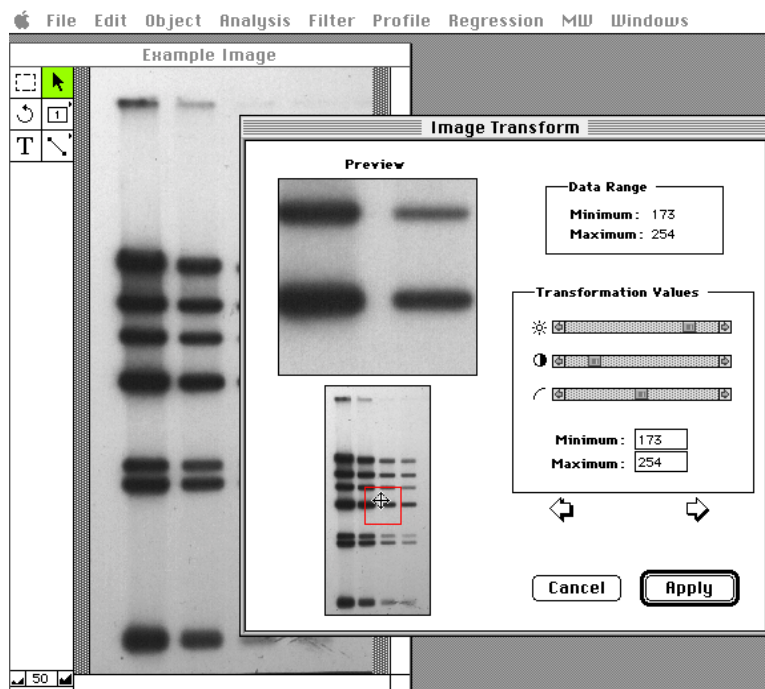
To display the example image provided, click on the FILE menu, and drag the mouse to *Open*. A standard file dialog box will appear showing the images available within that folder. Double-click on the image file "Example". The image will then load and display in the image window as in figure below. Alternatively, double clicking directly on "Example" image will also display the image. **Image files from previous versions of Molecular Analyst (1.1.1) can only be opened using the Open command from the FILE menu. These older images cannot be opened by double-clicking on them.**



*Displaying the Example Image*

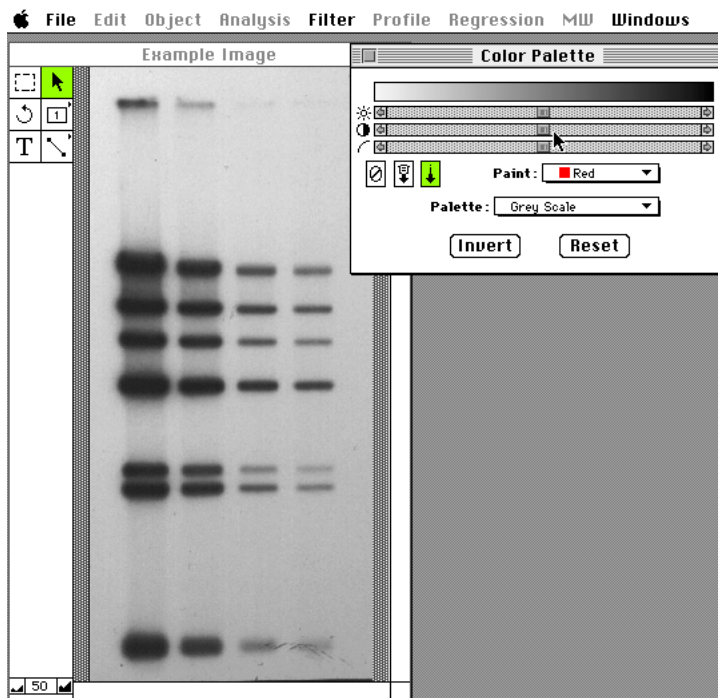
### **3) Gray level adjustment with Transformation**

Select *Transformation* from the ANALYSIS menu. This displays the transformation palette. This palette allows the image display to be improved in preparation for quantitation. Move the small preview box to any section of the image. Adjust the brightness, contrast, and curvature scroll bars until most of the bands are clear. Next, click on *Apply* to apply the transformation changes to the entire image. Transformation only modifies display data and not the underlying saved data. Furthermore, image modifications using the transformation palette can be printed to any Macintosh printer. Please see section 4.5 for more information on Transformation.



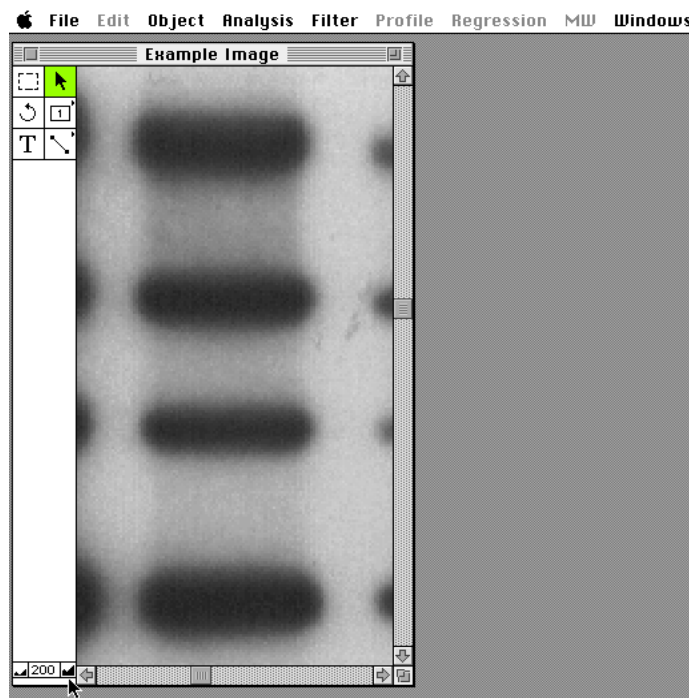
#### 4) Adjusting image brightness and contrast

The color palette is automatically displayed when an image is opened. Brightness, contrast, and color can be further adjusted by dragging the slider bars left or right until the desired view is displayed. The color palette changes display data and not underlying saved data and like transformation, color palette changes can also be printed to a Macintosh compatible printer.



#### 5) Magnifying the image

The example image is displayed at 50%. Click on the right or large mountains located at the bottom left of the image. The image will be zoomed out by one level to 100%, and the title bar will display the percentage of zoom. Clicking the large mountains again will zoom 200%. To reset the image display to its 100% size, click on the middle zoom box. To return to 50% click on the smaller mountains.



#### **6) Scrolling the image**

Scroll a zoomed image using the standard Macintosh scroll bars (the right and bottom of the image). The scroll bars will only be displayed if the image has been magnified.

#### **7) Reducing noise in the image**

To reduce noise you can use one of several filters which are available (Please see Section 4.6 for a description of the available filters). Select *Weighted Noise Reduction (5x5)* from the FILTER menu. To undo the filter and try another filter, select *Undo* from the EDIT menu. If you wish to change back to the smoothed image simply select *Redo*. Filtering only effects raw data if the *Filter Raw* command is active in the FILTER menu.

#### **8) Printing the image to a laser printer**

To print the image, first click on Preferences from the FILE menu.

- a. To print the actual size of the image, turn off "Print image on single page".
- b. To print the image, click on the image display window to make that window active.
- c. From the FILE menu select *Page Setup*. Print sizes may be varied by using the "% Reduce or Enlarge radio button. If the "Print image on single page" option is on, the image will be expanded or compressed to fit the page and ignores any change in the Reduce or Enlarge % window.
- d. Select *Print* from the FILE menu to send the image to the print spooler. Be sure that the Color/Grayscale option is chosen for halftone printing in the print window. For the best quality image it is recommend that the image be printed on an Apple LaserWriter IIx printer or better with the Apple Photograde feature enabled.

Refer to Section 4.2 "Acquiring an image from the Gel Doc 1000" for information on printing to a video printer.

#### **9) Saving the image**

Select *Save As...* from the FILE menu to save the modified example image under a different name and click OK.

### 3.2 Image Analysis: Volume Integration

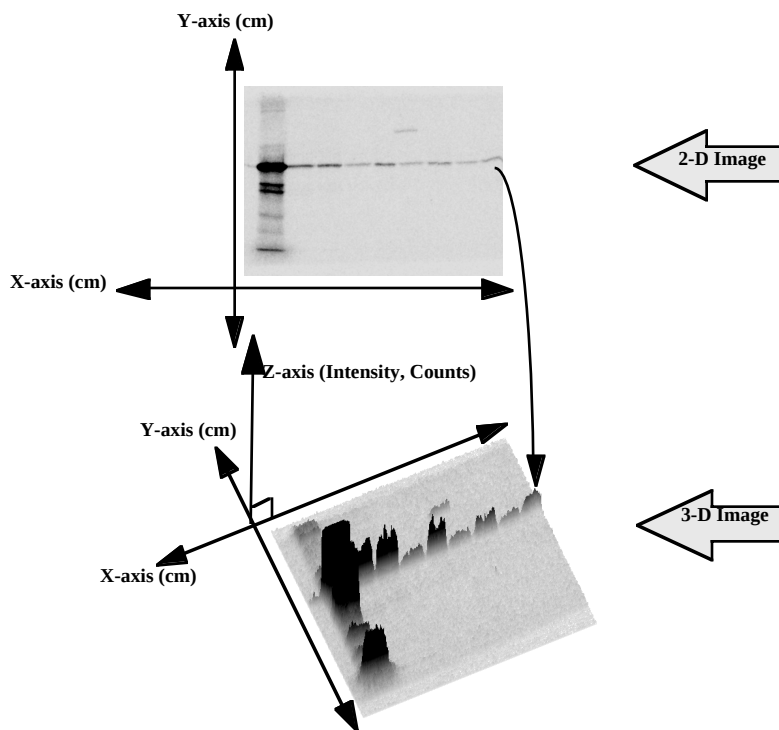
In this section you will be given an explanation of image analysis and then learn how to integrate image objects or "spots" using the volume integration tools.

- 1) First you will create objects to identify the areas you wish to integrate.
- 2) You will duplicate objects for quick analysis.
- 3) You will direct the program to calculate the volumes within the identified areas.
- 4) You will then subtract background values from your data
- 5) You will send the results to the "Results" window.
- 6) You will assign standard values
- 7) You will print your results
- 8) You will learn how to export your results
- 9) Finally, you will save your results to a file.

#### ***Overview of Image Analysis***

The Gel Doc 1000, Densitometer, and Molecular Imager systems are light or radiation detectors that are capable of converting biological signals into digital data. The digital data is then displayed on the computer in a 2-dimensional format using the Molecular Analyst software. The intensity of the various signals (i.e. bands or spots) is differentiated by the intensity of the gray levels displayed on the computer monitor. The stronger the signal, the darker the image on the monitor and vice versa.

Currently, most image analysis software displays data in a 2-dimensional format. However, the 2-dimensional display does not reflect all the information stored in the digital data. A 3-dimensional representation of the data (not available in Molecular Analyst) gives a more realistic and clear picture. (Figure 1).



**Figure 1.** Two and 3-dimensional representations of a digitized image.

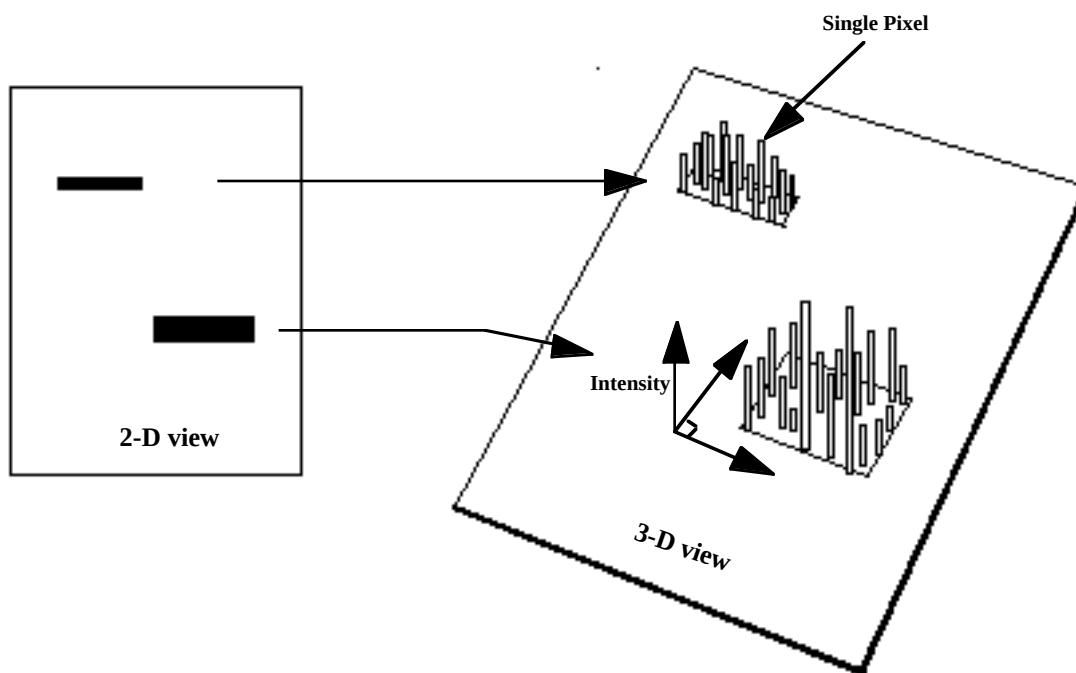
In a 3-dimensional view, signals are displayed as mountains or hills in a landscape surrounding. Therefore, the strength or intensity of the signal is represented by the volumetric capacity of the mountains or hills. It is essential to be aware that the 2-dimension image display represents 3-dimensional data. Understanding this concept is essential for accurate quantitative and qualitative analysis of the digital data.

The digitized data provides three important pieces of information necessary for qualitative and quantitative analysis (See Figure 1):

X-coordinate which indicates the signal horizontal position  
Y-coordinate which indicates the signal vertical position  
Z-coordinate which indicates the signal intensity

The x- and y- coordinates define the exact location of the signal, whereas the z- coordinate establishes the intensity of the signal. The units of the x- and y- coordinates is millimeters or centimeters. The units of the z-coordinate (intensity) is “Optical Density “ or O.D. when using the densitometer, and “Pixel Density Units” or PDU when using the Gel Doc 1000 or Molecular Imager systems.

The signal displayed on the monitor is generated from the assembly of hundreds of tiny individual screen pixels, each exhibits its own intensity level or count (Figure 2). In order for a signal to be visible, the counts of these congregated pixels must be higher than those pixels that represent the background. The total count of a signal is determined by summing up the individual intensity or count of all the pixels (Equation 1). The mean count or intensity unit of a signal is the total intensity unit divided by the total number of pixels (Equation 2).



**Figure 2. Pixel representation of digital signals**

Equation 1

**Total PDU or O.D.=  $\sum (I_1 + I_2 + \dots)$ , where I = intensity or count of each pixel**

Equation 2

**Mean PDU or O.D. = Total PDU or intensity / total number of pixels**

Accurate quantitation of the data is only possible if the signal lay within a maximum and minimum range of intensity level. This quantifiable range (which are often referred as the linear response or linear dynamic range) varies depending on the type of detectors. Table 1 summarize the theoretical and practical quantifiable range for three types of instruments. The Gel Doc 1000, GS-700 Densitometer and the GS-363 Molecular Imager have the maximum linear range of 0-256 counts ( $2^8$  bits), 0-4,096 OD ( $2^{12}$  bits), and 0-65,535 ( $2^{16}$  bits), respectively. However, in practice, there are other parameters that limits the quantifiable range. For example, the linear range of densitometer samples (i.e. x-ray film) limits the quantifiable range to approximately 2.0 O.D..

| Instrument              | Digital Gray Scale | Theoretical Linear Range | Limiting Factor |
|-------------------------|--------------------|--------------------------|-----------------|
| <b>Gel Doc</b>          | $2^8$ bit          | 256                      | Instrument      |
| <b>Densitometer</b>     | $2^{12}$ bit       | 4,096                    | Film & Samples  |
| <b>Molecular Imager</b> | $2^{16}$ bit       | 65,536                   | Instrument      |

Table 1. Theoretical and practical linear dynamic range

### 1) *Creating volume integration objects*

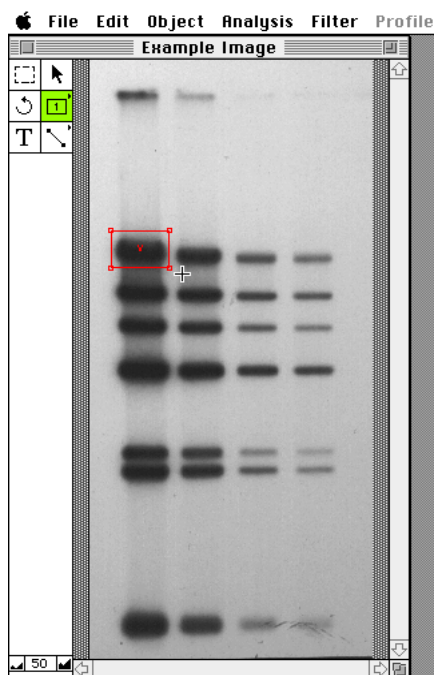
- a. To create volume objects, choose the rectangle  volume integration tool from the image display window tool palette.



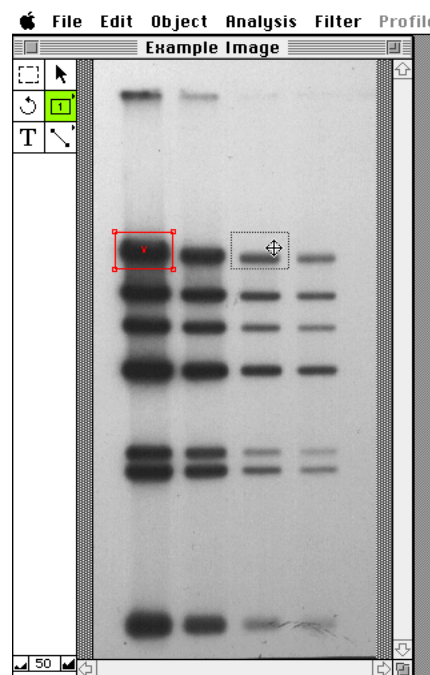
To integrate some of the bands within the example image you will use the rectangle volume tool as it most closely resembles the shape of the objects you wish to integrate.

- Click on the rectangle integration tool.
- Move the cursor into the image display window.
- Now move the cursor above and to the left of the first band which you wish to quantitate. Click and drag the mouse down and to the right of the band as shown in the figure below. The band should now be completely enclosed by the integration rectangle. Releasing the mouse will finish the spot.

If the spot is not in exactly the right place, move the cursor back over it. The cursor will change to a directional cross , indicating that you can now move the object. Click and drag the spot to the new area.



*Creating a volume object*



*Moving an object*

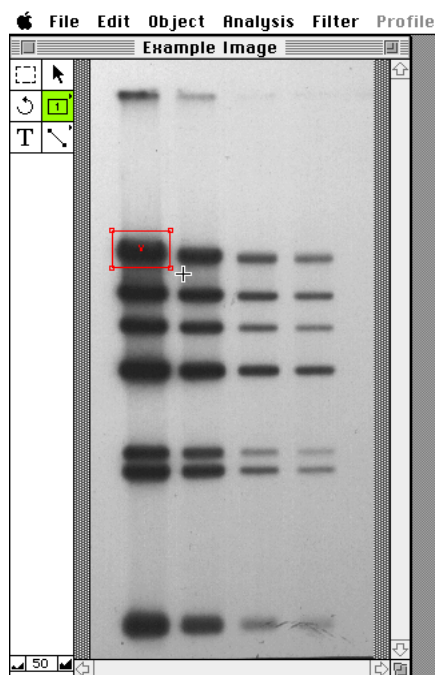
You can draw as many objects you want using the rectangle, oval or grid tools. However, note that only the most recent spot is displayed in red. This color indicates the active spot. Only the active spot can be moved, resized, erased or duplicated. To make any spot active, simply move the **ARROW** cursor (also called the positioning cursor) to it and click. It will change color to indicate its changed status. The arrow cursor can be used to duplicate and resize any object at any time. The volume tools only resize or duplicate current active objects. To resize an object, simply place the cursor on the corner of the object. The cursor will become an arrow. Next, click and hold the mouse button and adjust the size of the object. To delete an active object or objects, press the delete key.

## **2) Duplicating volume integration objects**

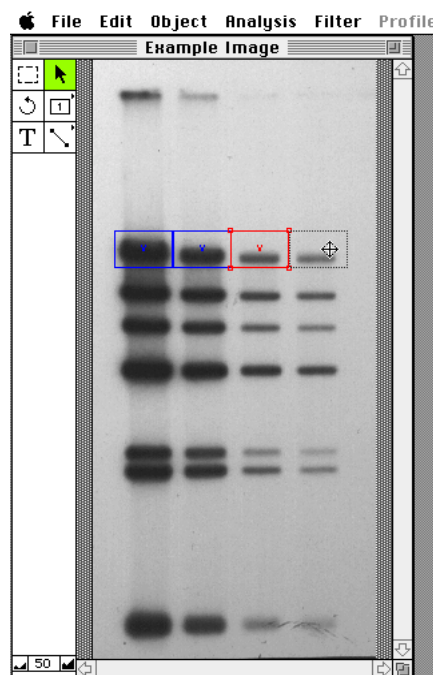
Often you want to obtain relative quantitation values within an image. The easiest way to compare values within different sections of the image is to use the same area for each spot. This simplifies interpretation of the resulting values. To duplicate the object which was drawn with the rectangle tool:

- a. simply move the cursor inside of the active object you wish to clone. If the object is not active, simply click on it first.
- b. Next hold down the *Option* key.
- c. Click and drag the duplicated object to the next band in the first row.

Continue duplicating the same object by clicking and dragging again to the next location. Continue this procedure until all bands in the first gel row are enclosed.



Creating a volume object



Duplicating volume objects

### 3) Integration of the bands

Now that all of the bands in the first gel lane have been identified, select *Auto Integrate* from the ANALYSIS menu. The objects will automatically be numbered, (in the order in which they were created), integrated, and the results sent to the Results window.

| untitled    |         |              |                 |             |             |                      |                           |          |               |             |
|-------------|---------|--------------|-----------------|-------------|-------------|----------------------|---------------------------|----------|---------------|-------------|
| Volume Name | Type    | Mean<br>(OD) | Area<br>(mm*mm) | Min<br>(OD) | Max<br>(OD) | Volume<br>(OD*mm*mm) | Adj. Volume<br>(OD*mm*mm) | % Volume | Concentration | Mol. Weight |
| 4 items     |         |              |                 |             |             |                      |                           |          |               |             |
| V1          | Unknown | 0.99         | 41.82           | 0.50        | 1.43        | 4.1364e+4            | 4.1364e+4                 | 33.20    | -             | -           |
| V2          | Unknown | 0.80         | 41.82           | 0.49        | 1.28        | 3.3325e+4            | 3.3325e+4                 | 26.75    | -             | -           |
| V3          | Unknown | 0.61         | 41.82           | 0.50        | 0.98        | 2.5697e+4            | 2.5697e+4                 | 20.62    | -             | -           |
| V4          | Unknown | 0.58         | 41.82           | 0.49        | 0.84        | 2.4213e+4            | 2.4213e+4                 | 19.43    | -             | -           |

### 4) Subtracting background from individual spots

Local or general background subtraction is available and is necessary to remove values contributed from background noise.

- To subtract local background from individual spots, choose *Local Background* from the ANALYSIS menu. If *Auto Integrate* is on, local background will be automatically subtracted. If *Auto Integrate* is not on, choose *Integrate Now* to calculate local background subtraction.

Outside and adjacent to each volume object, a two pixel width perimeter is then recorded. These values are then averaged to calculate as a local background for that volume object. This value is then used in the background subtraction of spot values and the results table will be updated to reflect the subtraction of background.

Alternatively, to perform general background subtraction, a volume object or several volume objects are designated as background in the volume object dialog box (double clicking on the object and selecting background from the *Type* pull-down menu). This value is then used in the background subtraction of spot values when *Auto Integrate* or *Integrate Now* is used.

### 5) Looking at results from individual spots

Double-clicking (with the arrow cursor) on an integrated spot, or selecting *Get Spot Info* will open up a volume object dialog box containing an objects calculated parameters. Alternately, double-click on any line in the Results window to also open up the same dialog window. Notice that you can give each spot a name in the expanded data dialog box, define its type (unknown, standard, or background), enter in a standard value (concentration, molecular weight, or pI), and add comments.

**U1**

Name: Band 1

Type: Unknown Shape: Rectangle

X-Center: 38.40 (mm) Mean: 0.94 (OD)  
Y-Center: 264.90 (mm) Area: 48.16 (mm\*mm)  
Width: 8.80 (mm) Minimum: 0.50 (OD)  
Length: 5.80 (mm) Maximum: 1.43 (OD)  
Aspect Ratio: 0.66 Porosity: 0.06

**Volume**

Volume: 4544.10 (OD\*mm\*mm)  
Adj. Volume: 1778.06 (OD\*mm\*mm)  
% Volume: 50.93

**Calibration**

Concentration: 0.00  
Molecular Weight: 0.00  
pI: 0.00

**Comments:**

### 6) Assigning standard values and generating a standard curve

Within this same dialog window you will now assign standard concentration values to three of the bands and calculate the unknown concentration of the fourth band. **It is critical to note the orientation of the standards in the image. The orientation of the standards (horizontal or vertical direction) are set within the image information dialog box which is accessible through the *Get Image Info* command in the FILE menu.**

**Example Image Info**

**Name:** Example Image  
**Kind:** Molecular Imager (Original)  
**Size:** 1.0 MB on disk (1,050,068 bytes used)  
**Where:** Macintosh HD:Desktop  
 Folder: Molecular Analyst f:  
**Width:** 400 (pixels) 4.0 (cm)  
**Length:** 976 (pixels) 9.8 (cm)  
**Bits/Pixel:** 16  
**Scan Res:** 100 (µm)  
**Scanned:** Tue, Mar 9, 1993, 12:35 PM  
**Created:** Wed, Jun 15, 1994, 11:18 AM  
**Modified:** Wed, Jun 15, 1994, 11:18 AM  
**Version:** 2.0b2  
**Direction:** ☐ Vertical ☒ Horizontal  
**Comments:**  
 \_\_\_\_\_  
**Scanner Info**  
**Type:** GS-250 Molecular Imager  
**Firmware Version:** Unknown  
**Hardware Version:** Unknown  
**Serial Number:** -

Setting the direction of the standards in the Image Info Dialog box.

Double click on the first row of the volume report to open the dialog window of V1.

| untitled 3             |         |              |                 |             |             |                      |                           |          |               |             |
|------------------------|---------|--------------|-----------------|-------------|-------------|----------------------|---------------------------|----------|---------------|-------------|
| Volume Name<br>4 items | Type    | Mean<br>(PD) | Area<br>(mm*mm) | Min<br>(PD) | Max<br>(PD) | Volume<br>(PD*mm*mm) | Adj. Volume<br>(PD*mm*mm) | % Volume | Concentration | Mol. Weight |
| V1                     | Unknown | 1773.83      | 23.12           | 43.00       | 4227.00     | 4.1011e+4            | 4.1011e+4                 | 39.60    | -             | -           |
| V2                     | Unknown | 1149.85      | 23.12           | 21.00       | 4058.00     | 2.6584e+4            | 2.6584e+4                 | 25.67    | -             | -           |
| V3                     | Unknown | 1013.66      | 23.12           | 28.00       | 3212.00     | 2.3436e+4            | 2.3436e+4                 | 22.63    | -             | -           |
| V4                     | Unknown | 542.42       | 23.12           | 31.00       | 2253.00     | 1.2541e+4            | 1.2541e+4                 | 12.11    | -             | -           |

Change the name from V1 to Band 1. Next, click on the *Type* pull-down menu and select Standard.

**Type:** Unknown  
☒ Standard  
 Background

This will activate the Calibration section of the dialog window.

**Calibration**

**Concentration:**

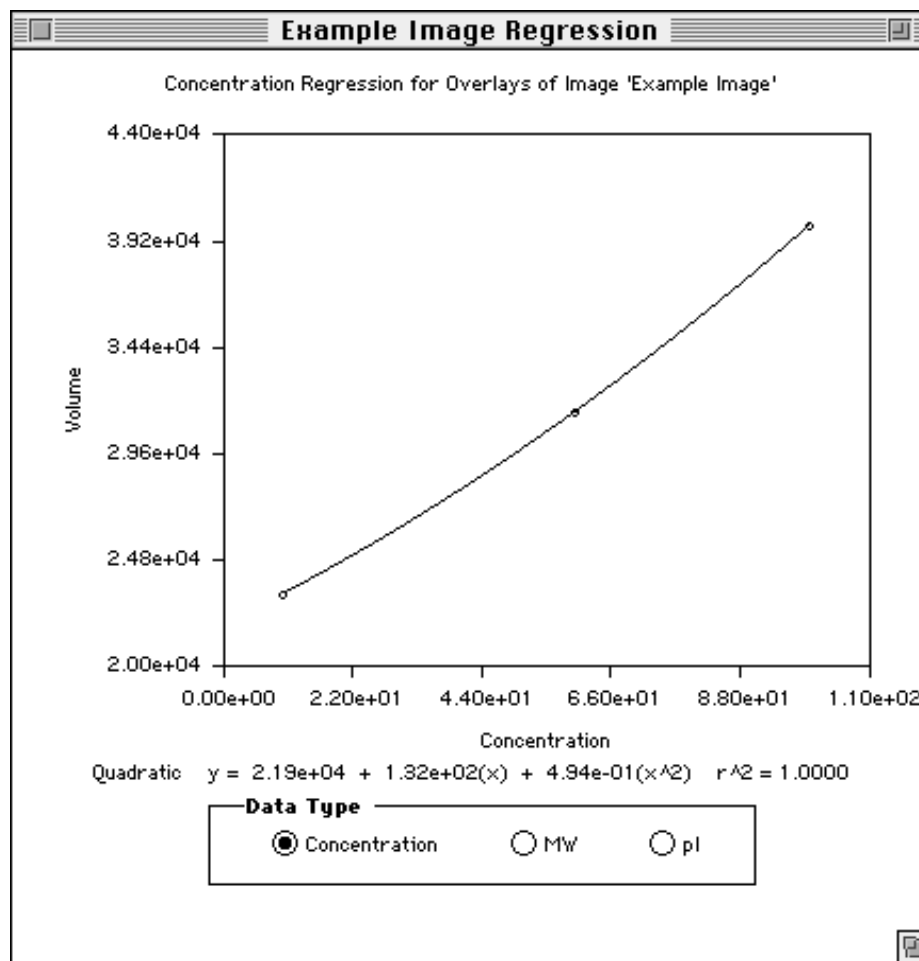
**Molecular Weight:**

**pI:**

Enter in the value 100.00 in the Concentration box and close the window. Notice that the new names have been placed on the image. To change the font size of the text, click on the image and select *Size* from the OBJECT menu. **Note: The font size for grids cannot be changed.** Repeat these same steps for row 2 and 4 of the volume report (names Band 2 and Band 4) and enter in the standard values of 60.00 and 10.00 respectively.

Now, select *Integrate Now* from the ANALYSIS menu. The concentrations, names, and types should automatically reflect the changes. To check the correlation coefficient of your standard data set, select *Regression* from the Windows menu. **This command is only active with the image in the foreground.** This will display the default, linear regression curve for the standard data points. The type of regression can be changed within the REGRESSION menu.

Select *Quadratic* from the REGRESSION menu and notice how the data for the unknown Band 3 within the volume report automatically changes.



Regression Window

### 7) Printing results

Next you will print the volume integration data in the Results window. Make the Results window the active window by clicking in the window to bring it to the foreground. Note that the width and position of the Results window can be adjusted using the *Page Setup* command. Next, select *Print* from the FILE menu to send the data to the printer spooler.

### 8) *Exporting results*

You will now export the information in the Results window to the clipboard. Once in the clipboard the results can be exported to other applications. Highlight all the rows in the result window (or use Select All) and select *Copy* from the EDIT menu. The data is now ready to be pasted to another application such as Microsoft Excel.

### 9) *Saving results*

You will now save the information in the Results window. Select *Export* from the FILE menu, and enter a name in the dialog, and click on the *Save* command. The title bar in the Results window will now update reflecting the new name.

## 3.3 Image Analysis: Area Integration/Molecular Weight Calculation

In this section, you will work with the example image provided with the Molecular Analyst/Macintosh software to create one-dimensional profiles.

- 1) You will use the profile tools to create a 1-dimensional profile.
- 2) You will remove profile background.
- 3) You will smooth profiles to reduce noise.
- 4) You will identify peaks.
- 5) You will save profile and peak results.
- 6) You will calculate a standard curve.
- 7) You will create a regression curve to calculate molecular weight.
- 8) You will generate a standard curve results table.
- 9) You will calculate the molecular weights of your unknowns.
- 10) You will print the profile graph and results.

### 1) *Creating profiles*

To create profiles you will use the profile tools from the image display window to identify the sections of the image for profile generation.



Click on the Profile tool icon and select the Rectangle tool.



- a. Move the cursor into the image window above and to the left of the first band in Lane 1 of the gel image.
- b. Click and drag the cursor down through all bands of the gel lane and toward the right side of the last band. If it is necessary to move the object, move the cursor into the profile object (the cursor will change to a arrow-star) and click the mouse and drag the cursor to the proper position, as shown below.
- c. If the enclosing rectangle is not the correct size, click and drag on any corner and readjust the size of the rectangle.

- d. If gel lanes are skewed, the rectangle object can be rotated using the rotate tool.

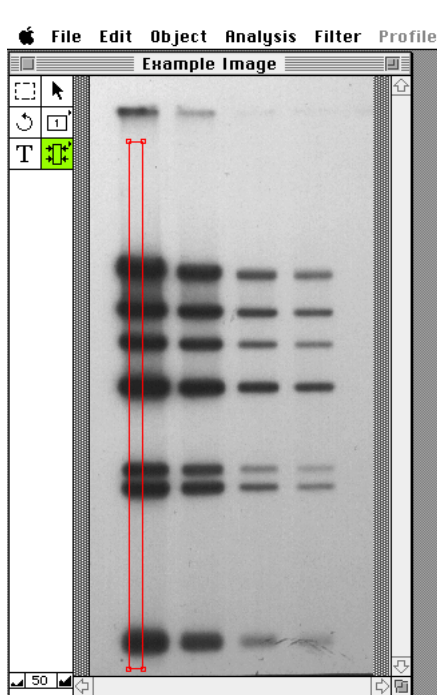


Profile objects can be duplicated for rapid analysis. To duplicate the object which was drawn with the rectangle tool:

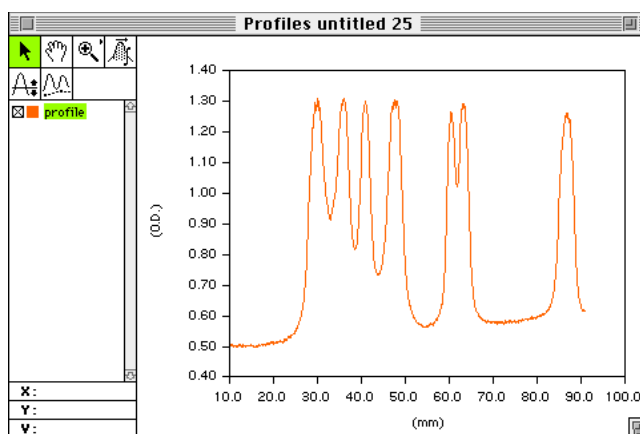
- simply move the cursor inside of the active object you wish to clone. If the object is not active, simply click on it first.
- Next hold down the *Option* key.
- Click and drag the duplicated object to the next lane.

Once the object or objects are in the proper position and are the correct size, select *Extract Profiles* from the ANALYSIS menu to send the column average plot to the Profile window. The profile is automatically plotted in the Profile window and the name will appear in the profile name list as "Profile".

Each profile name in the Name List has a check box and a color box in front of it. When the box is checked, the profile is shown on the Profile window, otherwise it is hidden. Selecting the profile names in the Name List (clicking to highlight the name) will select the corresponding profile as active (color) in the display window. Furthermore, any profile name that is underlined is designated as a baseline. However, only one profile can be designated as a baseline at a time.



Drawing a profile object

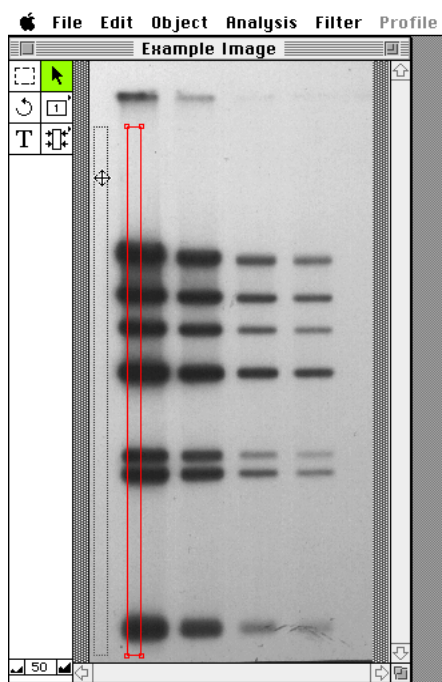


Generating a new profile

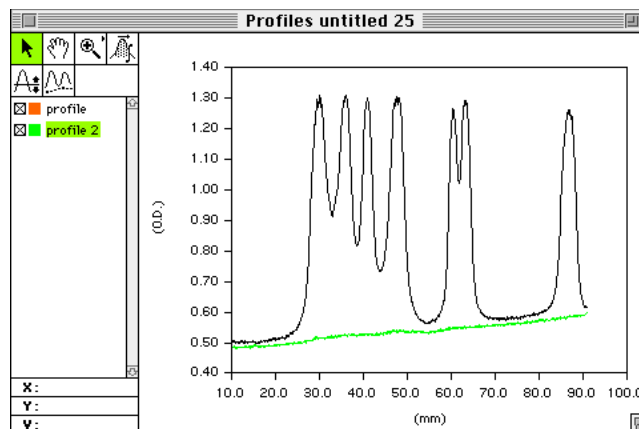
## 2) Removing Background Area from Profiles

To subtract actual background data, you will first select an area of the image to use for background removal. You must create a background profile of a length identical to that of the sample lane profiles you will be subtracting the background data from.

- Move the cursor inside the Profile object created earlier.
- Holding the <Option> key down (to duplicate the object) and while holding the <Shift> key down (to constrain X and Y movement), click and drag the new profile object to the left side of lane 1.



Duplicating a profile object and using it to define background



Generating a background profile

- c. Releasing the mouse button will create the background object.
- d. Select *Extract Profiles* from the ANALYSIS menu. The background profile will be sent to the Profile window and is automatically named "Profile 2".
- e. Next, again using the <Shift-Option> keys, duplicate another object and place it on the second lane of data.
- f. Select *Extract Profiles* from the ANALYSIS menu to send this profile to the profile window. This profile is automatically named "Profile 3".
- g. Double click on the name "Profile" in the list. The profile information dialog will open with the profile name highlighted. Type "STD" to replace the name "Profile". Close the dialog box. Repeat this process with "Profile 3" and type the name "Unknown".
- h. Double click on the name "Profile #2" in the title bar. The profile information dialog will open with the profile name highlighted. Type "BKGD" to replace the name "Profile #2". Close the dialog box and the name in the profile title bar will reflect the change.
- i. Next, with the background profile active, select the command *Use as Baseline* from the PROFILE menu. The name "BKGD" will then be underlined to indicate a background profile.
- j. With the "STD" and "Unknown" profile names highlighted (click on each while holding the <Shift> key) and BKGD **not highlighted** (use <Command> key for discontinuous selection), select the *Subtract Baseline...* command from the PROFILE menu. The area under the background profile will then be subtracted from the area under the "STD" and "Unknown" profiles. **Note: Only active profiles (highlighted profile names) may have functions performed on them.**

### 3) Reducing Profile Noise

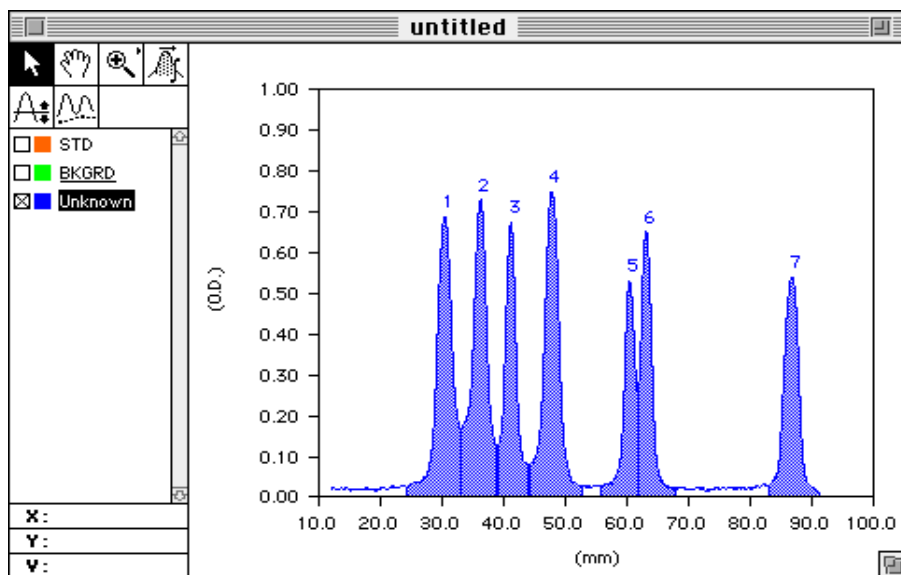
You will now reduce high frequency noise within the “STD” and “Unknown” profiles by applying a smoothing option to them.

- To smooth profiles, smooth criteria is first set in by opening the *Smooth Options* command in the PROFILE menu. For this tutorial you will use the default values.
- Next, with the “STD” and “Unknown” profiles active, go to the PROFILE menu and select *Smooth Profiles*. The profile will be smoothed in three cycles, according to default setting.

### 4) Automatic Peak Identification

Now you will automatically identify peaks in our “Unknown” profile.

- First, hide the “STD” profile by clicking in the box, and removing the “x”, next to the name “STD”.
- Next, you will set the automatic peak identification criteria. Under the PROFILE menu select *Peak Criteria*. Set the criteria units to “Absolute Units” by clicking on the radio button.
- Click all three minimum criteria check boxes and set the following minimum criteria values: peak height = .4 O.D. units, peak width = 0 mm, peak area = 0 O.D.•mm. **(Peak height assumes that the baseline has been subtracted from the profile).**
- Next, turn off “Smooth First” from the Options list.
- From the PROFILE menu select *Find Peaks*. Approximately seven large peaks should be identified and shaded. If more peaks were identified, select a larger peak height criteria.
- To see a list of these peaks, including their areas, go under the PROFILE menu and choose *List Peaks*. The results window is then automatically displayed. To view additional information about each peak, double click on the appropriate line in the list.



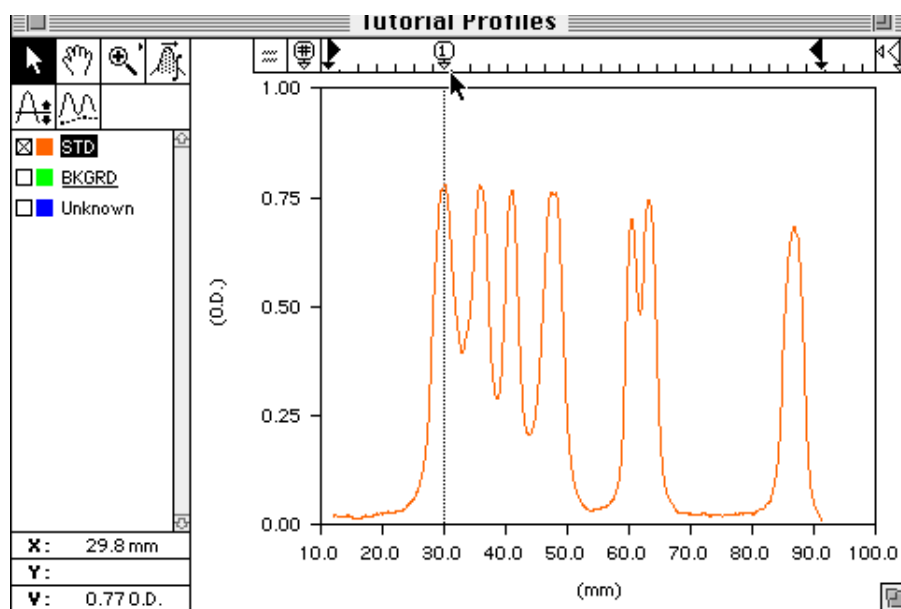
| untitled 2        |       |       |       |        |          |             |            |           |  |
|-------------------|-------|-------|-------|--------|----------|-------------|------------|-----------|--|
| Profile Area Name | From  | To    | Width | Height | Location | Area        | % of Total | Detection |  |
| 7 items           | (mm)  | (mm)  | (mm)  | (O.D.) | (mm)     | (O.D. * mm) | (%)        |           |  |
| Unknown.area 1    | 24.10 | 33.00 | 8.90  | 0.67   | 30.30    | 2.0787e+0   | 16.68      | Automatic |  |
| Unknown.area 2    | 33.10 | 38.80 | 5.70  | 0.60   | 36.10    | 2.0487e+0   | 16.44      | Automatic |  |
| Unknown.area 3    | 38.90 | 44.00 | 5.10  | 0.60   | 41.10    | 1.4851e+0   | 11.92      | Automatic |  |
| Unknown.area 4    | 44.10 | 52.60 | 8.50  | 0.72   | 47.70    | 2.2224e+0   | 17.84      | Automatic |  |
| Unknown.area 5    | 55.80 | 61.70 | 5.90  | 0.51   | 60.40    | 1.1028e+0   | 8.85       | Automatic |  |
| Unknown.area 6    | 61.80 | 67.80 | 6.00  | 0.63   | 63.10    | 1.3811e+0   | 11.09      | Automatic |  |
| Unknown.area 7    | 83.00 | 91.20 | 8.20  | 0.53   | 86.90    | 1.5181e+0   | 12.18      | Automatic |  |

### 5) Saving Profile and Results

To save the profiles, go to the FILE menu and select *Save*. A dialog box will appear prompting for the name of the file you want to save the profile under. Save the profile as "Tutorial Profiles". To save the information contained in the Results window go to the FILE menu and select *Export*. A dialog box will appear prompting for the name of the file you want to save the results data under. Save the results as "Unknown Profile Areas". To close the Results window, click in the small box in the upper left corner of the list box.

### 6) Calculating a Standard Curve

- Hide the "Unknown" profile by clicking on the box next to the name "Unknown" to remove the "X".
- Next, display and make active the hidden "STD" profile by clicking on the box next to the name "STD" and clicking on the name.
- Go under the MW menu and select *New Ruler*. A ruler will appear at the top of the graph with two flags for identifying the profile origin and front. The flags should be in the appropriate position and require little movement. However, if not aligned, click and drag the origin flag to the right until it is aligned with the top/origin of the gel. Now drag the front flag to the left until it is aligned with the bottom/front of the gel.
- Next, drag molecular weight markers (from the marker well in the ruler) and align them with the peaks. Move the first molecular weight marker until it is aligned with the center of the first peak. Enter the name "Fragment 1" and size of 23130 at the dialog that appears and click OK.



**MW Marker 1 Info**

**Name:** 1

**MW:** 100000.00

**Units:** kilobases

**Ruler:** untitled 36

**Created:** Fri, Jun 17, 1994, 2:57:33 PM  
**Modified:** Fri, Jun 17, 1994, 2:57:35 PM

**Comments:**

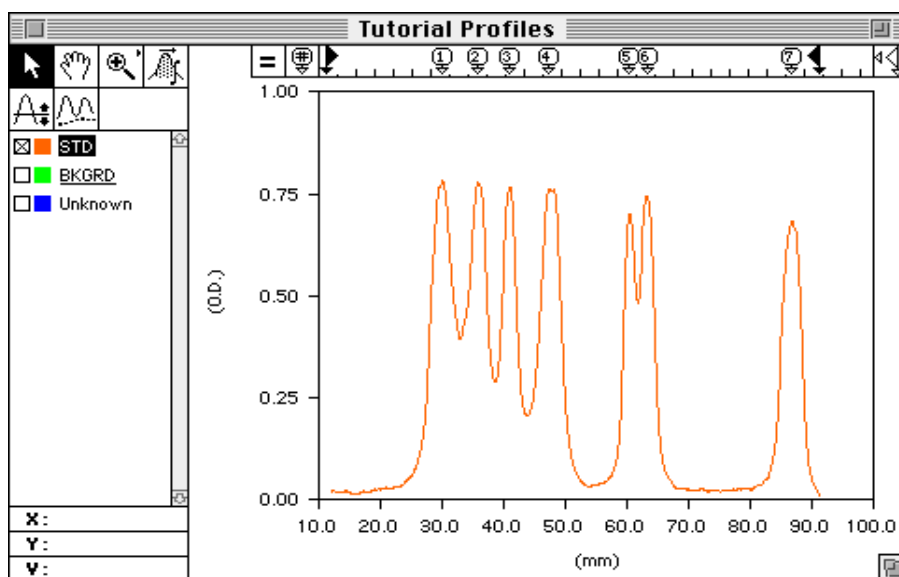
Cancel OK

- e. Continue this process until the remaining six fragments of the DNA size standards have been identified with the following information:

| <u>Name</u> | <u>Size</u> |
|-------------|-------------|
| Fragment #1 | 23130       |
| Fragment #2 | 9416        |
| Fragment #3 | 6557        |
| Fragment #4 | 4361        |
| Fragment #5 | 2322        |
| Fragment #6 | 2027        |
| Fragment #7 | 546         |

**Note: Do not abbreviate the molecular weight numbers of standards (i.e. kb)**

- f. Save the molecular weight set for future use by selecting *Save Set As...* under the MW menu.
- g. Now click on the calculate symbol (=) within the ruler. **(Any change in molecular weights or names will be implemented when the calculate symbol (=) is used.)** Notice that the X-scale of the graph window and the coordinate box are now in units molecular weight (or size).



*Alignment of DNA size markers*

- h. Scale the size of the window using the "Window Icon"(located in the lower right corner of the profile window) to view the x- scale better.
- i. To check the goodness-of-fit or correlation coefficient between the markers and the molecular weights, you can now look at the regression curve by selecting *Regression* from the WINDOW menu.
- j. From the Regression Window, select *Logistic*. Notice that the curve and regression formula change according to the type of regression analysis is selected. **The default regression parameter is always linear and should be changed to a higher level of regression (Logistic) for more accurate analysis.** The type of regression analyses can be changed between Linear, Quadratic, Cubic, Logistic, and Spline by selecting the respective command in the REGRESSION menu.

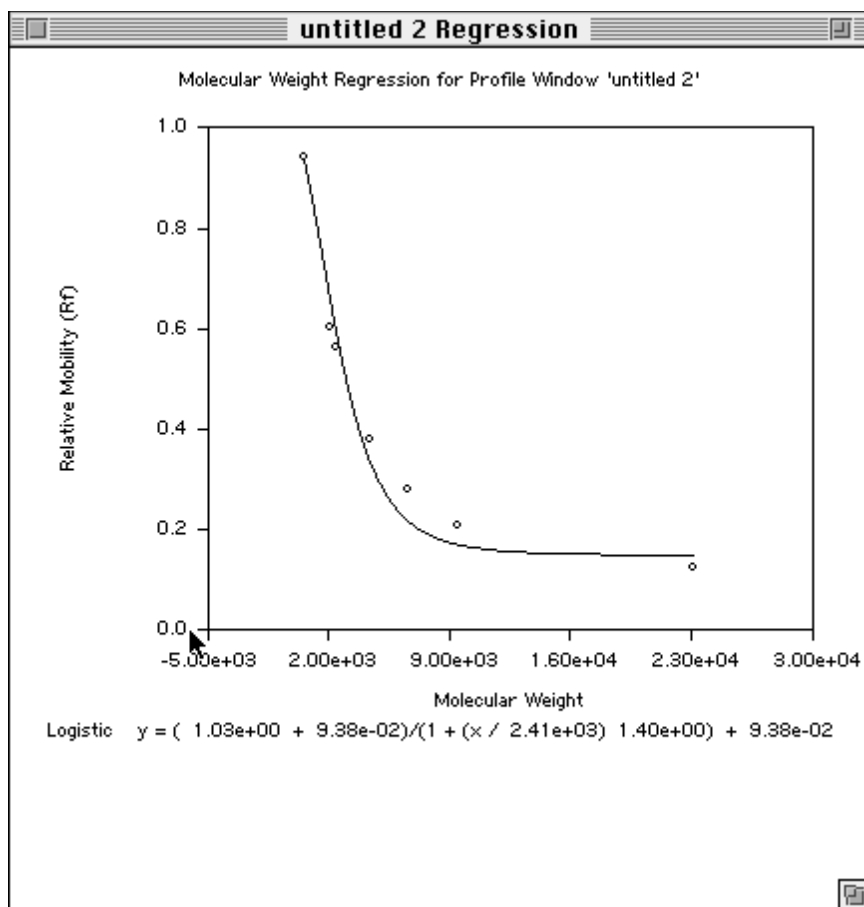
## 7) Regression Overview

The Regression Window provides an interactive graphical representation of the "goodness-of-fit" of your molecular weight, and concentration standards. All methods that extrapolate the molecular weight or concentration of an unknown substance require the use of a series of standards together with an equation that models their behavior. These equations or methods are generally based on the behavior of proteins or nucleic acids in specific separation media, such as polyacrylamide or agarose gels. The underlying assumptions of these methods may or may not hold for a particular experimental protocol. The best way to evaluate the suitability of a particular method is to determine the error implicit in its use. Although this can be done numerically, it is far easier to visually evaluate the fit between the equations and the standards.

The Regression window in the Molecular Analyst\Macintosh program provides a direct, visual representation of the fit between your molecular weight, names, or concentration standards. It also shows the percent error between the standards and the predicted molecular weight or concentration at that position. Since the Molecular Analyst\Macintosh program is multitasking, changes in the molecular weight ruler (in the Profile window), or STD concentration values in the profile window will be reflected in the Regression window after the calibration sign "=" is used.

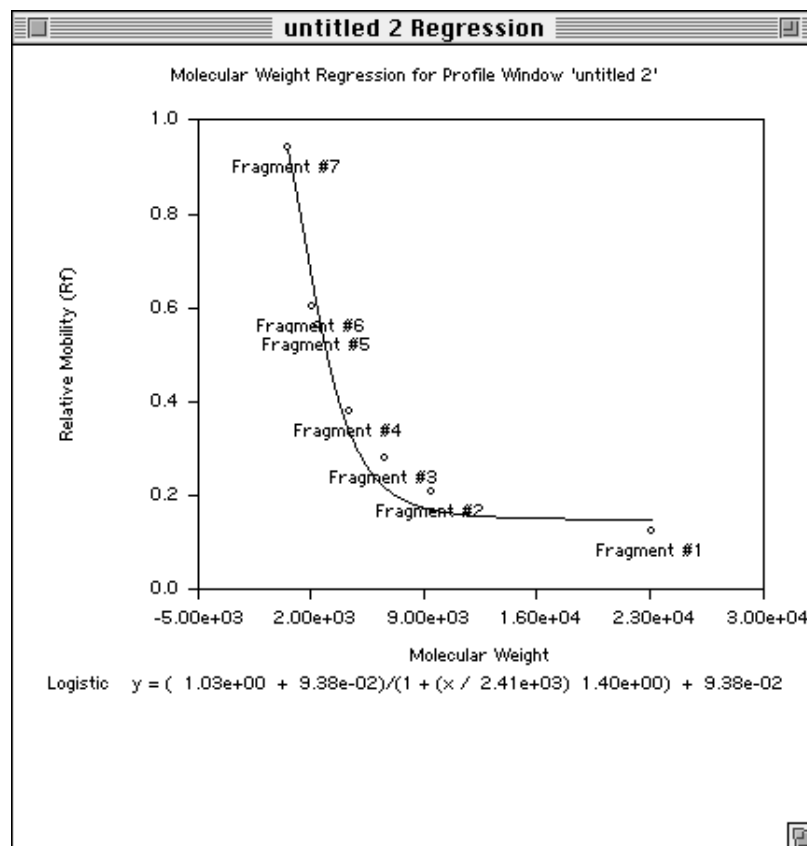
## 8) Using the Regression Window for MW Determination

When you open the Regression window, the regression curve will look similar to the following displayed below. The regression window can only be active when there is (1) an active molecular weight ruler in the Profile window, or (2) when volume standard concentrations have been entered in the volume object dialog window.



*The Regression window*

The y-axis displays the relative mobility (Rf) of the standard, and the x-axis displays its size or molecular weight. The Rf is the ratio between the distance from the top/origin of the gel to any given marker and the distance to the dye front. The mobility of any marker is determined by its placement on the ruler relative to the origin and front markers therefore, the placement of the origin and front markers is critical. The size of the markers is determined by the information entered in the Marker dialog box. The names of each marker can be displayed next to each point on the regression curve by selecting *Show Names* from the REGRESSION menu.



### 9) Changing the regression curve display

To display the errors associated with each marker, choose the Show Errors command from the REGRESSION menu. Both the names and errors can be hidden by selecting the *Hide Names* or *Hide Errors* commands. Additionally, names next to the regression points can be changed by double-clicking on the markers in the profile window, changing the name, and clicking “=”. The regression curve will be automatically updated.

These two menu options alternate at the same place in the REGRESSION menu. The errors are shown on a percent scale. They are defined as the percentage difference between the calculated molecular weight at the Rf position and the molecular weight of the marker.

### 10) Interactive regression

Because of the multitasking nature of the Molecular Analyst/Macintosh program, you can check the regression as you create a ruler. This can be very helpful if you are not sure which standard a particular band on a gel corresponds to. Simply adjust the size of both the Profile window or Display window and the Regression window so that you can see both at once. As you adjust the markers in the ruler, names, or the standard spot concentration values, the regression display will adjust to show the current regression when the calibrate sign “=” is used.

### 11) Creating a Standard Results Table

To visualize the standard data within a results table, select the appropriate regression curve and select *List* from the MW menu. **The *List* command is only active when the Profile window is in the foreground.** The current regression analysis will then be used to create its own standard results table. If the type of regression is changed, a new standard results table needs to be created using the *List* command from the MW menu.

| untitled 13    |       |                  |      |                   |                              |                         |         |          |         |         |         |
|----------------|-------|------------------|------|-------------------|------------------------------|-------------------------|---------|----------|---------|---------|---------|
| Standards Name | Index | Mobility<br>(mm) | Rf   | Mw<br>(kilobases) | Calculated Mw<br>(kilobases) | Residual<br>(kilobases) | % Error | Coeff 0  | Coeff 1 | Coeff 2 | Coeff 3 |
| 7 items        |       |                  |      |                   |                              |                         |         |          |         |         |         |
| 1              | 1     | 29.94            | 0.17 | 23130.00          | 19690.55                     | 3439.45                 | 0.17    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 2              | 2     | 36.12            | 0.25 | 9416.00           | 11524.89                     | -2108.89                | -0.18   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 3              | 3     | 40.96            | 0.32 | 6557.00           | 7570.25                      | -1013.25                | -0.13   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 4              | 4     | 47.78            | 0.41 | 4361.00           | 4418.44                      | -57.44                  | -0.01   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 5              | 5     | 60.64            | 0.58 | 2322.00           | 1907.35                      | 414.65                  | 0.22    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 6              | 6     | 63.53            | 0.62 | 2027.00           | 1619.80                      | 407.20                  | 0.25    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 7              | 7     | 86.88            | 0.94 | 546.00            | 530.73                       | 15.27                   | 0.03    | 32203.56 | 2.52    | 0.20    | -99.94  |

Standards result table

## 12) Saving a Standard Results Table

To save the information contained in the Standard Results window go to the FILE menu and select *Export*. A dialog box will appear prompting for the name of the file you want to save the results data under. Results are saved as a Text file.

## 13) Calculating unknown Molecular Weights

- To obtain the molecular weights of the "Unknown" profile, display and make active the "Unknown" profile. The standard profile may be visible, but not active.
- Move the top/origin marker until it aligns with the profile origin, and bottom/front marker until it aligns with the profile front as was done with the standard profile.
- Now click on the calibrate calculate symbol (=) within the ruler. Notice that the X-scale of the graph window, and the coordinate box are now in units of molecular weight (or size).
- Move the interactive arrow anywhere within the graph and read the calculated molecular weight values right off the screen, in the X-axis coordinate box. Alternatively, select *Find Peaks* and then *List Peaks* from the PROFILE menu to send the calculated molecular weights to a results window. This results window is the same as the data previously saved with the addition of the molecular weight data.

| untitled 6        |       |       |       |        |          |             |            |           |               |  |
|-------------------|-------|-------|-------|--------|----------|-------------|------------|-----------|---------------|--|
| Profile Area Name | From  | To    | Width | Height | Location | Area        | % of Total | Detection | Calculated Mw |  |
| 7 items           | (mm)  | (mm)  | (mm)  | (O.D.) | (mm)     | (O.D. * mm) | (%)        |           | (kilobases)   |  |
| profile.area 1    | 24.10 | 33.00 | 8.90  | 0.66   | 30.30    | 2.0783e+0   | 16.84      | Automatic | 22168         |  |
| profile.area 2    | 33.10 | 38.80 | 5.70  | 0.60   | 36.10    | 2.0380e+0   | 16.51      | Automatic | 9436          |  |
| profile.area 3    | 38.90 | 43.90 | 5.00  | 0.59   | 41.10    | 1.4765e+0   | 11.96      | Automatic | 6518          |  |
| profile.area 4    | 44.00 | 52.70 | 8.70  | 0.72   | 47.80    | 2.2348e+0   | 18.11      | Automatic | 4356          |  |
| profile.area 5    | 55.70 | 61.70 | 6.00  | 0.50   | 60.40    | 1.1021e+0   | 8.93       | Automatic | 2347          |  |
| profile.area 6    | 61.80 | 67.70 | 5.90  | 0.62   | 63.10    | 1.3765e+0   | 11.15      | Automatic | 2070          |  |
| profile.area 7    | 83.00 | 91.00 | 8.00  | 0.53   | 86.90    | 1.5108e+0   | 12.24      | Automatic | 545           |  |

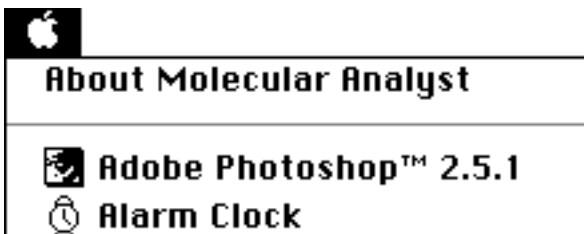
## Printing the Graph and Results

When the *Print* command is selected under the FILE menu, Profile, Regression, or Results windows which are in the foreground will be printed. To print any results list that are not currently displayed, go to the FILE menu and select Results from the pull-down menu. Next, select the appropriate Results file, and click *Open*. To print the results, go under the FILE menu and select *Print*.

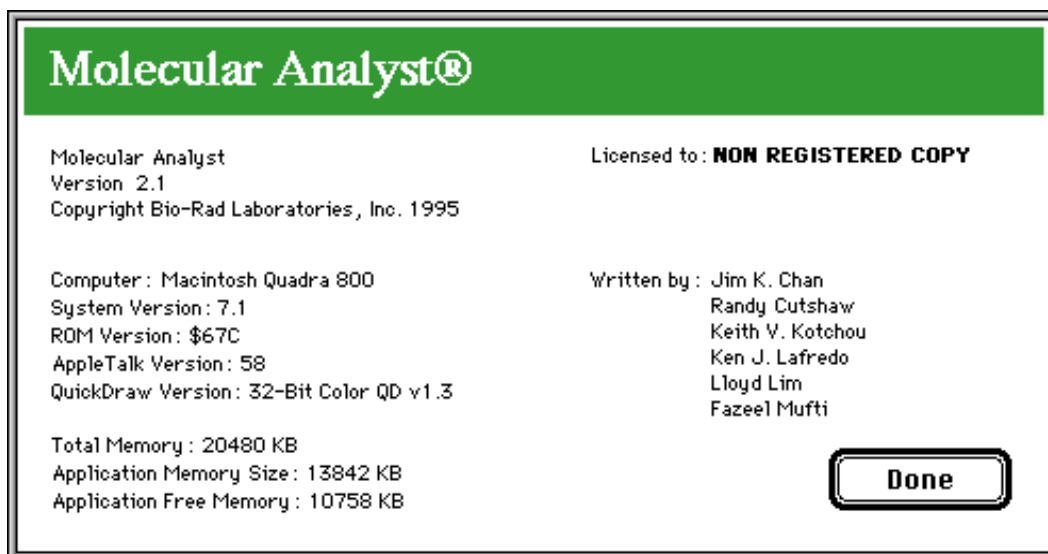
## Section 4 Description of Menus

### 4.1 Apple Menu

The *Apple* menu contains the About Molecular Analyst menu item along with the standard Apple Menu Items.



Selecting the About Molecular Analyst... menu item will display the about box. This dialog contains software version, system configuration, memory, and license information.

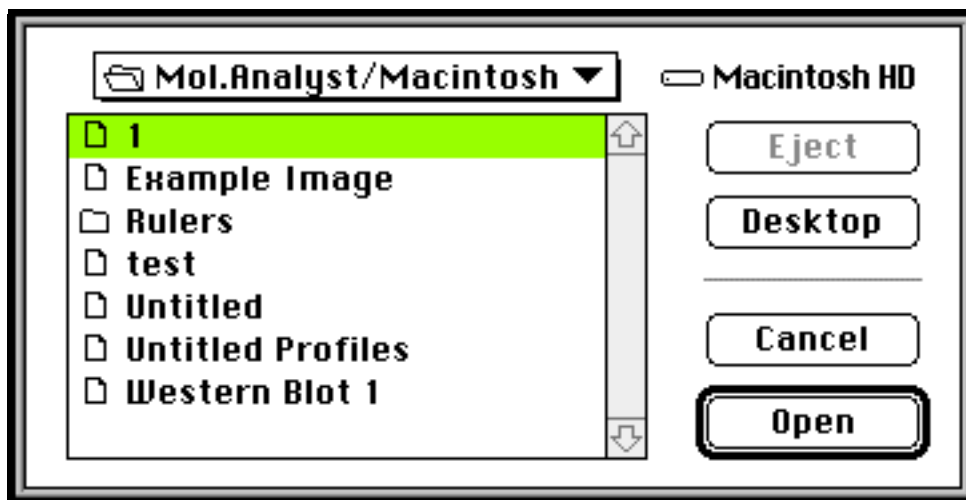


## 4.2 File Menu

| File            |    |
|-----------------|----|
| Open...         | ⌘O |
| Close           | ⌘W |
| Save            | ⌘S |
| Save As...      |    |
| Revert to Saved |    |
| Acquire...      | ▶  |
| Import...       | ▶  |
| Export...       |    |
| Get Image Info  |    |
| Preferences...  |    |
| Page Setup...   |    |
| Print...        | ⌘P |
| Quit            | ⌘Q |

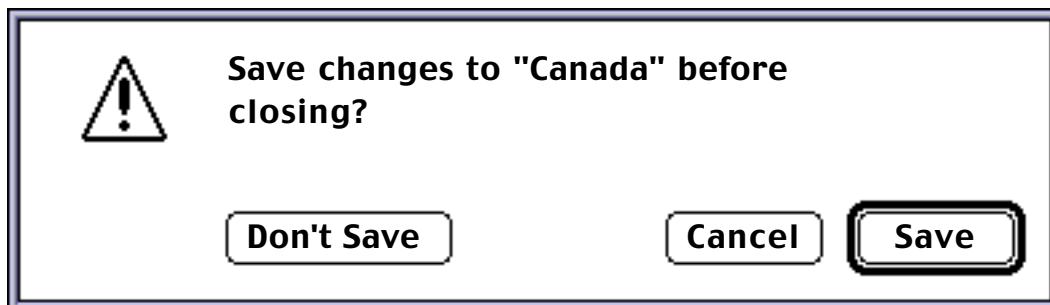
### Open

The *Open...* menu item allows the user to retrieve previously saved data. A preview of the saved image can be shown as the user selects each file name from the list of files available. The types of data which can be retrieved through the *Open...* menu item are: image data in BRIFF format, images saved from previous versions of Molecular Analyst, profile data, and molecular weight ruler data.



### Close

Selecting the *Close* menu item will close the currently selected window. If any savable changes have been made since the last save, the user will be prompted to save changes before closing.



### Save

The *Save* menu item saves data within the current window. If the current window is named, then the data will be saved in the file associated with that window. If the current window is unnamed then the *Save As...* operation will be invoked.

### Save As...

The *Save As...* command allows the user to save the data associated with the current window to a file. This operation will work with named and unnamed windows.

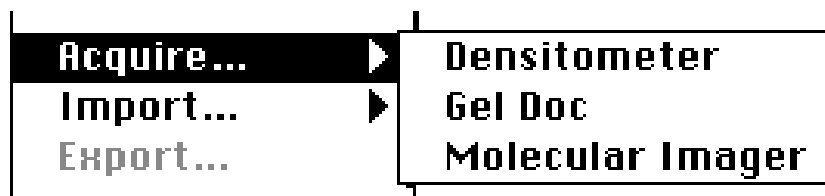
### Revert to Saved

The *Revert to Saved* command allows the user to discard all modifications since the last save. The user will be prompted to verify the revert command. If the user selects to revert, all changes since the last save are discarded.



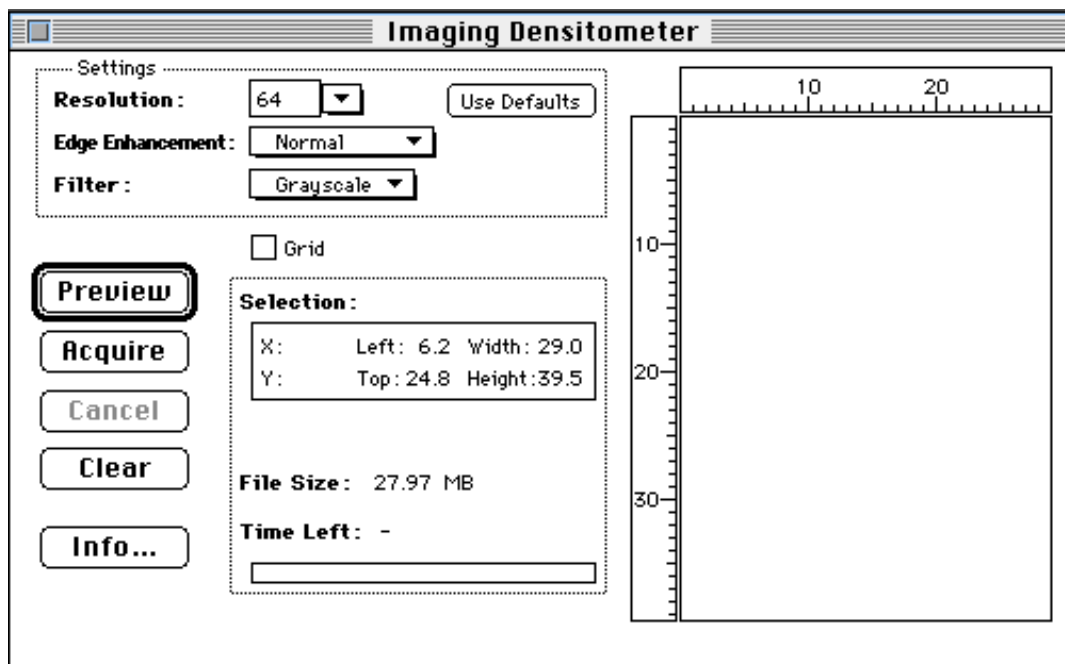
### Acquire

The *Acquire* menu is a hierarchical menu containing the available scanning modules. Selecting one of these menu items will display the scanning window associated with the respective instruments.



### **Acquiring an Image from the GS-670 Densitometer**

To display the acquire window for the densitometer, click and hold on the *Acquire* command from the FILE menu. Next, select Densitometer from the pop-up menu. The Acquire window should appear as shown below:



GS-670 Densitometer Acquisition Window

#### **Select Scanning Area**

The scan window is initially blank with its dimensions representing the full size of the scanning area (Transmittance: 29cm x 39.5cm, Reflectance: 29cm x 42cm).

- Move the cursor onto the grid area.
- Using the mouse, identify the area where your sample(s) by clicking and dragging the mouse from the upper left to the lower right coordinates of the sample. Release the mouse button. The resulting dashed rectangle identifies the area which will be scanned. The *Selection Box* will show the coordinates of the area selected, the file size of the image and the time left to complete scanning.
- Next, select the type of edge enhancement desired. The GS-670 densitometer has built-in image enhancement filters. From the *Edge Enhancement* pull down menu, select Normal to provide no enhancement to the image (Sharp = a fraction of the surrounding pixels are subtracted from the image edges, Extra Sharp = a larger a fraction of the surrounding pixels are subtracted, Soften = a pixel is averaged into the surrounding pixels).
- Lastly, select the proper filter for your sample. Each filter corresponds to a wavelength range which is sensitive to specific dyes and stains.

| <b><u>Filter Color</u></b> | <b><u>Wavelength</u></b> | <b><u>Application Examples</u></b>                       |
|----------------------------|--------------------------|----------------------------------------------------------|
| Red                        | (595nm-750nm)            | Coomasie G-250, BCIP /NBT, Fast GreenFCF, Methylene Blue |
| Green                      | (520nm-570nm)            | Coomasie R-250, Basic Fuchsin                            |
| Blue                       | (400nm-530nm)            | Crocein Scarlet                                          |
| Gray Scale                 | (400nm-750nm)            | Silver Stains, Copper Stains, Film                       |

### ***Preview Scanning Area***

If desired you can preview the scanning area by performing a low resolution (800 $\mu$ ) scan. **The preview scan is designed to give you a rapid preview of the sample and its location, and always scans at 800 $\mu$ .** No data is saved during previewing, but then, again no data is lost.

- a. If you do not know the exact position of your sample, simply select the whole scanning area with the mouse. Scanning areas may be modified and moved easily with the mouse.
- b. Click on the *Preview* command button to initiate scanning. As scanning progresses, you can adjust the brightness, contrast, or gamma of the image with the color palette window. This allows you to view faint samples which might not be visible using the full default view range (0-255). Because the software is processing extremely large amounts of data during the acquisition process, response to mouse clicks tend to be somewhat slower. When the preview scanning is complete you can rescan specific areas of the sample at higher resolution using the *Acquire* button.
- c. Scanning may be stopped with the *Cancel* button, and another preview scan or higher resolution scan performed.
- d. The *Clear* button can also be used to clear any image in the acquisition window in preparation for further scanning.

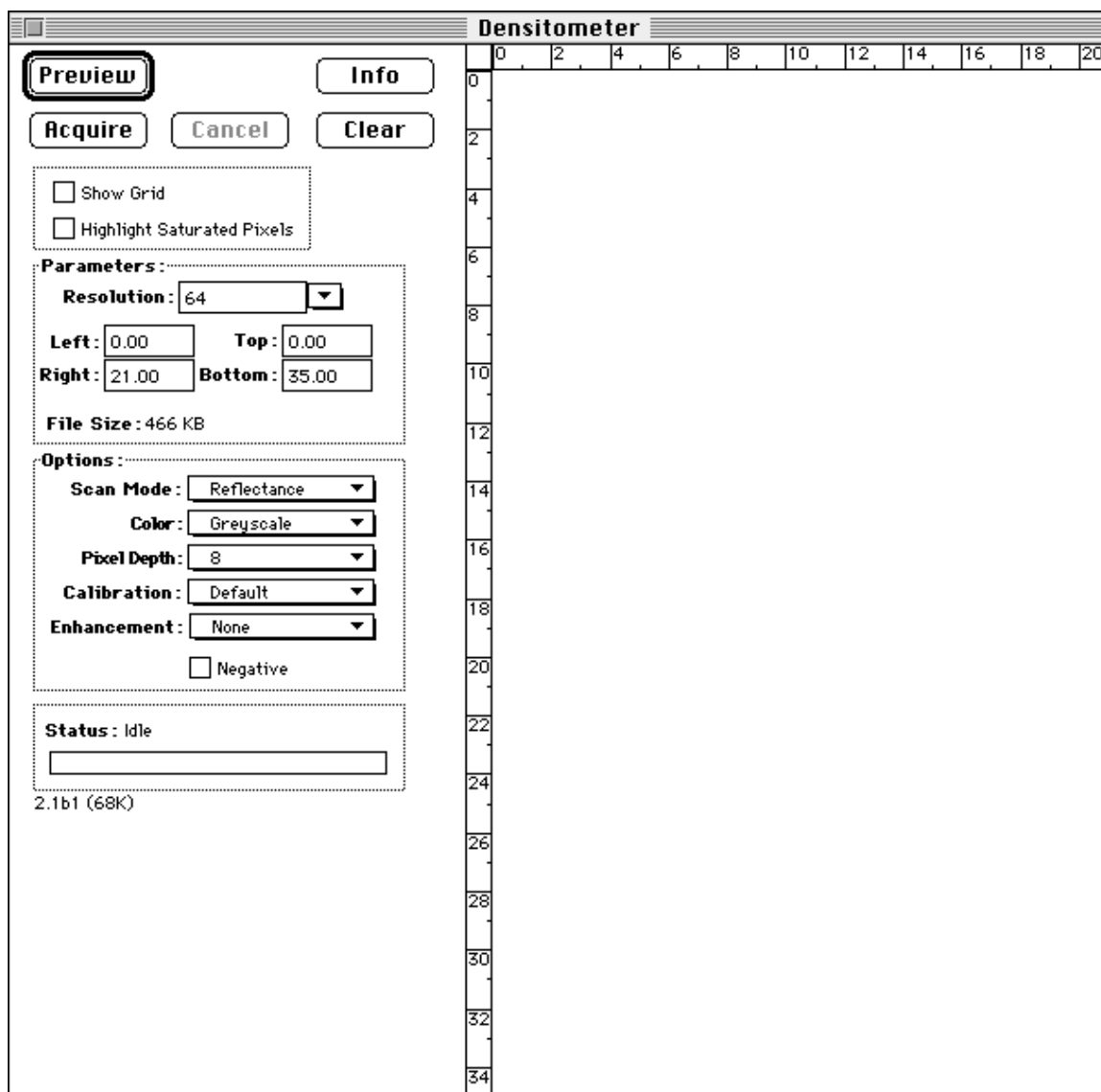
### ***Initiating Image Acquisition***

Images are acquired and analyzed at any desired resolution from 64 $\mu$  (best) to 800 $\mu$ .

- a. Using the mouse, outline a specific area on the preview scan to scan at a higher resolution.
- b. Enter in 64 $\mu$  (default) and then click on the *Acquire* command button. The scanner will immediately start scanning. Images may also be lightened or darkened at higher resolutions using the above mentioned color palette window. Simply scroll the brightness, contrast, or gamma tools until the desired display is achieved.
- c. Once scanning has completed, the program will automatically display the image in an Untitled display window.
- d. Save the image before beginning analysis. From the *FILE* menu select *Save*. Enter a name in the save file dialog box which appears and click *OK*.

### ***Acquiring an Image from the GS-700 Densitometer***

To display the acquire window for the densitometer, click and hold on the *Acquire* command from the *FILE* menu. Next, select *Densitometer* from the pop-up menu. The *Acquire* window should appear as shown below:



### Select Scanning Area

The scan window is initially blank with its dimensions representing the full size of the scanning area in reflectance mode

- Select the appropriate *Scan Mode*. Reflectance mode allows scanning of opaque samples such as blots, gels dried on filter paper, and photographs and has dimensions of 21 cm x 35 cm. Transmittance mode allows the scanning of transparent samples such as films, gels, and slides and has dimensions of 20 cm x 25.5 cm.
- Select the proper filter for your sample from the *Color* option. Each filter corresponds to a wavelength range that is sensitive to specific dyes and stains.

| <u>Filter Color</u> | <u>Wavelength</u> | <u>Application Examples</u>        |
|---------------------|-------------------|------------------------------------|
| Red                 | (595nm-750nm)     | Coomasie G-250, BCIP /NBT          |
| Green               | (520nm-570nm)     | Coomasie R-250, Basic Fuchsin      |
| Blue                | (400nm-530nm)     | Crocein Scarlet                    |
| Gray Scale          | (400nm-750nm)     | Silver Stains, Copper Stains, Film |

- c. Select the appropriate *Pixel Depth* (8-bit or 12-bit). 8-bit mode allows you to scan samples as 256 gray level images. This mode provides 8 bit precision and generates 8 bit image data files. 12-bit mode allows you to scan samples as 4,096 gray level images. This mode provides greater 12-bit precision and generates 12-bit image data files. The 12-bit image files will be larger than the 8 bit files.
- d. If scanning in transmittance mode for wet gels and the diffuser plate, choose *Calibration*. *Calibration* is only available under Transmittance mode when using the diffuser plate. This calibration compensates for the density of the diffuser plate. *Enhancement* and *Negative* are two other available options. *Negative* allows the scanning of negative transparent samples such as a negative film strip or photograph of an ethidium gel. *Enhancement* filters are not available with the GS-700 Densitometer. Finally, *Show Grid* and *Highlight Saturated Pixels* are options that provide grid overlay and saturated pixel indication respectively.
- e. Next, move the cursor onto the grid area.
- f. Using the mouse, identify the area where your sample(s) by clicking and dragging the mouse from the upper left to the lower right coordinates of the sample. Release the mouse button. The resulting dashed rectangle identifies the area which will be scanned. The *Selection Box* will show the coordinates of the area selected, the file size of the image and the time left to complete scanning.

### ***Preview Scanning Area***

If desired you can preview the scanning area by performing a low resolution (24 dpi) scan. **The preview scan is designed to give you a rapid preview of the sample and its location, and always scans at 24 dpi.** No data is saved during previewing, but then, again no data is lost.

- a. If you do not know the exact position of your sample, simply select the whole scanning area with the mouse. Scanning areas may be modified and moved easily with the mouse.
- b. Click on the *Preview* command button to initiate scanning. As scanning progresses, you can adjust the brightness, contrast, or gamma of the image with the color palette window. This allows you to view faint samples which might not be visible using the full default view range (8-bit, 0-255 or 12-bit, 0- 4,096). Because the software is processing extremely large amounts of data during the acquisition process, response to mouse clicks tend to be somewhat slower. When the preview scanning is complete you can rescan specific areas of the sample at higher resolution using the *Acquire* button.
- c. Scanning may be stopped with the *Cancel* button, and another preview scan or higher resolution scan performed.
- d. The *Clear* button can also be used to clear any image in the acquisition window in preparation for further scanning.

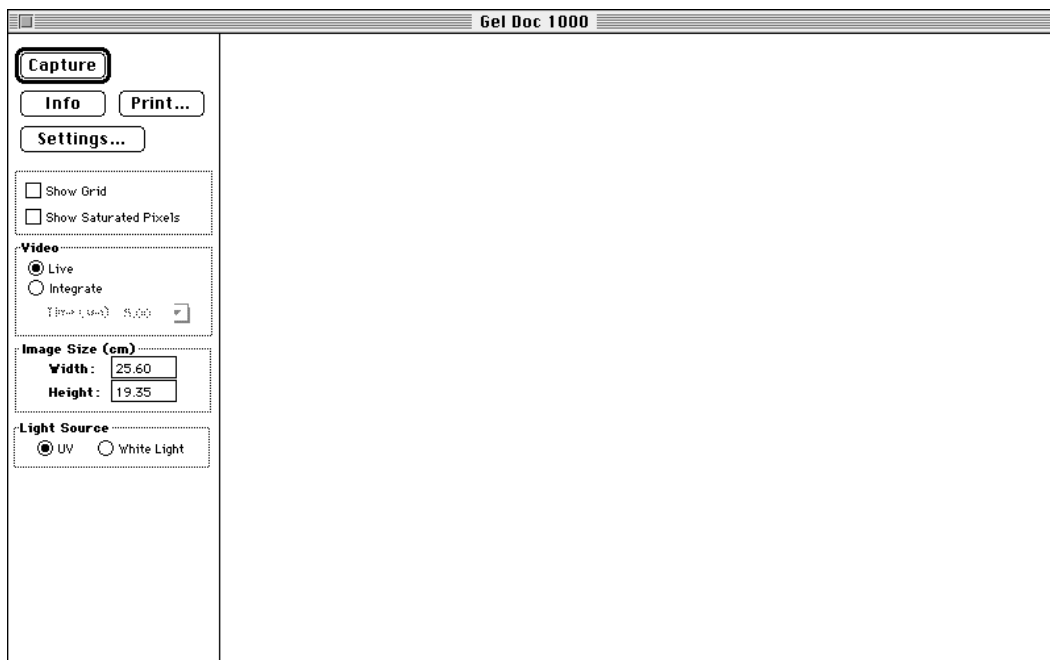
### ***Initiating Image Acquisition***

Images are acquired and analyzed at any desired resolution.

- a. Using the mouse, outline a specific area on the preview scan to scan at a higher resolution.
- b. Type in a high resolution value (i.e. 400 dpi) and then click on the *Acquire* command button. The scanner will immediately start scanning. Images may also be lightened or darkened at higher resolutions using the above mentioned color palette window. Simply scroll the brightness, contrast, or gamma tools until the desired display is achieved.
- c. Once scanning has completed, the program will automatically display the image in an Untitled display window.
- d. Save the image before beginning analysis. From the FILE menu select *Save*. Enter a name in the save file dialog box which appears and click *OK*.

### **Acquiring an Image from the Gel Doc 1000**

To display the acquire window for the Gel Doc 1000, click and hold on the *Acquire* command from the FILE menu and select Gel Doc from the pop-up menu. The Acquire window should appear as shown below in LIVE mode displaying a real-time image.



Gel Doc 1000 Acquisition Window

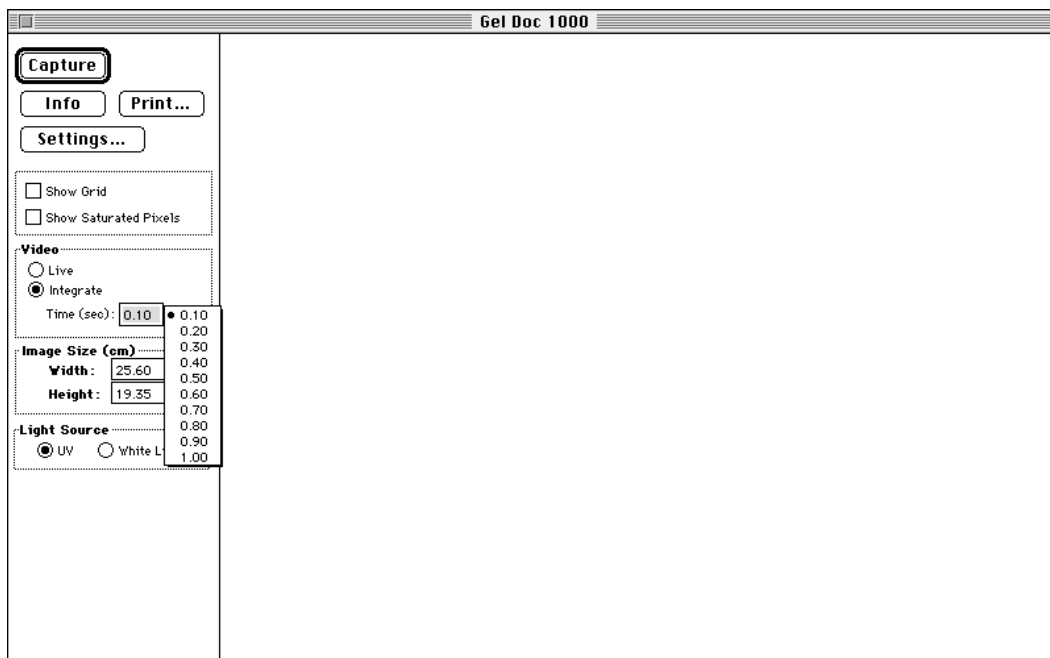
### ***Displaying “Live” Images -***

When the scan window is opened, it will automatically be in “LIVE” mode and will display a live video image allowing you to zoom and focus on the sample area. If no image is visible, check cable connections from the frame grabber board to the camera and that the protective cap is off the lens. Also check that the transilluminator is on and working. The image should stay in focus while zooming from high to low magnification. If focus is not maintained, refer to Appendix D to correct the camera focus.

### ***Integrating -***

To identify faint areas of fluorescence, click on the “Integrate” radio button. Integration time is any number from 0.1 - 60 seconds. (Integration has the same effect as exposure time). The integration pull-down menu only displays up to 1 second, however any number up to 60 seconds may be entered manually.

To integrate, click on the integration time pull-down menu and select the time for integration. To integrate for longer periods of time, highlight the integration number inside the *Integration Time* window (see 0.10 in the above figure) and using the <Enter> key, type in another integration time up to 60 seconds using the keyboard.



### ***Printing a live or integrated image from the Acquisition Window-***

Live or integrated images from the acquire window are only printed to a thermal printer.

#### ***Printing an image to a Mitsubishi video printer-***

To print a live or integrated image to a video printer, select *Print* from the acquisition window. The video printer will automatically print the image. Please see Appendix E to adjust the Mitsubishi P68U Thermal Printer. Tables cannot be printed to a video printer and must be printed on a laser printer using the *Print* command from the FILE menu.

**Print...**

### ***Capturing or “Freezing” the image-***

To quickly freeze or capture the image for observation or analysis, click on the *Capture* button. The image will be displayed in an untitled display window. It is recommended that the image be saved by using the *Save* command from the FILE menu.

**Capture**

### ***Saving the Image***

Save your image as an image file by choosing *Save* from the FILE menu.

### ***Printing captured or saved images-***

Captured and saved images can also be printed either to a laser or thermal printer.

#### ***Printing captured images to a Mitsubishi video printer-***

To print saved or captured images to a video printer, select *Video Print* from the FILE menu. If there is no Bio-Rad frame grabber board present, this command will not be available. Data cannot be printed to a video printer and must be printed on a laser printer using the *Print* command from the FILE menu.

#### ***Printing to a laser printer-***

To print saved or captured images to a laser printer, select *Print* from the FILE menu.

## Other Acquisition Parameters-

### Info

The Info button displays current available information on the version of the Molecular Analyst : Gel Documentation software

Info

### Settings

When this button is selected, Range and Offset controls are displayed. The effects of Range and Offset on the acquired image are similar to brightness and contrast controls on a video monitor. This command is rarely used because it alters the raw signal coming from the camera. Please use the default values for Range and Offset and adjust brightness and contrast with the modify view or transformation palettes.

Settings...

### Show Grid & Show Saturated Pixels

If Show Grid is checked, a grid is displayed over the image.

If *Show Saturation* is checked, all saturated values will be displayed in red. This feature is useful when quantitating bands on a gel. For accurate quantitation, reduce integration time or close down the aperture until all red saturated values are white.

☐ Show Grid  
☐ Show Saturated Pixels

### Light Source

This command changes the units from "Counts" to "O.D.". Select "Counts" when working with fluorescent or UV transilluminated samples. Select "O.D." when working with white light transilluminated samples.

Light Source  
☒ UV    ☐ White Light

### Change Image Size...

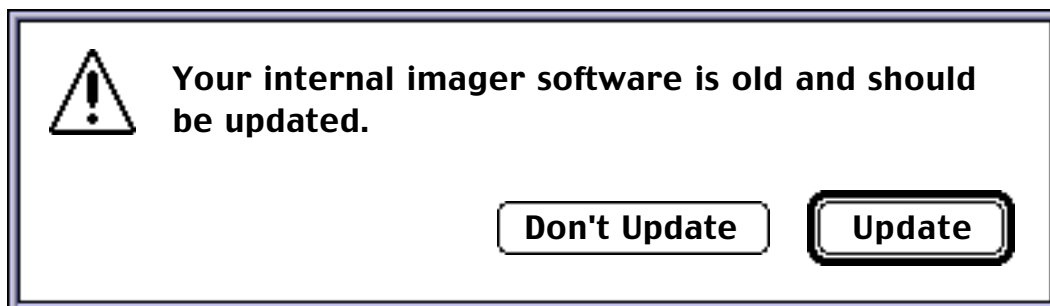
Image size always defaults to the largest possible viewing area (zoomed out). For very accurate quantitation it is necessary to enter in the scan dimensions taken from the imaging area. This is done by simply using a fluorescent ruler to measure the imaging area and entering the information.

Image Size (cm)  
Width: 25.60  
Height: 19.35

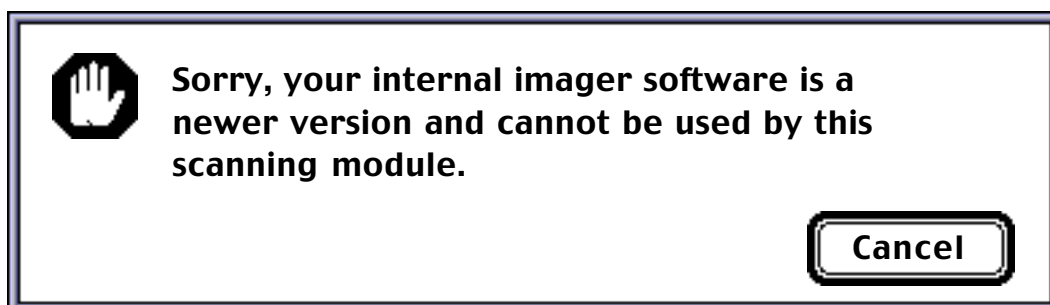
### **Acquiring an Image from the Molecular Imager**

To display the acquire window for the Molecular Imager, click and hold on the *Acquire* command from the FILE menu. Next, select Molecular Imager from the pop-up menu.

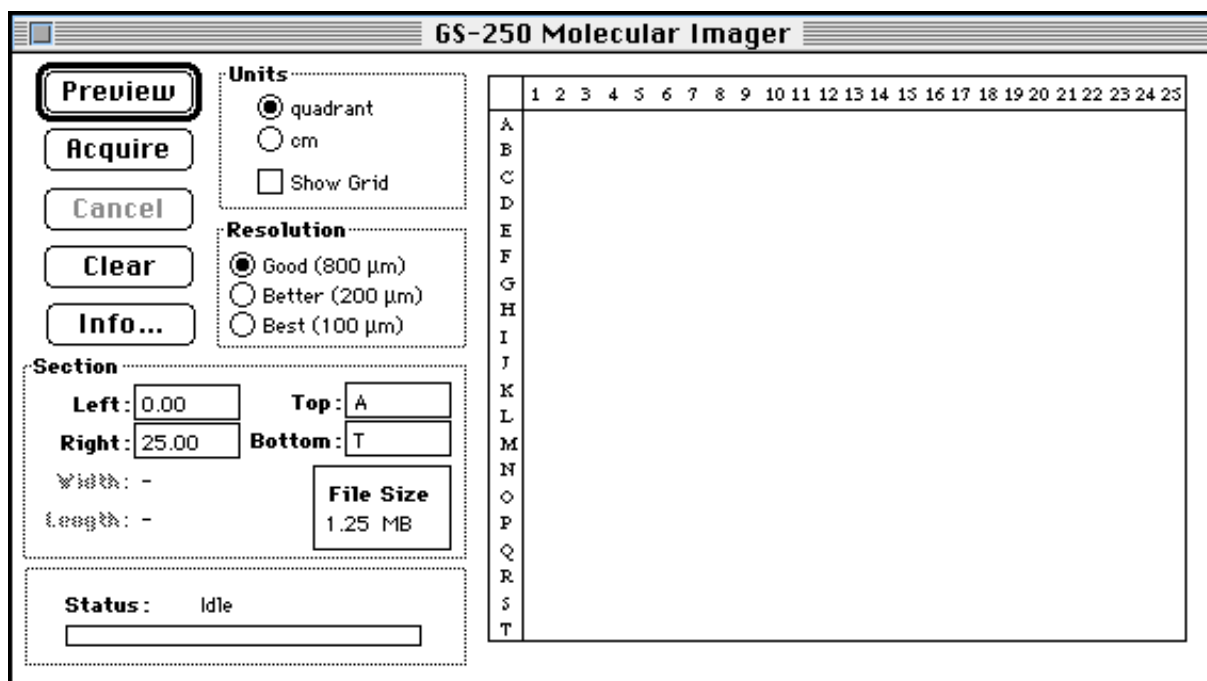
When the Imager scan window is first displayed it queries the imager for the internal software version. If this version is older than the version contained in the scanning module the user will be prompted to update. A dialog will be displayed while the Imager is being updated.



If however the Imager internal software is newer than the version contained in the scanning module or cannot be used by the application the user will not be able to scan and will be asked to contact Bio-Rad Laboratories.



If the internal software is correct or is updated, the Acquire window should appear as shown below:



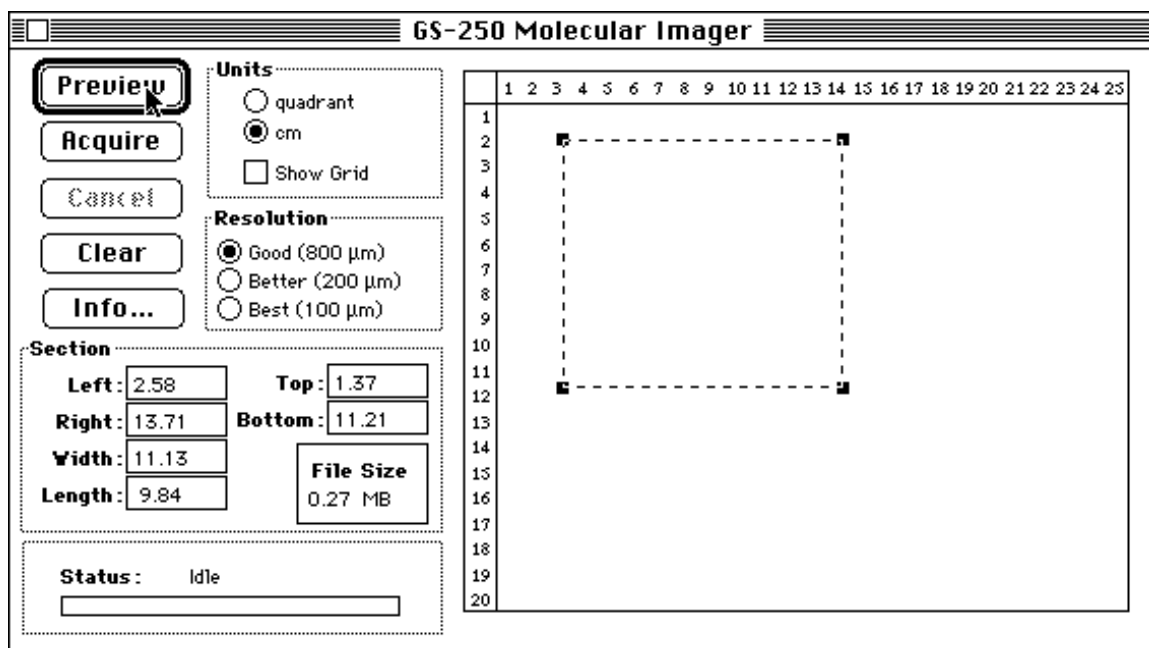
Molecular Imager Acquisition Window

### ***Select Scanning Area***

Identify the area where your sample(s) was located on the loading dock's sample placement grid. Move the cursor to the sample's upper left coordinates within the grid area. Press the mouse button and drag the mouse to the lower right coordinates of the sample and release the mouse button. The resulting dashed rectangle identifies the area which will be scanned. Additionally, the coordinates of the area that are selected, the file size of the image, and the time left to complete scanning are also displayed.

### ***Preview Scanning Area***

If desired you can preview the scanning area by performing a low resolution scan. **The preview scan is designed to give you a rapid preview of the sample and its location and always scans at 800μ.** No data is saved during previewing, but then, again no data is lost. If you do not know the position of your sample, simply select the whole scanning area with the mouse. Click on the *Preview* command button to initiate scanning. As scanning progresses, you can adjust the brightness, contrast, or gamma of the image with the color palette window. This allows you to view faint samples which might not be visible using the full default view range (0-65,536).



Preview Scan at 800 $\mu$

When the preview scanning is complete you can rescan specific areas of the sample at Better (200 $\mu$ ) or Best (100 $\mu$ ) resolution, as well as, indicating whether or not "Image Processing is necessary.

### ***Initiating Image Acquisition***

Images are acquired inside the preview image and analyzed at either Good (800 $\mu$ ), Better (200 $\mu$ ), or Best (100 $\mu$ ) resolution. Because the files generated at Better resolution are only 25% the size of Best files, this choice is preferred for general purposes.

### ***Molecular Imager Image Processing from the Acquisition Window***

Images can be processed during acquisition. Normal, ANR, and None image processing options are available from the Preferences dialog box from the FILE menu or in the Information dialog box from the Acquisition Window. Normal provides simple processing (for acquired and saved images) with median filter and auto-transformation adjustment, ANR provides "Adaptive Noise Reduction filtering for acquired images (See FILTER menu), and None provides no default image processing. Before selecting a higher resolution, click on the Info button and select the appropriate image processing.

|                                                                                                                         |                                           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Scanner :                                                                                                               | Bio-Rad Model GS-250<br>Molecular Imager. |
| Serial Number :                                                                                                         | 0                                         |
| Hardware version :                                                                                                      | 0.00                                      |
| Software version :                                                                                                      | 0.00                                      |
| Minimum Scanning Area :                                                                                                 | 0.16 × 0.16 cm                            |
| Maximum Scanning Area :                                                                                                 | 25.00 × 20.00 cm                          |
| Maximum Resolution :                                                                                                    | 100 μm                                    |
| Pixel Density :                                                                                                         | 16 bit precision                          |
| Module Version :                                                                                                        | 2.1b1                                     |
| <b>Image Processing</b><br><input type="radio"/> Normal <input type="radio"/> ANR <input checked="" type="radio"/> None |                                           |
| <input type="button" value="OK"/>                                                                                       |                                           |

**Note:** To process a previously acquired Molecular Imager image with the “Normal” image processing option, click on the  icon that is displayed under the tool palette in the display window. To undo the processing, simply click on the  icon.

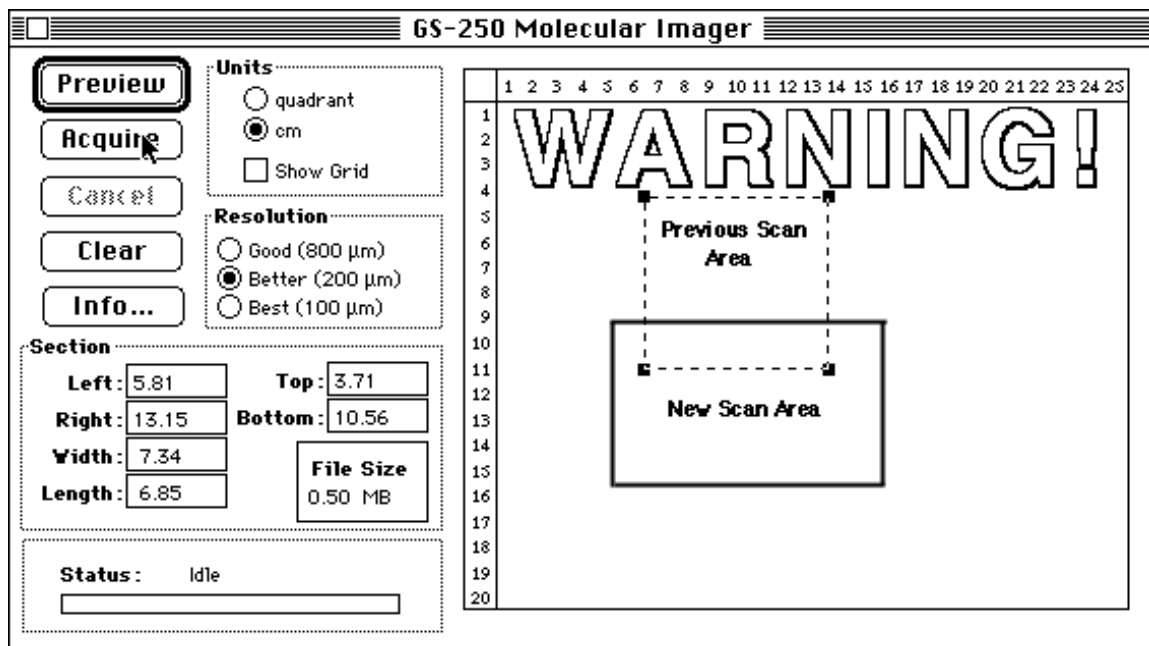
Next, click on either the Better or Best resolution radio button and then click on the *Acquire* command button.

| GS-250 Molecular Imager                                                                                                                                                                                                |  |                                                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| <b>Resolution</b><br><input type="radio"/> Good (800 μm)<br><input checked="" type="radio"/> Better (200 μm)<br><input type="radio"/> Best (100 μm)                                                                    |  |                                                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| <b>Section</b><br><b>Left:</b> 5.81 <b>Top:</b> 3.71<br><b>Right:</b> 13.15 <b>Bottom:</b> 10.56<br><b>Width:</b> 7.34<br><b>Length:</b> 6.85<br><b>File Size:</b> 0.50 MB                                             |  |                                                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| <b>Status:</b> Idle                                                                                                                                                                                                    |  |                                                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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High resolution scan inside the preview image

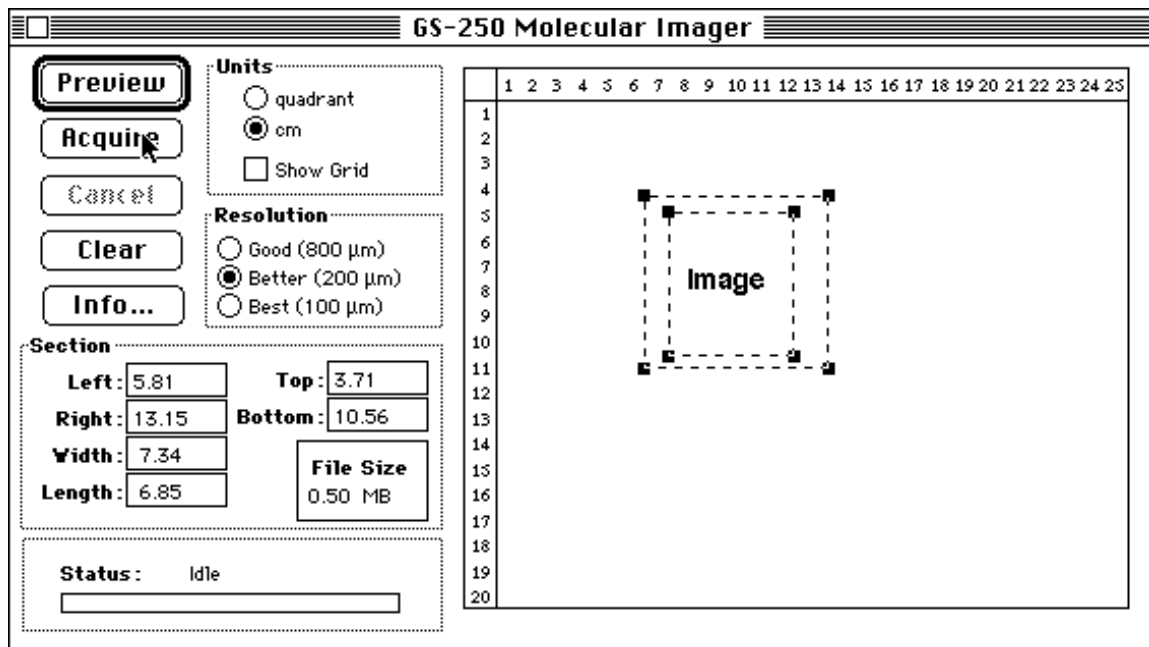
The higher resolution rescan (200μ or 100μ) or new scan must be completely inside or outside the preview scan area. There cannot be any overlap of previewed and non-scanned areas.

**WARNING! In the Molecular Imager, a new or higher resolution scan area cannot overlap any previously scanned area. This will result in an error!**



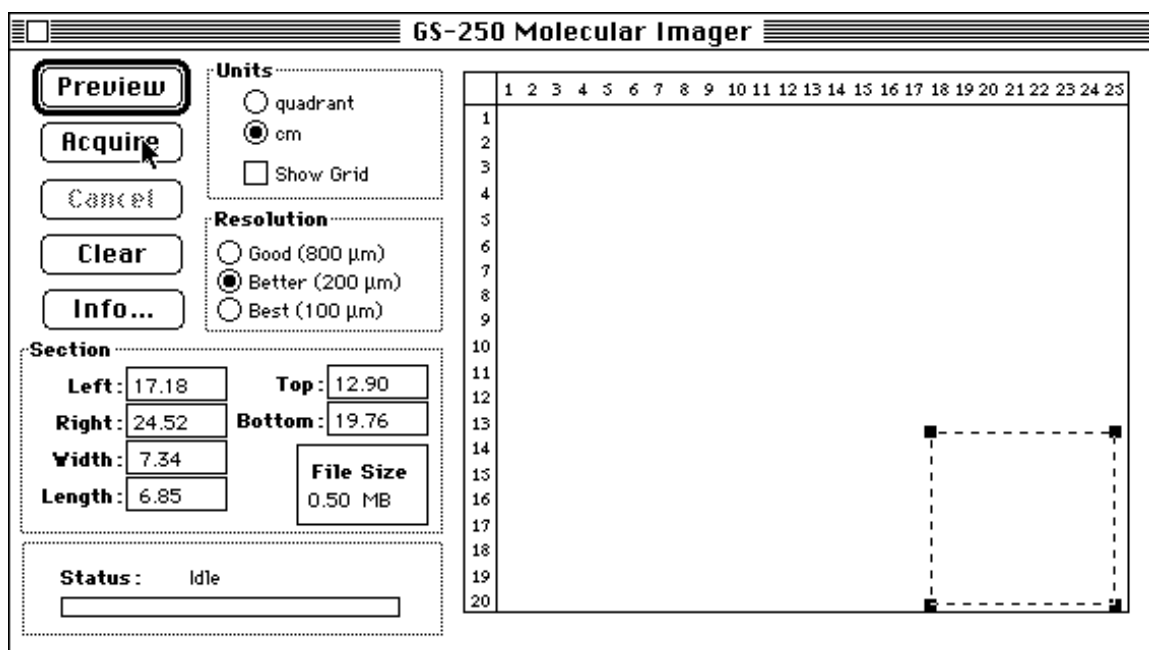
Overlapping scan areas will result in a warning message and a failure to scan

To reduce scan times with the Molecular Imager, once the desired image has been displayed, scanning can be stopped anytime by selecting Cancel. The image can then be saved by mousing out a smaller section WITHIN the previously scanned area and clicking on Acquire. The untitled image will be displayed.



Saving images when stopping a scan

When scanning multiple samples, new sample areas can be quickly scanned by drawing a new scanning square on other areas and selecting Acquire.



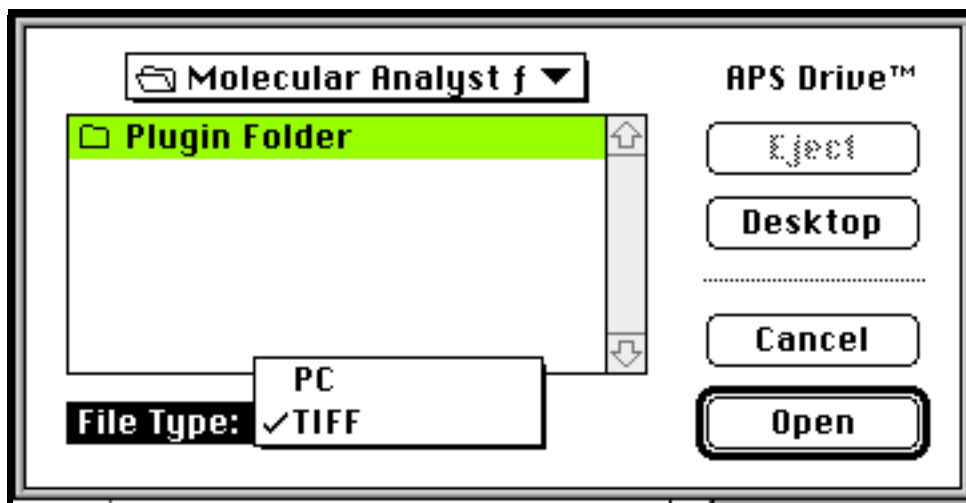
Once scanning has completed, the program will automatically display the image in an Untitled display window. **The image should now be saved before beginning analysis.** From the FILE menu select *Save* . Enter a name in the save file dialog box which appears and click OK. Close the Acquire window by clicking on the upper left square to begin analysis.

### Import

The *Import...* command allows the user to import TIFF and MA/PC images or to add individual profiles from a previously saved profile file to the current Profile window.

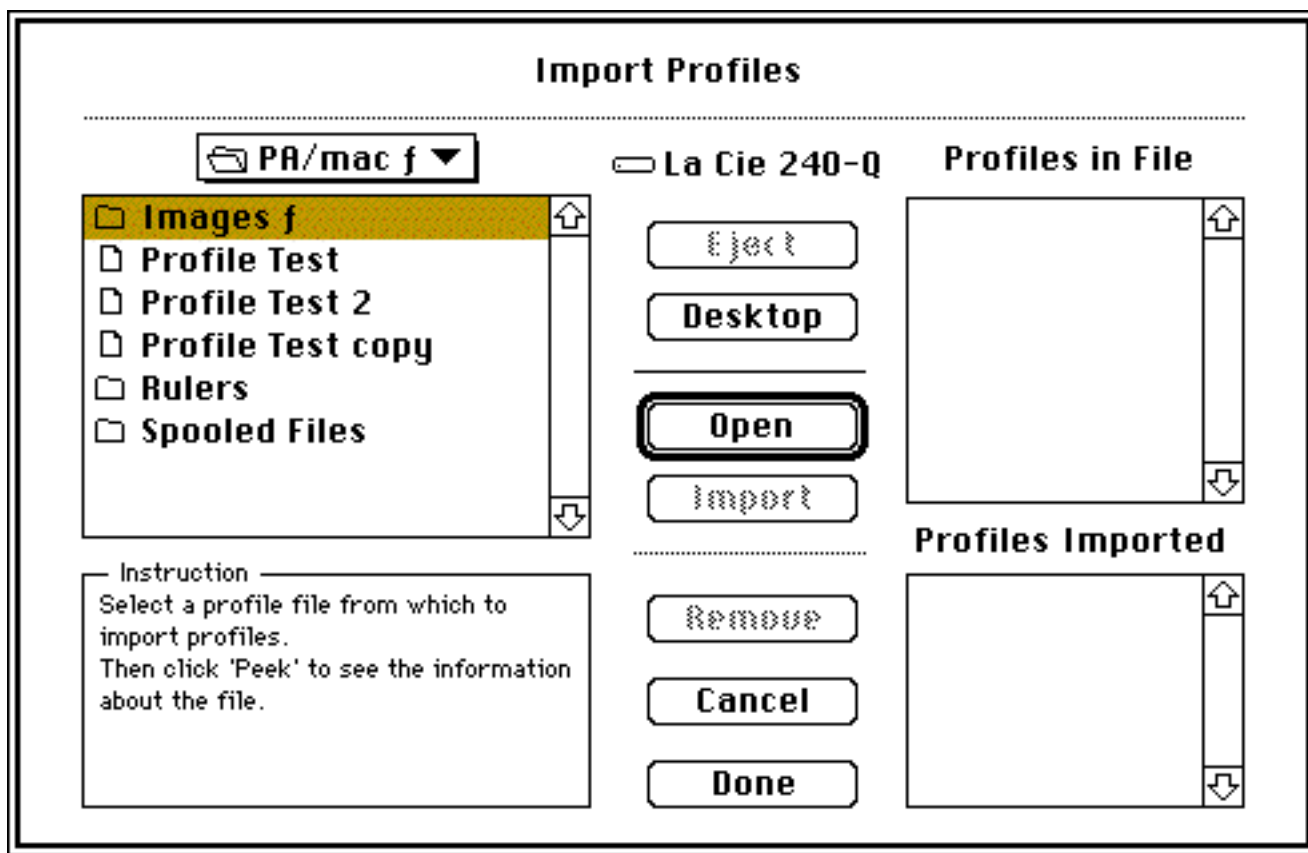
#### Import Images

The *Import Image* command allows TIFF and PC images acquired from the densitometer or Molecular Imager to be displayed and manipulated by the Macintosh software.



### **Import Profiles**

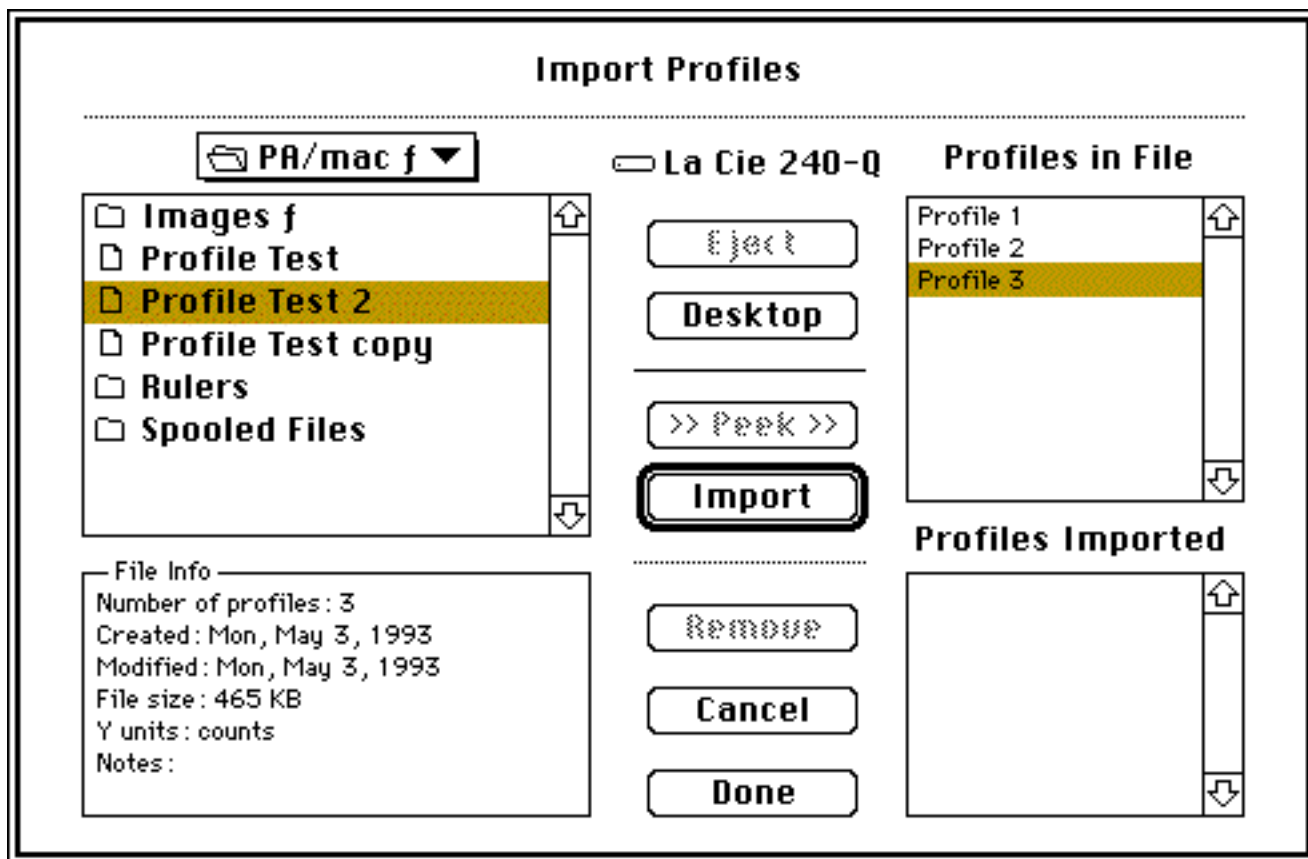
Only when a Profile window is the front window, the *Import...* menu item in the FILE menu is activated. The Import dialog window contains a superset of the features available from the familiar Macintosh Open File dialog window.



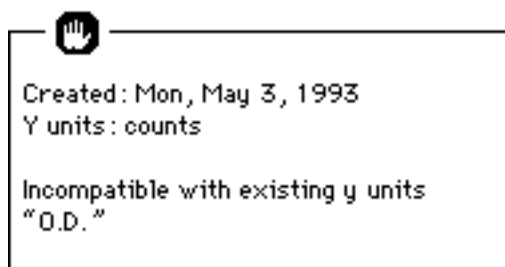
To import a profile, the user first selects a profile file from which to import profiles.. Once a profile file is selected, the Open button will change to an outlined ">> Peek >>" button and the "File Info" box will display the information on the profile file. Pressing the Peek button or double clicking on a profile file will display the profile names contained in the selected file.

Only one profile name in the "Profiles in File" list can be selected at a time. If a profile name is selected, the Peek button will be dimmed and the Import button will be outlined and become the default button. The corresponding profile information is displayed in the "Profile Info" box.

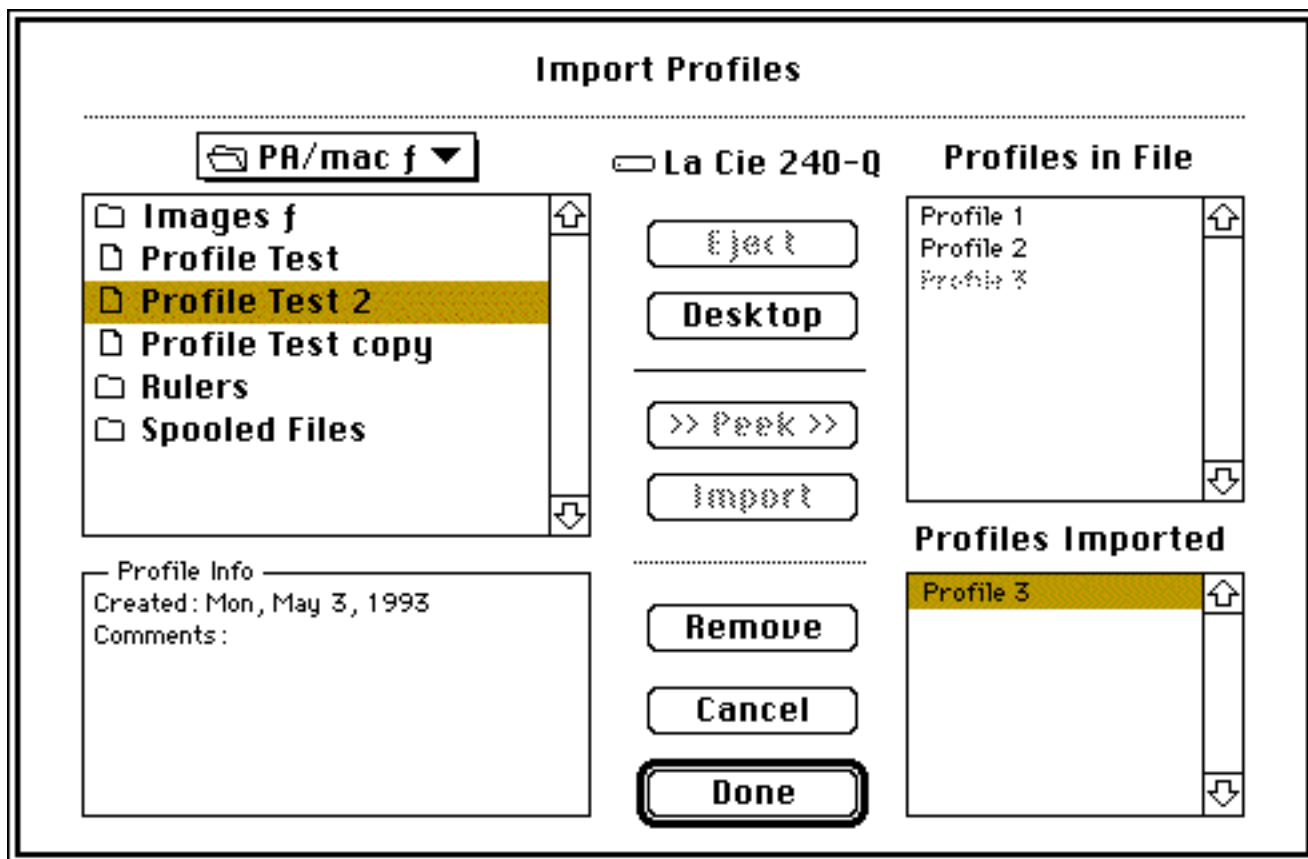
The Peek button is useful because reading a file is a time consuming process. The user may switch between selected profile files.



If the y units of the profiles in the profile file selected are not compatible to that of the existing profiles in the Profile window, the Import button will be dimmed and no profile import is allowed. The Notes in the "File Info" box will show this fact.

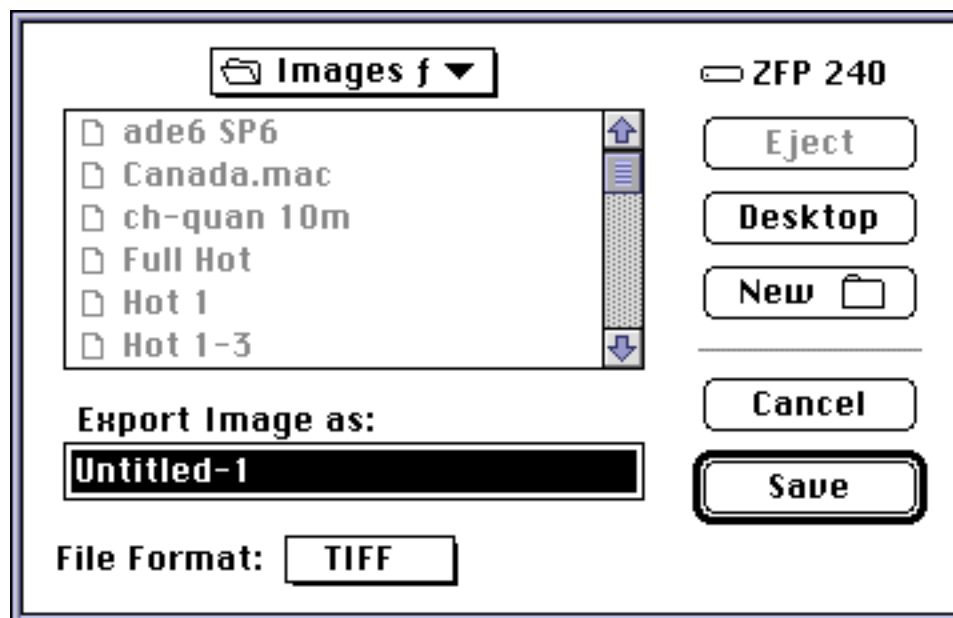


Pressing the Import button with a highlighted profile name or double clicking on a profile name will import the selected profile into the "Profiles Imported" list. The Import button will be dimmed and the Done button will be outlined. The imported profiles in the "Profiles in File" list will be dimmed so that no more selection is allowed.



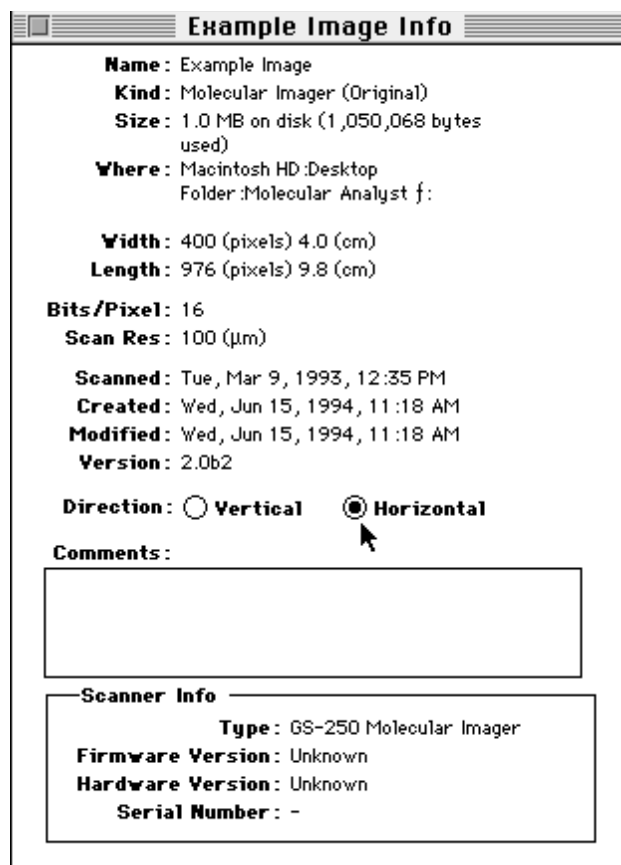
### Export

The *Export...* command allows the user to save result windows and image data to a file in a format other than the standard Bio-Rad BRIFF format. Images can be saved as raw data, PC, or 8 or 16 bit TIFF files.



## Get Image Info

The *Get Image Info* menu item displays information about the associated data. It also indicates the orientation of the image and standards.



## Preferences

The *Preferences* command displays default processing options for the Molecular Imager, and printing and saving options. Clicking on each icon (Saving, Printing, Imager) displays the preferences command options.

### Imager

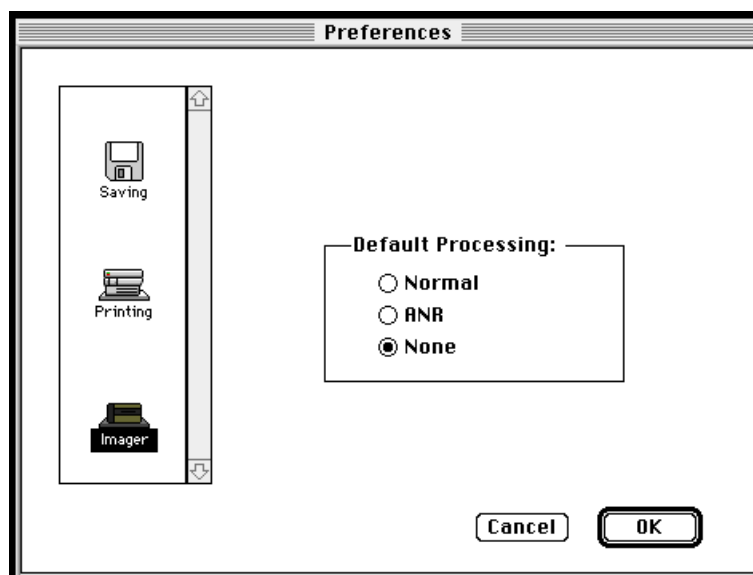
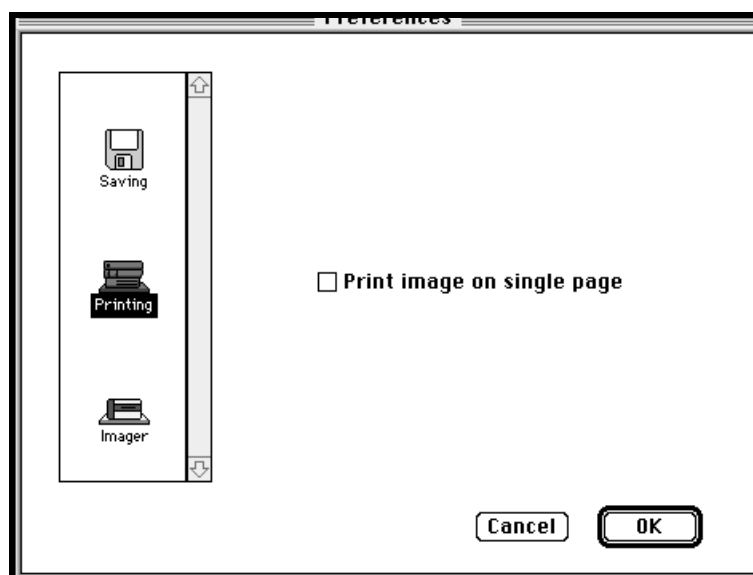
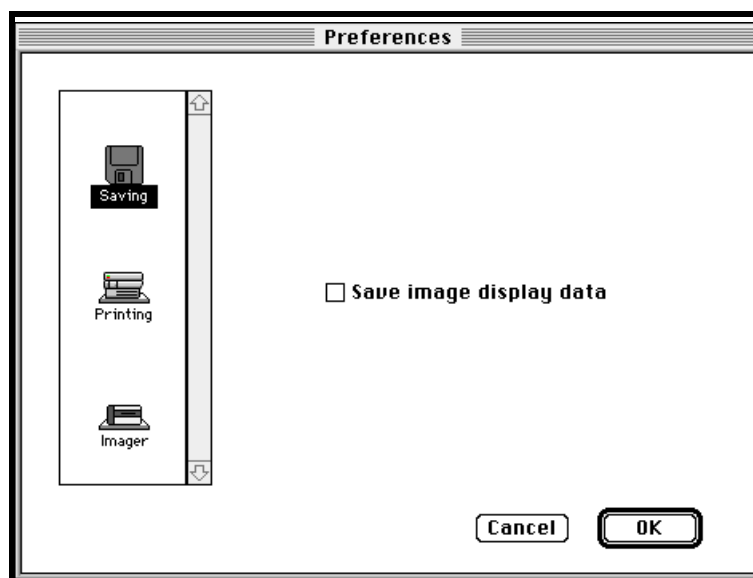
There are three default image processing options for the Molecular Imager: Normal, ANR, and None. Normal provides simple processing (for acquired and saved images) with median filter and auto-transformation adjustment, ANR provides "Adaptive Noise Reduction filtering for acquired images (See FILTER menu), and None provides no default image processing.

### Saving

If "Save image display data" is on display data will be saved. This option uses more hard disk space. It is recommended that this be turned off.

### Printing

If "Print image on single page" is on, it will expand or compress the image to fit the page. If "Print image on single page" is off, it will print the image at actual size.



### Page Setup

The *Page Setup...* command uses the standard Macintosh Page Setup dialogs and routines. It is through the scaling feature within Page Setup that allows users to print magnified images.

### Print

The *Print...* command uses the standard Macintosh print dialogs and routines. This will allow the user to obtain a hard copy of the image and result data associated with the currently selected window. If the Gel Doc frame grabber board is present in the Macintosh, the command *Video Print* will also be available for printing saved images to a video printer.

## 4.3 Edit Menu

The EDIT menu allows the use of standard edit functions on the image data, objects, and profiles. The information is stored in the clipboard and can be shared among other applications. Image data however cannot be pasted in from any other application.

| Edit             |    |
|------------------|----|
| Can't Undo       | ⌘Z |
| Cut              | ⌘H |
| Copy             | ⌘C |
| Paste            | ⌘U |
| Clear            | ⌘B |
| Duplicate        | ⌘D |
| Select All       | ⌘A |
| Crop             |    |
| Extract          |    |
| Hide All         | ⌘H |
| Flip Horizontal  |    |
| Flip Vertical    |    |
| Restore Selected |    |

### Undo

The *Undo* command allows the user to undo a filtering or cropping operation. When an operation is Undone, the *Undo* command will change to *Redo*, which will allow the user to redo the last operation before the undo was performed. When an operation cannot be Undone, the *Undo* command will change to *Can't Undo*.

### Cut

The *Cut* command performs a copy and clear on objects. This option is not available for image selection and is disabled when the image selection tool is in use.

### **Copy**

The *Copy* command copies the current results selection and profile data (text format) to the clipboard. This command is not active with objects or images.

### **Paste**

The *Paste* command pastes the current information from the clipboard to the image window. Only objects can be pasted on to an image. Image data cannot be pasted. This command is disabled when there is data which is not recognized by the application.

### **Clear**

The *Clear* command clears all active objects displayed on an image. The Delete key will also clear all active objects.

### **Duplicate**

The *Duplicate* command quickly copies and pastes active profiles within the profile window. This command allows the user to select a profile and then quickly make a copy without effecting the clipboard. The copy will be place down and to the right of the original profiles. The operation cannot be performed on image data and therefore this command is disabled when the image selection tool is in use.

### **Select All**

The *Select All* command will select all overlay objects when the image window is the active window and the image selection tool is not selected. If the image window is selected and the selection tool is active, *Select All* will select the entire image.

### **Crop**

The *Crop* command cuts an image to any desired size which has been pre-determined using the selection tool. This cropped image then becomes the new image and retains the name of the original image. The original size of the image is lost unless the *Crop* command is undone. The *Crop* command can be undone using the *Undo* command.

### **Extract**

The *Extract* command cuts an image to any desired size which has been pre-determined using the selection tool. However, the new image is displayed in a new untitled image window, while the old image remains unchanged.

### **Flip Horizontal**

The *Flip Horizontal* command flips an active profile in the horizontal direction. This command is not functional with images.

### **Flip Vertical**

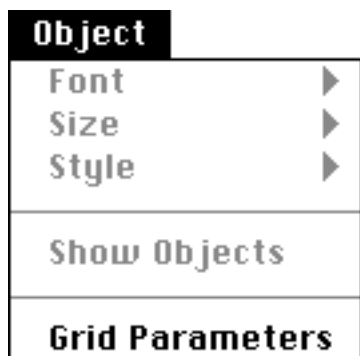
The *Flip Vertical* command flips an active profile in the vertical direction. This command is not functional with images.

### **Restore Selected**

The *Restore Selected* command will restore an active profile to its original, unmodified form.

#### 4.4 Object Menu

The OBJECT menu is used to perform operations on image objects. This menu allows the user to show or hide objects, select fonts and sizes, and to define grid parameters.



##### Font

The *Font* command is a hierarchical menu containing all of the currently available fonts in the system.

##### Size

The *Size* command is a hierarchical menu containing the font sizes for the current font. The pre-defined sizes will be: 9, 10, 12, 14, 18, and 24. The *Other...* menu item is used to selected a size not in the pre-defined list. **The font size for grids cannot be changed.**

##### Style

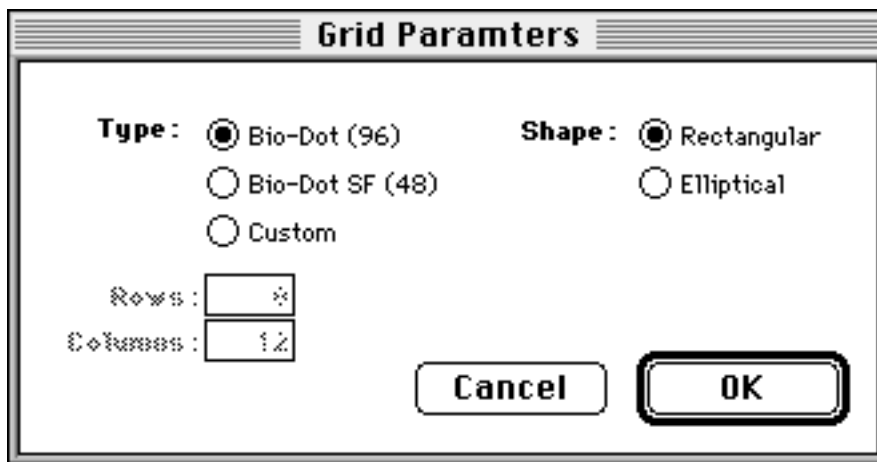
The *Style* command is a hierarchical menu containing the font styles. The pre-defined styles will be: Plain, **Bold**, *Italic*, Underline, Outline, and Shadow.

##### Show Object - Hide Objects

The *Show Objects - Hide Objects* command allows the user to show or hide overlay volume objects on an image.

##### Grid Parameters

The *Grid Parameters...* menu item allows the user to change the values used to draw grids. The user can select either a 96 cell, 48 cell, or custom grid.



## 4.5 Analysis Menu

| Analysis         |    |
|------------------|----|
| Find Spots       | ⌘F |
| Spot Parameters  |    |
| Get Overlay Info | ⌘I |
| Auto Integrate   |    |
| Integrate Now    | ⌘= |
| Local Background |    |
| Transformation   | ⌘T |
| Extract Profiles | ⌘E |

### Find Spots

The *Find Spots* command will invoke the Automatic Spot Finding Algorithm. This algorithm searches through the current image selection or the entire image and automatically finds spots based upon the current parameters. The algorithm searches across each row of the image until it finds a pixel that is larger than the minimum height. This point becomes the spot's left edge. The search continues until it finds a pixel smaller than the minimum height. The point before this point becomes the spot's right edge. The spot is then added to a list. This continues until the image is completely processed. Note that each of these spot's top and bottom elements are the height of the row on which they are found. The list is then compressed, by finding any spots in adjacent rows that overlap along the row. The top and bottom of each spots are found as the spots are compressed into a single spot.

The remaining compressed spots are kept or discarded by comparing them against the remaining selection criteria. These criteria are the minimum spot volume, minimum spot width and length, maximum spot width and length, minimum and maximum spot aspect ratio, minimum porosity.

The spot finding parameters can be changed through the *Spot Parameters...* menu item. A progress dialog is displayed to the user during the spot finding. A progress dialog is displayed to show the percentage of the image which has been searched. Selecting *Integrate Now* will display all found spots in the results window and are marked on the image just as if the user defined them manually. The user can change the name later through the *Get Overlay Info...* menu item.

| Results untitled |         |              |                 |             |             |                      |                           |          |               |             |
|------------------|---------|--------------|-----------------|-------------|-------------|----------------------|---------------------------|----------|---------------|-------------|
| Volume Name      | Type    | Mean<br>(OD) | Area<br>(mm*mm) | Min<br>(OD) | Max<br>(OD) | Volume<br>(OD*mm*mm) | Adj. Volume<br>(OD*mm*mm) | % Volume | Concentration | Mol. Weight |
| 4 items          |         |              |                 |             |             |                      |                           |          |               |             |
| V1               | Unknown | 1.04         | 37.44           | 0.54        | 1.43        | 3.8802e+3            | 3.8802e+3                 | 33.75    | 0.0000e+0     | 0.0000e+0   |
| V2               | Unknown | 0.83         | 37.44           | 0.50        | 1.28        | 3.0930e+3            | 3.0930e+3                 | 26.90    | 0.0000e+0     | 0.0000e+0   |
| V3               | Unknown | 0.62         | 37.44           | 0.50        | 0.98        | 2.3361e+3            | 2.3361e+3                 | 20.32    | 0.0000e+0     | 0.0000e+0   |
| V4               | Unknown | 0.58         | 37.44           | 0.49        | 0.84        | 2.1884e+3            | 2.1884e+3                 | 19.03    | 0.0000e+0     | 0.0000e+0   |

**Spot Parameters**

The *Spot Parameters...* command will display the Automatic Spot Detection Parameters dialog. This dialog allows the user to set the values needed for the spot detection algorithm. The spot size parameters determine the minimum and maximum length and width needed to be a valid spot. The Search Type allows the user to set the internal parameters for specific types of spots. The spot parameters dialog will contain simple parameters to set to find spots on different types of images. The user can set additional parameters by selecting the More Options... check box.

The dialog box is titled "Automatic Spot Detection Parameters". It contains a section titled "Spot Size (mm)" with two rows of input fields: "Width" and "Length". Each row has "Min:" and "Max:" labels followed by text boxes containing "1.00" and "10000.00" respectively. Below this section is a checkbox labeled "More Options..." which is currently unchecked. At the bottom are "Cancel" and "OK" buttons.

This option will expand and contract the parameters dialog accordingly. This allows the user to specify more detailed parameters.

The dialog box is titled "Automatic Spot Detection Parameters". It contains the same "Spot Size (mm)" section as the first image. Below it, the "More Options..." checkbox is now checked, and a new section titled "Spot Criteria" has appeared. This section includes "Volume:" and "Height:" labels, each with a "Min:" label and a text box containing "0.00" and "0.50" respectively. To the right of these are labels "(OD x mm x mm)" and "(OD)". Below these are three radio buttons: "%Total", "%Max", and "Abs. Units", with "Abs. Units" selected. At the bottom of the dialog is a section titled "Artifact Exclusion Criteria" with "Aspect Ratio:" and "Porosity:" labels, each with a "Min:" label and a text box containing "0.00". To the right of the "Aspect Ratio" text box is a "Max:" label and a text box containing "10000.0". At the bottom are "Cancel" and "OK" buttons.

The spot criteria parameters determine the minimum volume and height which a spot needs to be in order to be valid. The % Total, % Max, and Absolute Units options are used to determine the units used for the height spot criteria.

### Get Overlay Info

The *Get Overlay Info* command will display information about the selected spots. The user can also double click on an object on the image or on the report window to display this information window. If the user has many spots selected, opening many windows could take a long time. So while the windows are opening, the user can press command-period at any time to cancel the opening of the rest of the windows. Within this window the user can change the spot name, type, concentration, and comment. The type available are: Standard, Unknown, and Background.

**Spot Info**

**Name:**

**Type:**  **Shape:** Ellipse

**X-Center:** 100 (mm) **Mean:** 2.5 (O.D.)

**Y-Center:** 100 (mm) **Area:** 425.00 (mm x mm)

**Width:** 50 (mm) **Minimum:** 0.0 (O.D.)

**Length:** 50 (mm) **Maximum:** 3.0 (O.D.)

**Aspect Ratio:** 0.1 **Porosity:** 1.0

**Volume**

**Volume:** 1024.00 (O.D. x mm x mm)

**Adj. Volume:** 987.00 (O.D. x mm x mm)

**% Volume:** 100

**Calibration**

**Concentration:**

**Molecular Weight:**

**pI:**

**Comments:**

The user can change the name of the spot by simply selecting the Name field and typing in a new name.

**Name:**

The type determines how the spot is used in other calculations.

**Type:**

|            |
|------------|
| Unknown    |
| ● Standard |
| Background |

If the user selects Unknown, then the Concentration, Molecular Weight, and Isoelectric Point items become static text. The will contain the correct values based upon the spots defined as Standards and their associated values.

**Calibration**

**Concentration:** 1000.00

**Molecular Weight:** 1000.00

**pI:** 1000.00

If the user selects Standard, then the Concentration, Molecular Weight, and Isoelectric Point items become editable. The user can then enter values to be used in the calibration for the unknowns.

**Calibration**

**Concentration:**

**Molecular Weight:**

**pI:**

If the user selects background, then all of the spots of type background are used to calculate an average background value. This value is then used to calculate the adjusted volume and mean. If the Use Local Background option is selected, then a parameter around the object is used to calculate a local background. This local background is then used instead of the defined backgrounds. When Background is selected calibration is not applicable therefore is disabled.

**Calibration**

**Concentration:** N/A

**Molecular Weight:** N/A

**pI:** N/A

The volume section contains spot volume, adjusted volume from calibration, background subtraction, and the percent volume as compared with the other spots.

**Volume**

**Volume:** 1024.00 (O.D. x mm x mm)

**Adj. Volume:** 987.00 (O.D. x mm x mm)

**% Volume:** 100

The user can also enter comments.

### Comments:

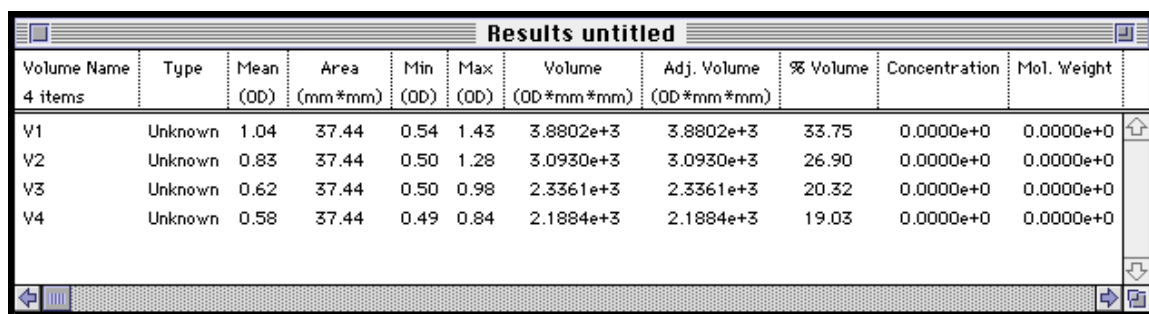
This is my comment for this particular spot that was found.

### Auto Integrate

The *Auto Integrate* command determines whether or not the results list window is automatically updated when spot values change. A spot is changed when it is re-sized, re-positioned, added, deleted, or the image changed (e.g. filtered). When this option is off the user can update the spot information by selecting the *Integrate Now* menu item.

### Integrate Now

The *Integrate Now* command is used in conjunction with the *Auto Integrate* command. This allows the user to re-calculate the results window information at a particular point in time, such as local background subtraction. When the Auto Integrate option is on, the Integrate Now option is of no use therefore its menu item is disabled.



| Volume Name | Type    | Mean<br>(OD) | Area<br>(mm*mm) | Min<br>(OD) | Max<br>(OD) | Volume<br>(OD*mm*mm) | Adj. Volume<br>(OD*mm*mm) | % Volume | Concentration | Mol. Weight |
|-------------|---------|--------------|-----------------|-------------|-------------|----------------------|---------------------------|----------|---------------|-------------|
| 4 items     |         |              |                 |             |             |                      |                           |          |               |             |
| V1          | Unknown | 1.04         | 37.44           | 0.54        | 1.43        | 3.8802e+3            | 3.8802e+3                 | 33.75    | 0.0000e+0     | 0.0000e+0   |
| V2          | Unknown | 0.83         | 37.44           | 0.50        | 1.28        | 3.0930e+3            | 3.0930e+3                 | 26.90    | 0.0000e+0     | 0.0000e+0   |
| V3          | Unknown | 0.62         | 37.44           | 0.50        | 0.98        | 2.3361e+3            | 2.3361e+3                 | 20.32    | 0.0000e+0     | 0.0000e+0   |
| V4          | Unknown | 0.58         | 37.44           | 0.49        | 0.84        | 2.1884e+3            | 2.1884e+3                 | 19.03    | 0.0000e+0     | 0.0000e+0   |

**Name** - Initially identifies the object's label as it is displayed on the screen (V=Volume, G=Grid). The object name displayed within the image and in the results window can be changed in the Overlay Information dialog box. The Overlay Information dialog box can be displayed by double-clicking on a volume object, double-clicking on a row in the results window, or selecting *Get Overlay Info* from the ANALYSIS menu.

**Type** - Sample types include *Unknown*, *Standard*, or *Background*. The sample type displayed in the results window can be changed in the Overlay Information dialog box.

**Mean** - Mean is the average O.D. or Counts of the pixels within the object, where O.D. and Counts = Volume/Area.

**Area** - Area of the identified object is in units of mm<sup>2</sup>.

**Minimum** - Minimum O.D. or Counts value within the object.

**Maximum** - Maximum O.D. or Counts value within the object.

**Volume** - The integrated volume of the object (O.D. or Counts \* Area).

**Adj Volume** - The integrated volume of the object adjusted for background removal.

% Vol - The percent volume of the object in relation to the total volume of all objects identified

Concentration - Concentration of unknowns is calculated from the standard curve, created by specifying standard concentrations and regression analysis.

Molecular Weight - Molecular weight of unknowns is calculated from the standard curve, created by specifying standard molecular weights and regression analysis

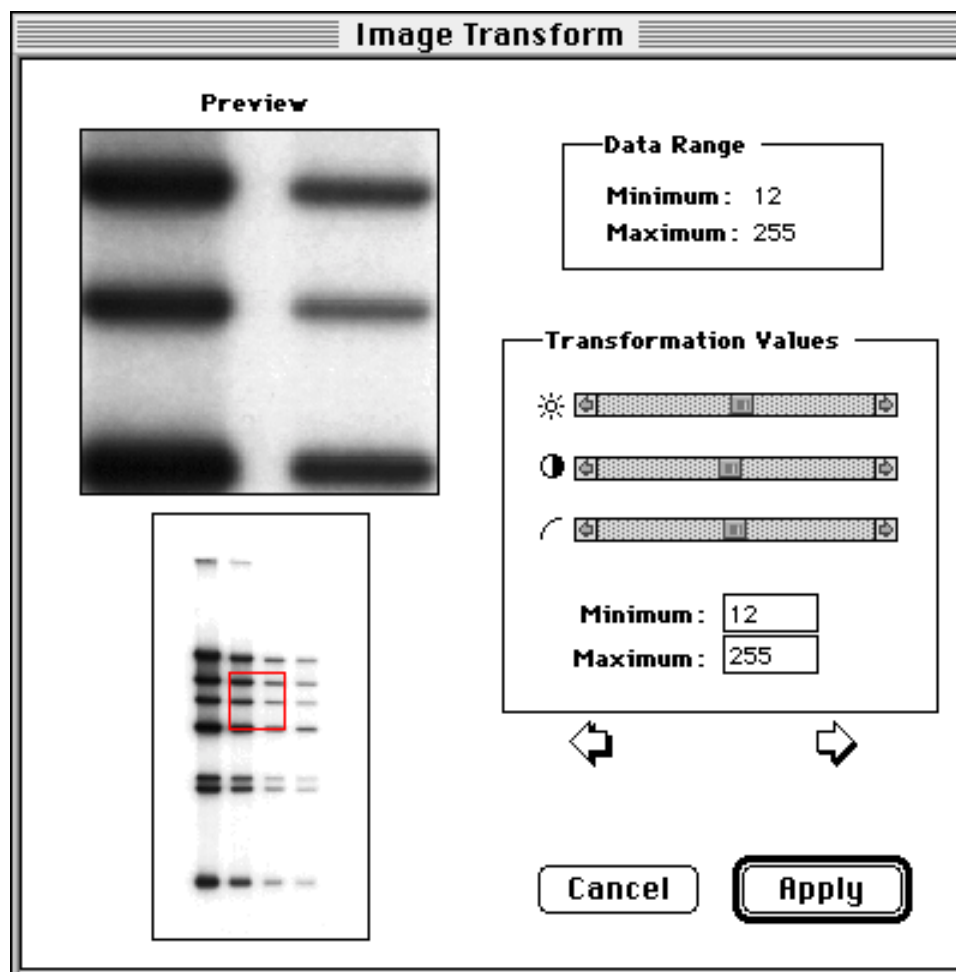
pI - The iso-electric point of unknowns is calculated from the standard curve, created by specifying standard pI and regression analysis.

### Local Background

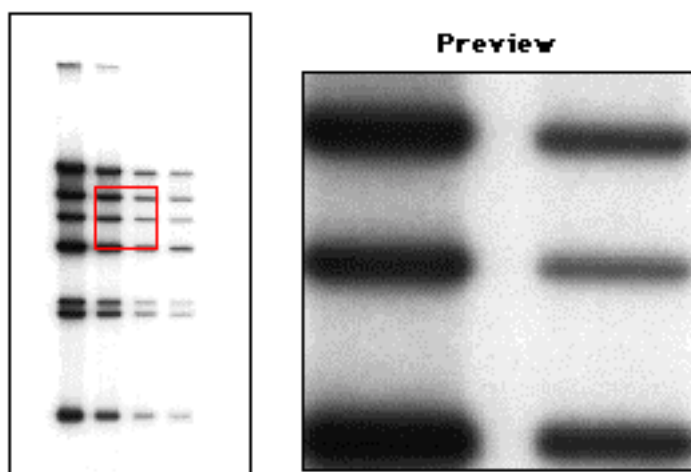
The *Local Background* command determines the type of background calculation used. When this option is on, a parameter two pixels wide is taken around each spot. These values are then averaged to calculate a local background for that spot. This value is then used in the calculation of spot values. When this option is turned off then the defined background (see spot info) is used.

### Transformation...

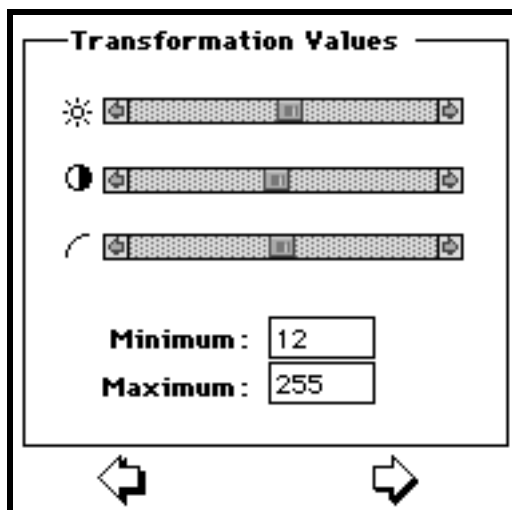
The *Transformation...* menu item allows the user to re-map the 16-bit or 8-bit raw data into the 8-bit view data. You can use this tool to examine how gray levels are distributed in your image. This basic information may help you determine how successfully an image can be enhanced.



This dialog contains a view of the entire image with a selection rectangle. The user can drag this rectangle to the position of the image to be previewed. This portion of the image is then displayed in the Preview section of the dialog. Any change in transformation will be first previewed in the Preview window.



Once the desired transformation is achieved, clicking on Apply will then affect the transformation changes.



The *Curvature* control   allows the user to adjust the curvature (also known as gamma correction) of non-linear curve types for the displayed image. When linear curve types are selected this control is disabled. This control operates like a standard scroll bar.

The *Contrast* control   allows the user to adjust the contrast of the displayed image. This control operates like a standard scroll bar. Scrolling to the left increases contrast or sharpness of the image and raises the minimum while lowering

the maximum. Scrolling to the right decreases contrast, blurring the image, and lowers the minimum while raising the maximum.

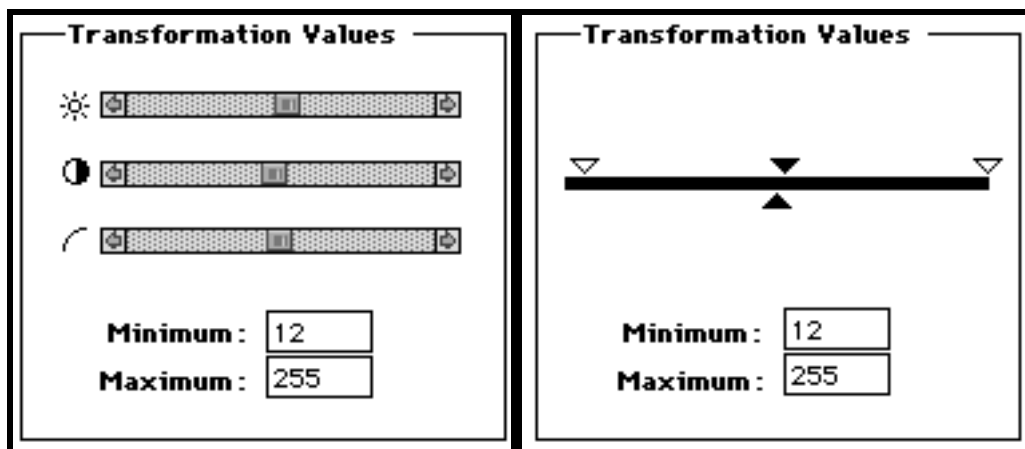
The *Brightness* control  allows the user to adjust the brightness of the displayed image. This control operates like a standard scroll bar. Scrolling to the left makes the image darker and lowers both the minimum and the maximum. Scrolling to the right makes the image lighter and raises both the minimum and the maximum.

Minimum:   
Maximum:

The *Minimum and Maximum* controls allow the user to visualize different sections of 8 bit, 12 bit, or 16 bit image data as the brightness and contrast controls are changed. Because display data on the computer monitor is limited to 0-255 gray levels, the minimum and maximum controls provide a way to squeeze 8 bit (0-255) or 12 bit (0-4,096) data from the densitometer and 16 bit (0-65535) data from the Molecular Imager into 0-255 gray levels.



The *Scrolling* control allows the user to scroll between types of transformation displays. The current displays available are the three scroll bar display and the single scroll bar display. Each display performs the same function.



Within the single scroll bar display, the upper markers control brightness and contrast and the lower marker controls curvature.

To reset the transformation, click on the Reset button. This will display the original range corresponding to the entire dynamic range of the image. Clicking on the Apply button will apply the transformation selection to the image.

Note: Transformation changes are not applied to images that have been cropped or extracted. All transformation changes to an image can be printed.

### Extract Profiles

The *Extract Profiles* command is used to transfer the object data to the profile analysis window and to create a profile. Each time this command is selected, a profile is created from all active profile objects that are displayed on an image.

## 4.6 Filter Menu

The FILTER menu is used to perform various types of filtering on images. These options allow the user to filter all or part of the image with pre-defined, as well as, user defined filters.



### Filter Raw

The *Filter Raw* command determines whether or not filtering is applied to the raw data as well as the view data. When this option is selected all filtering will be applied to the raw data. A new view data will then be created from this filtered raw data for display. Some filters cannot be applied to the raw data (e.g. Weighted Noise Reduction), these menu items will be disabled when the Filter Raw option is on.

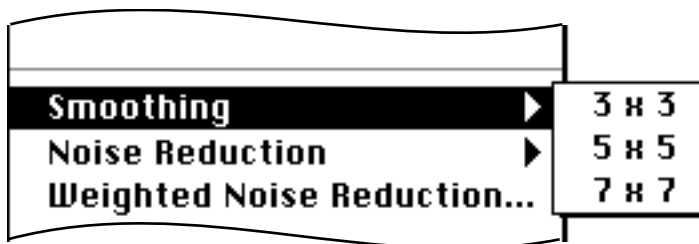
### Last Filter

The *Last Filter* command changes the filter to the name of the last filter operation performed. This will allow the user to quickly reapply a particular filter type.

### Smoothing

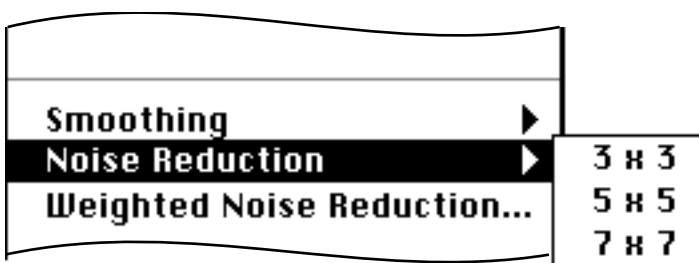
The *Smoothing* command is a hierarchical menu containing a sub menu with 3 item. This option will perform a simple averaging filter of 3x3, 5x5 or, 7x7 upon the image. This type of filtering passes a 3x3, 5x5 or 7x7 grid over the image and sums up all the pixel

values (counts) within the grid. Next, it then averages the summed value, and then places the value in the center of the grid. The end result is a smoother looking image.



### Noise Reduction

The *Noise Reduction* command is a hierarchical menu containing a sub menu with 3 item. This option will perform a simple median filter of 3x3, 5x5 or, 7x7 upon the image. This type of filtering passes a 3x3, 5x5 or 7x7 grid over the image and takes the median value of the pixel values (counts). It then places this median value in the center of the grid. The end result of this filter is an enhancement of features and a reduction in noise.

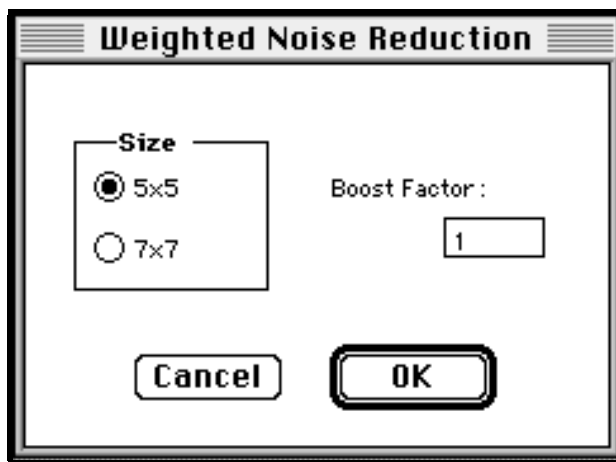


### Weighted Noise Reduction

The *Weighted Noise Reduction...* menu item applies a technique useful in bringing out details in extremely noisy images. It tends to bring out features in an otherwise murky background. This makes it appealing in cases where a good publication print is desired. In doing so, it destroys the quantitative nature of an Image.

User input: Boost Factor (selectable from 1 - 20)

User input: Filter size used in weighting(an odd number selectable from 5-7)



### Sharpen

The *Sharpen* and *Sharpen In Place* commands apply a sharpening filter upon the image. This filter tends to sharpen the image.

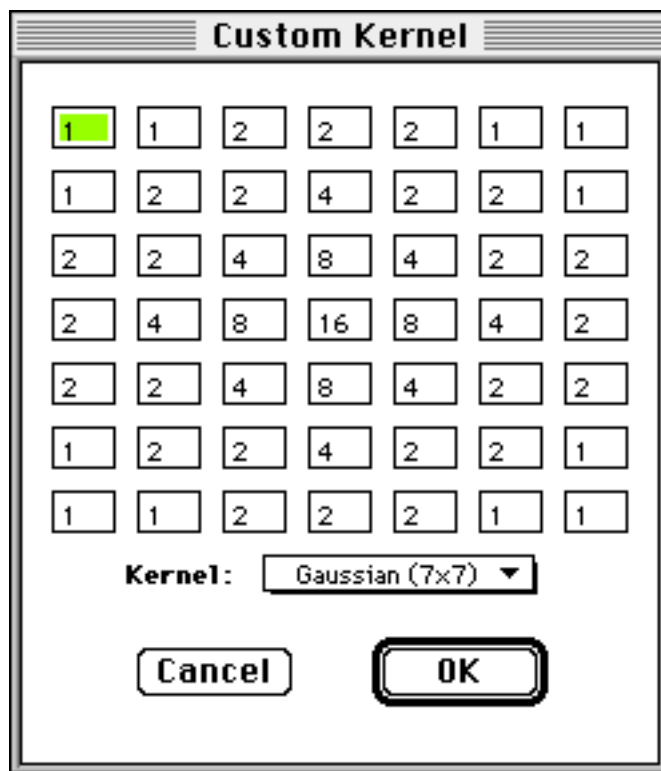
### Sharpen In Place

The *Sharpen* and *Sharpen In Place* commands apply a sharpening filter upon the image. This filter also tends to sharpen the image.

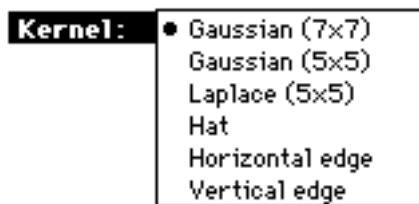
### Convolve

The *Convolve...* command allows the user to use a pre-defined or a user defined convolution kernel to filter the image. Since this process can take some time to perform, the user may define a small region of the image (using the selection tool) to preview the effects of the kernel.

Upon selecting this menu item a dialog as follows will be displayed.



The user can then select from the kernel pop-up menu, one of the pre-defined kernels and scaling factor. The kernel definition is displayed to the right of this menu. The user can also define a kernel by manually entering kernel values.



Once the kernel has been defined the user can select small portions of the image (using the select tool) to try with different kernels until they are satisfied with the image quality. When satisfied, the user can select the whole image, at which point the kernel will be applied to the entire image.

### DW-MTM

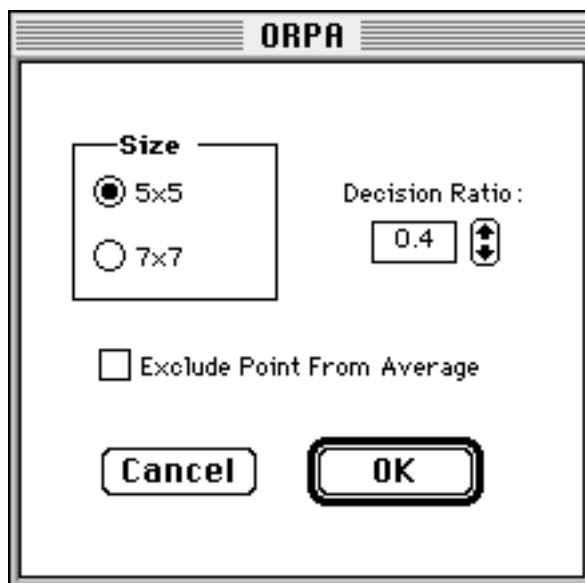
The *DW-MTM* command invokes the Double Window Modified Trimmed Mean Filter. This filter smoothes the image while retaining image edges.

### ORPA...

The *ORPA...* command invokes the Out of Range Pixel Averaging Filter.

This technique can be used to reduce the overall "grain" in images (useful in images that originate from the Molecular Imager). The algorithm should not degrade the quantitative information present in the original images, nor produce an unreasonable reduction in perceived resolution. Since this process can take some time to perform, the user may define a small region of the image (using the selection tool) to preview the effects of the kernel. The "Exclude Point from Average" option may include or exclude the center kernel point in the smoothing calculation.

User input: Decision ratio (selectable from 0.3-0.7)  
User input: Filter size used in weighting (an odd number selectable from 5-7)  
User input: Exclude Point from Average



### ANR

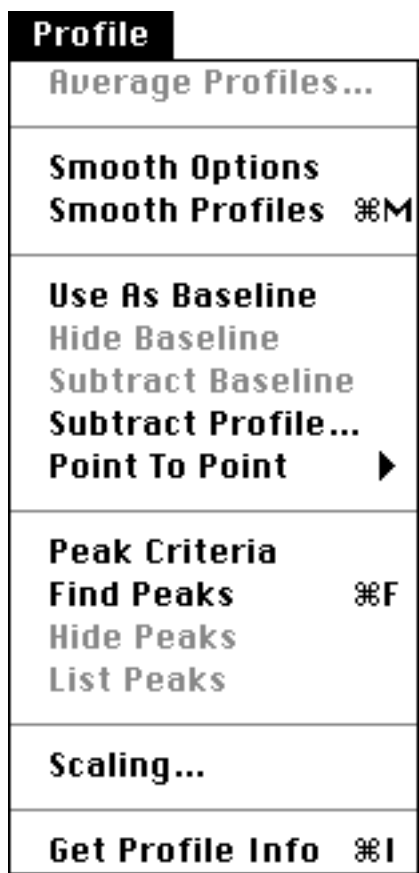
The *ANR* command invokes the Adaptive Noise Reduction Filter.

This technique is operational with Molecular Imager images and actively removes additive noise from images. It does not soften or blur the edges and is useful when visualizing images with poor signal-to-noise ratios.

This command is inactive if the command *Filter Raw* is not selected from the FILTER menu.

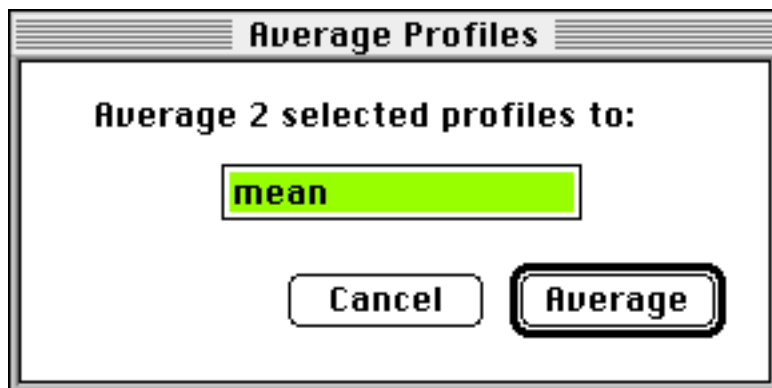
#### 4.7 Profile Menu

The *PROFILE* menu contains operations that will actually modify profile data. If there is no Profile window displayed, all items in the *PROFILE* menu will be disabled.



##### Average Profiles...

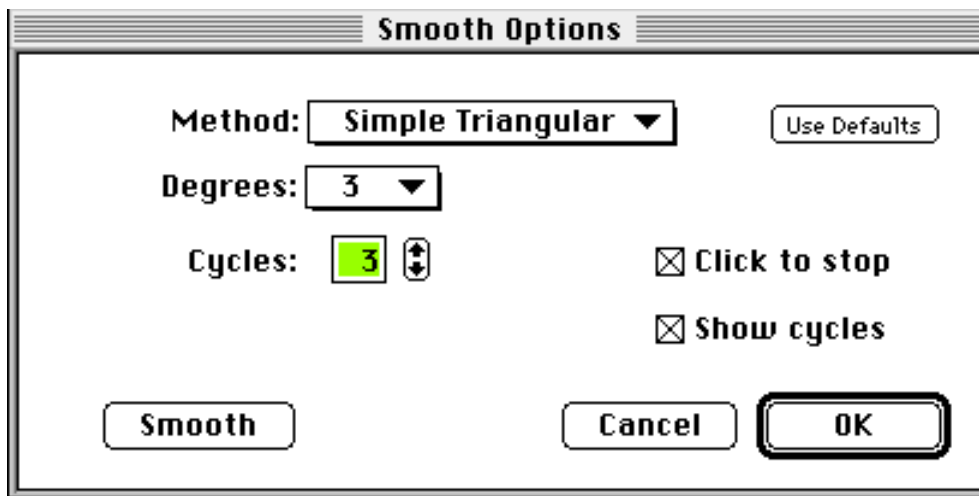
The *Average Profiles...* command will average any number of profiles that are present in the Profile window.



After averaging, the new averaged profile will be plotted, selected, and added to the existing selected profiles.

### Smooth Options

The *Smooth Options* command will display a dialog where the parameters used in the smoothing of profiles can be set. The window of smoothing refers to the number of adjacent data points that will be averaged in the smoothing operations. Simple Triangular provides one window for smoothing. DW-MTM provides two windows for more powerful smoothing while retaining profile edges. Enhancement will expand active profiles according to the boost factor. Enhancement is achieved by taking profile frequencies and multiplying them by some "Boost" factor.



### Smooth Profiles

The *Smooth Profiles* command will smooth the currently selected profiles using the options set in the "Smooth Options" criteria. After smoothing is performed, the smoothed data replaces the original data. The area under curves is not altered.

### Use As Baseline

Any currently selected profile can be designated as the baseline. The profile will be displayed in particular color in the Profile window and its name will be underlined in the profile name list. After specifying a profile as a baseline, it can then be subtracted from a profile by using the *Subtract Baseline* command. The *Use As Baseline* toggles from and to *Unused As Baseline*.

### Hide Baseline

The *Hide Baseline* command toggles between *Hide Baseline* and *Show Baseline*. If a baseline has been created, then this option allows the baseline to be either shown or hidden. If a profile is chosen as a baseline, as indicated by an underline of the profile name in the profile name list, then this option allows the profile curve display to be either shown or hidden. Note that the show/hide box of the profile in the profile name list will be changed accordingly.

### Subtract Baseline

The *Subtract Baseline* command will subtract the current baseline from the currently selected profiles. There are three methods of background subtraction: horizontal baseline, interactively derived (point-to-point) baseline, and actual background.

#### Horizontal Baseline Subtraction

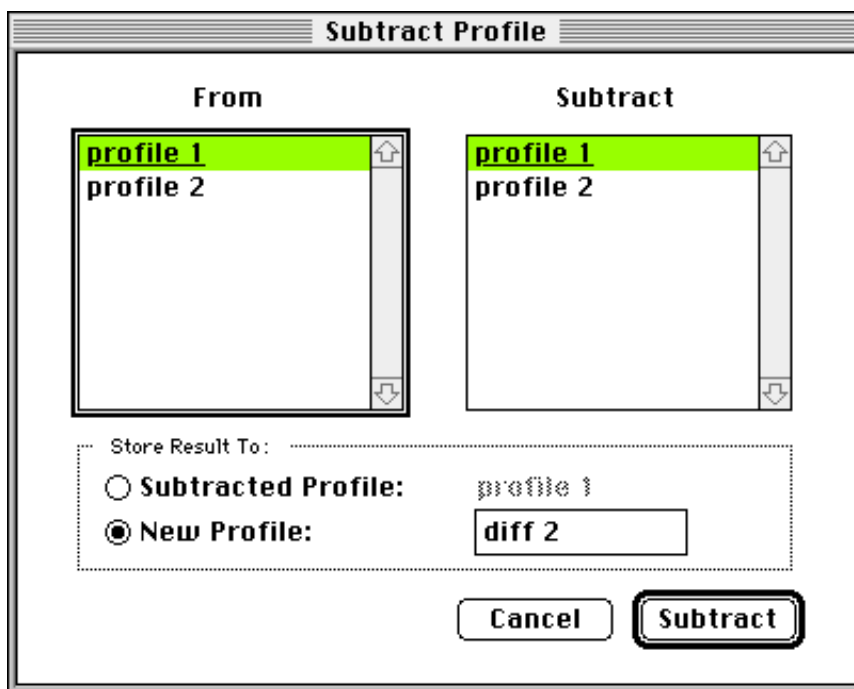
A horizontal baseline is created by using the horizontal baseline tool. Once a baseline is created and at least one profile is selected in the profile window, the *Subtract Baseline* command in the *PROFILE* menu will subtract the baseline from every selected profile.

#### Interactively Derived (Point-to-Point) Baseline Subtraction

See "Tool Palette" section for details on generating the Point-to-Point baseline.

#### Actual Background Subtraction

*Actual Background Subtraction* allows actual image background to be subtracted from other profiles in order to achieve a more accurate horizontal baseline. The subtracting profile should be a sample of image background. Many profiles can be selected in the "Subtract..." list but only one profile can be selected from the "From..." list.



### Subtract Profile...

The *Subtract Profile...* command will find the difference between any two profiles.

### Point To Point

The *Point To Point* command will popup another menu for selecting the method to be used for creating the interactively derived baseline. This baseline links the points generated by using the Point-to-Point tool on the selected profiles.

### Peak Criteria

The *Peak Criteria* command produces a dialog that allows the criteria to be set to find the peaks of the currently selected profile.

The screenshot shows a dialog box titled "Peak Criteria". It is divided into two main sections: "Minimum Criteria" and "Options".

**Minimum Criteria:**

- Units:** A dropdown menu currently set to "Absolute Units".
- Peak Height:** A checked checkbox followed by a text input field containing "0.00" and the unit "counts".
- Peak Width:** An unchecked checkbox followed by a text input field containing "0.00" and the unit "mm".
- Peak Area:** An unchecked checkbox followed by a text input field containing "1.00" and the unit "counts \* mm".

**Options:**

- Highlight Peaks:** A checked checkbox.
- Number Peaks:** A checked checkbox.
- Smooth First:** A checked checkbox.

At the top right of the Options section is a button labeled "Use defaults". At the bottom of the dialog are three buttons: "Find Peaks", "Cancel", and "OK".

Peaks are identified through height or amplitude (**Peak height criteria assumes that the baseline has been subtracted from the profile**), width (at the base), and area. The criteria units dictate how the minimum criteria are interpreted. Percent total refers to the total area of all peaks. So setting peak area to 5% total means that peaks containing at least 5% of the total area of all peaks will be flagged. Percent maximum refers to the peak with the greatest magnitude with respect to the criterion selected. Thus, setting minimum criteria to 20% maximum peak height means that peaks that are at least 20% of the amplitude of the highest peak (in absolute units) will be flagged. The absolute units are those which are current for the active profile.

You can also set options to determine how the found peaks will be visualized. If peak highlighting and numbering is selected, the found peaks will be marked as indicated below. Automatic smoothing is a very useful option. Generally, you will always want to smooth your data before finding peaks since any noise will be flagged as a peak. Smoothing will use the currently specified *Smooth Options* (found under the PROFILE menu). If the current profile is already smoothed, then the Auto-Smooth selection will have no effect.

### Find Peaks

The *Find Peaks* command analyzes the selected profile and identifies peaks that match the set of criteria previously entered in the Peak Criteria dialog. Each peak found is shaded and peak numbers are drawn on the top of the peaks.

### Hide Peaks

The *Hide Peaks* command toggles between *Hide Peaks* and *Show Peaks*. This function will either hide or show the identification flags specified in the Peak Criteria dialog options. If no "Find Peaks" has been done, there will be no action and the *Hide Peaks* menu item is dimmed.

### List Peaks

Profile peaks are generated by choosing the *Find Peaks* command. They are then listed in this results window by choosing the *List Peaks* command in the *Profile* menu. The results can then be displayed, edited in the corresponding information dialog, saved, and printed. Results in the window include profile points, profile peaks, profile areas, and molecular weight results from standards.

| Results untitled 3 |       |       |       |        |          |             |            |           |  |
|--------------------|-------|-------|-------|--------|----------|-------------|------------|-----------|--|
| Profile Area Name  | From  | To    | Width | Height | Location | Area        | % of Total | Detection |  |
| 6 items            | (mm)  | (mm)  | (mm)  | (O.D.) | (mm)     | (O.D. * mm) | (%)        |           |  |
| profile 1.area 1   | 10.80 | 17.10 | 6.30  | 0.45   | 16.00    | 1.3638e+0   | 21.47      | Automatic |  |
| profile 1.area 2   | 17.20 | 20.60 | 3.40  | 0.36   | 18.70    | 9.2584e-1   | 14.58      | Automatic |  |
| profile 1.area 3   | 20.70 | 24.70 | 4.00  | 0.35   | 23.10    | 9.1941e-1   | 14.47      | Automatic |  |
| profile 1.area 4   | 27.10 | 30.80 | 3.70  | 0.35   | 29.10    | 9.8113e-1   | 15.45      | Automatic |  |
| profile 1.area 5   | 33.40 | 36.90 | 3.50  | 0.41   | 34.70    | 7.3688e-1   | 11.60      | Automatic |  |
| profile 1.area 6   | 37.80 | 41.80 | 4.00  | 0.31   | 38.90    | 4.6366e-1   | 7.30       | Automatic |  |

Clicking each row will highlight the data. This is useful when exporting to other applications. Pressing the keyboard Shift key while dragging the mouse will contiguously select rows. Pressing the keyboard Option key while clicking will select individual rows. Highlighted rows can be cut, copied, or cleared using the *Edit* menu. The Undo command will reverse the action. Double clicking on a row in the Results window will bring up an information dialog.

| Profile Integration Info |                           |
|--------------------------|---------------------------|
| <b>Name:</b>             | Profile 1.Area 1          |
| <b>Profile:</b>          | Profile 1                 |
| <b>File:</b>             | Untitled Profiles         |
| <b>From:</b>             | 116 mm <b>To:</b> 320 mm  |
| <b>Area:</b>             | 6.062e+6 00 x mm          |
| <b>% of Total:</b>       | 58.84 %                   |
| <b>* of Peaks:</b>       | 34                        |
| <b>Created:</b>          | Thu, Jun 3, 1993, 9:20 AM |
| <b>Modified:</b>         | Thu, Jun 3, 1993, 9:21 AM |
| <b>Comments:</b>         | <div></div>               |

The Results window has adjustable tabs between the names of different fields. Move the mouse cursor to the line separating the field names, and the cursor will change to a tab adjustment tool . Click on the separating line and drag it to its desired position. The screen will be redrawn to reflect the new tab spacing.

| From: |  | To    |
|-------|-------------------------------------------------------------------------------------|-------|
| 3 mm  |                                                                                     | 7 mm  |
| 7 mm  |                                                                                     | 11 mm |
| 11 mm |                                                                                     | 15 mm |
| 16 mm |                                                                                     | 20 mm |
| 25 mm |                                                                                     | 29 mm |
| 29 mm |                                                                                     | 32 mm |
| 48 mm |                                                                                     | 52 mm |


 Tab will be set here

The tabs are used for printing as well as display on the Macintosh screen. The relative spacing of the different records during printing is based on their relative placement in the Results window. If two records are too close to each other in a printout, adjusting the tab between the two records should correct the problem.

### Scaling

Scales of the x and y axes are enlarged or contracted when the profile window is resized. That is, the axes will always scale to fit the window.

Double clicking on the x or y axis numbers will bring up a movable Scaling dialog. An alternative way is to select the *Scaling...* menu item from the *Profile* menu.

**Scaling**

---

**Scales**

|           | Min   | Max   | Units  | Label |
|-----------|-------|-------|--------|-------|
| <b>X:</b> | 10.00 | 45.00 | mm ▼   |       |
| <b>Y:</b> | 0.00  | 0.70  | O.D. ▼ |       |

**Set To:** Fixed Scale ▼

**All Profiles**

|           | Min   | Max   | Units |
|-----------|-------|-------|-------|
| <b>X:</b> | 11.60 | 44.60 | mm    |
| <b>Y:</b> | 0.02  | 0.66  | O.D.  |

Scaling refers to the size that the plotted profiles have in the display screen. Profiles automatically use full scaling (*Scale to Fit All Visible Option*). A profile that takes up the entire display window is said to be in full scale (*Scale to Fit Selected Option*). However, if comparing multiple profiles it is more useful to have the display window correspond to an absolute measurement (*Fixed Scale Option*). Note that profiles can be selected only if they are active. *Scale to Fill All Option* will scale to include all profiles (visible, selected, and hidden) in the Profile window.

The pop-up menus inside the x and y boxes display different units for x and y axes respectively. X units are defaulted to millimeters (mm). All profiles show the same x-y units and labels. Y units are displayed in either Absorbance (O.D.), % Transmission / Reflectance, % Absorbance, or Counts. Selecting *Absorbance* converts the Y axis of the graph to O.D.. The OD scale is used for concentration quantities and represents  $-\log_{10}$  (data/calibration). Selecting *% transmission* on the Y axis is useful for analyzing data whereby the bands in the photograph are white, and the background is black. This data is represented as negative peaks under absorbance (O.D.). Such data can be from photographs of ethidium bromide gels. Selecting *Absorbance\*cm* on the Y axis represents absorbance (O.D.) multiplied by the width of the profile object.

**Scales**

|           | Min | Max   | Units  | Label          |
|-----------|-----|-------|--------|----------------|
| <b>X:</b> | 3.8 | 29.9  | mm ▼   | Profile Length |
| <b>Y:</b> | 0   | 23789 | O.D. ▼ |                |

If the *Fixed Scale* option in the Scaling dialog is not selected, the minimum and maximum values of the X and Y axes are dimmed and are not selectable. However, the labels and the units can still be modified.

### Get Profile Info

The Get Profile Info command allows the user to enter a profile name and comments and display relevant profile information

**Profile Peak Info**

**Name:** Profile 1.Pk 1

**Profile:** Profile 1  
**File:** Untitled Profiles

**Center:** 131 mm  
**Height:** 3038 O.D.  
**Area:** 1.961e+6 O.D. x mm  
**% of Total:** 0.14 %

**From:** 62 mm    **To:** 766 mm  
**Detection:** Automatic

**Created:** Thu, Jun 3, 1993, 9:20 AM  
**Modified:** Thu, Jun 3, 1993, 9:21 AM

**Comments:**

### 4.8 Regression Menu

The *Regression Parameters* menu item allows the user to change the calibration parameters which are used for concentration, molecular weight, and isoelectric point calibration.

| Regression                                |
|-------------------------------------------|
| ✓MW vs Rf<br>Log MW vs Rf                 |
| Linear<br>✓Quadratic<br>Cubic<br>Logistic |
| % Volume<br>✓Abs Volume                   |
| Show Names<br>Show Errors                 |

### **MW vs Rf**

The *MW vs Rf* command is a method to calculate molecular weights from the current set of volume or profile standards. This method uses the molecular weight (dependent variable) against the relative mobility (Rf dependent variable) of the standard.

### **Log MW vs Rf**

The *Log MW vs Rf* command is a method to calculate molecular weights from the current set of volume or profile standards. This method uses the log of the molecular weight (dependent variable) against the relative mobility (Rf dependent variable) of the standard.

### **Linear**

The *Linear* command item uses a Linear regression method to calculate a standard curve and requires a minimum of two points. This is the default for the regression calculation.

### **Quadratic**

The *Quadratic* command item uses a Quadratic regression method to calculate a standard curve and requires a minimum of three points.

### **Cubic**

The *Cubic* command item uses a Cubic regression method to calculate a standard curve and requires a minimum of four points.

### **Logistic**

The *Logistic* command item uses a Logistic regression method to calculate a standard curve and requires a minimum of five points.

### **% Volume**

The *% Volume* command item changes the x axis of the volume analysis regression curve to % volume and is active during concentration determination.

### **Abs Volume**

The *Abs Volume* command item changes the x axis of the volume analysis regression curve to absolute volume and is active during concentration determination.

### Show Names

The *Show Names* command item toggles between “Show Names” and “Hide Names”. When activated, it displays the names of the profile or volume standards at their plotting position on the regression curve.

### Show Errors

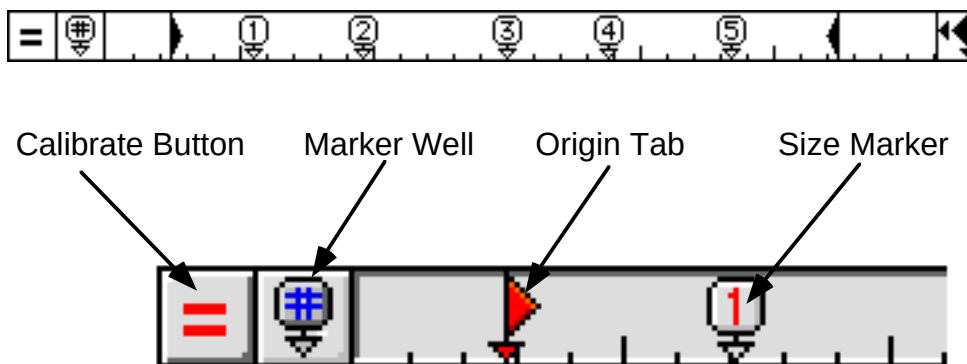
The *Show Errors* command item toggles between “Show Errors” and “Hide Errors”. When activated, it displays the percent error of the profile or volume standards relative to their regression positions.

## 4.9 MW Menu



### New Ruler

This feature places a new molecular weight standard ruler at the top of the current Profile window. This molecular weight ruler defines a series of standards as tab positions.



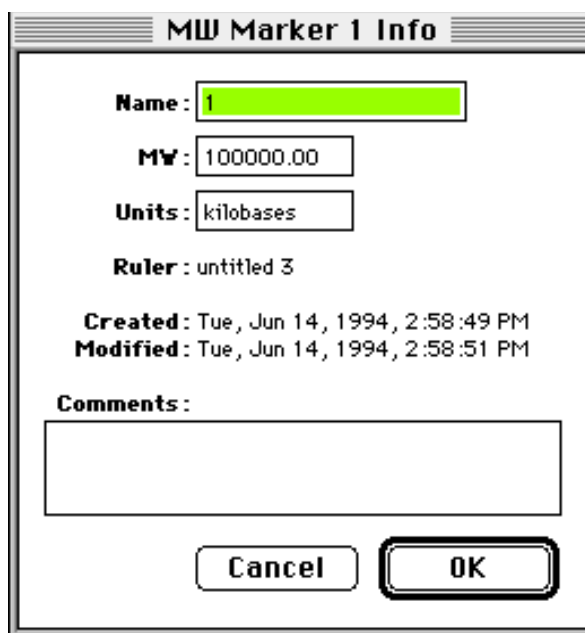
The ruler extends across the top of the graph window and consists of three areas. The area directly above the graph window is the ruler itself where the positions of peaks and fronts will be indicated. Since molecular weight determinations are based on the migration of a sample from the top (origin) of a gel relative to the bottom (ion front), these points must be observable on the profile itself or measured directly from the gel. The left and right triangles can be clicked on and dragged within the MW ruler to indicate the origin and front respectively. These must be set for each profile before clicking on the calibrate button.

Immediately to the left of the ruler is the marker well. Molecular weight markers can be created by dragging from the marker well into the ruler, over the peak of interest, and entering a name and weight. Conversely, dragging a marker into the well will dispose the

marker. The maximum limit of markers (or tabs) in a MW ruler is twenty (20). Immediately to the left of the marker is the calibration button, indicated by the = sign. When the markers are set, pressing this button will calibrate the standard set. This calibration button is also used to update marker names and weights before the regression curve can be updated.

Pressing the calibration (=) button when at least two markers are set will calibrate the standard set. At least two markers are needed in order to perform a linear calibration. The x-axis and the status box will be changed to reflect the scale in molecular weight units. The profile window can then be sized accordingly to accommodate all molecular weight values. The MW units will not be changed back to normal units unless the MW ruler is hidden or another profile curve is selected.

Double clicking one of the markers on a MW ruler will bring up the following info window for display or modification:

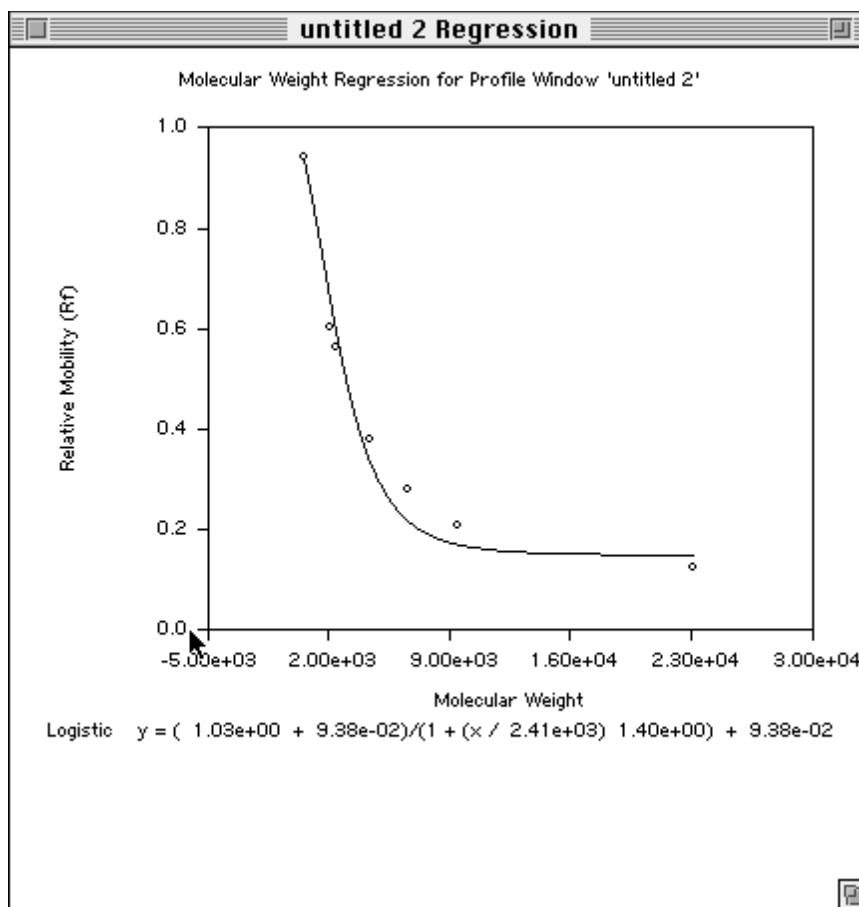


The image shows a dialog box titled "MW Marker 1 Info". It contains several input fields and labels: "Name:" with a text box containing "1"; "MW:" with a text box containing "100000.00"; "Units:" with a text box containing "kilobases"; "Ruler:" with a text box containing "untitled 3"; "Created:" with a timestamp "Tue, Jun 14, 1994, 2:58:49 PM"; "Modified:" with a timestamp "Tue, Jun 14, 1994, 2:58:51 PM"; and "Comments:" with a large empty text area. At the bottom are "Cancel" and "OK" buttons.

Once a ruler has been calibrated, the molecular weights of the standard values can be sent to a results window by selecting List from the MW menu.

Although the molecular weight ruler allows direct reading of molecular weights from profiles, it does not provide an assessment of the goodness-of-fit between the markers and the extrapolated molecular weights. This function is provided by the Regression Window. Select *Regression* from the WINDOW menu to display a regression curve plotted in the appropriate format. This allows a direct visualization of the correspondence between the markers and the estimated molecular weights.

The Regression Window is semi-dynamic, so that any change in the molecular weight ruler (names or weights) will update the regression display when the calibration sign (=) is used.



After a standard profile containing molecular weight markers has been calibrated, the calibrated ruler can be applied to other profiles for the calculation of unknown molecular weights. The profile you want to evaluate in terms of molecular weight is plotted in the foreground of the Profile window. Note that although the marker tabs become inactive, the top and bottom markers remain active. Move the top marker (left-most) to the part of the profile corresponding to the gel origin. Move the bottom marker (right-most) to the part of the profile corresponding to the gel front. Then press the Calibrate (=) button. This profile is now calibrated and molecular weights can either be read directly off the screen by moving the pointer to the desired peak and reading the X coordinate box, or the peaks in the unknown profile can be sent to the Results Window using the *List Peaks* command from the PROFILE menu.

### Open Set

These rulers can be stored on a disk and later recalled. This allows you to rapidly determine the molecular weights of a series of profiles by loading a ruler and adjusting the position of the tabs to correspond to the migration position of standards. A MW ruler stored in the BRIFF format can be read through the *Open...* menu item of the FILE menu. Protein standards ruler information from files which have been pre-made or have been previously made by the user can be selected and opened.

A pre-existing molecular weight ruler file can be opened by selecting the *Open Set...* option. The ruler is found and loaded using the standard Macintosh File dialog.

### Save Set

When created, a ruler can be saved to disk by selecting the *Save Set...* command from the MW menu.

### Show Ruler

The *Show Ruler* command item can be toggled to *Hide Ruler*. It will either show or hide the MW ruler in the currently displayed Profile window. If a ruler has not been created in the Profile window, this menu item is dimmed.

### List

The *List* command item will place the name, size,  $r^2$ , and errors of all the profile standard molecular weight markers of the current ruler into a results window which will then be displayed.

If the ruler has been calibrated, then the relative mobility and calculated error of the standards are also provided. Errors are calculated as the difference between the actual position of the marker and the best position of the marker as determined by the method of molecular weight estimation. Information on the molecular weight markers can be printed from the results window. As the regression for the standards is changed, a new results window needs to be created.

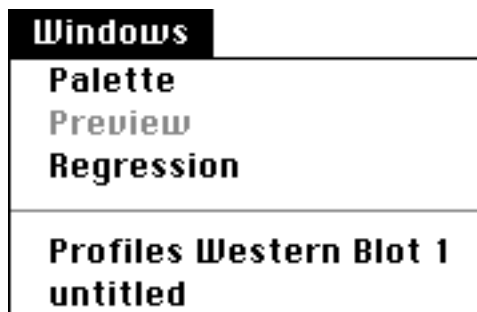
| untitled 13    |       |                  |      |                   |                              |                         |         |          |         |         |         |
|----------------|-------|------------------|------|-------------------|------------------------------|-------------------------|---------|----------|---------|---------|---------|
| Standards Name | Index | Mobility<br>(mm) | Rf   | MW<br>(kilobases) | Calculated MW<br>(kilobases) | Residual<br>(kilobases) | % Error | Coeff 0  | Coeff 1 | Coeff 2 | Coeff 3 |
| 7 items        |       |                  |      |                   |                              |                         |         |          |         |         |         |
| 1              | 1     | 29.94            | 0.17 | 23130.00          | 19690.55                     | 3439.45                 | 0.17    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 2              | 2     | 36.12            | 0.25 | 9416.00           | 11524.89                     | -2108.89                | -0.18   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 3              | 3     | 40.96            | 0.32 | 6557.00           | 7570.25                      | -1013.25                | -0.13   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 4              | 4     | 47.78            | 0.41 | 4361.00           | 4418.44                      | -57.44                  | -0.01   | 32203.56 | 2.52    | 0.20    | -99.94  |
| 5              | 5     | 60.64            | 0.58 | 2322.00           | 1907.35                      | 414.65                  | 0.22    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 6              | 6     | 63.53            | 0.62 | 2027.00           | 1619.80                      | 407.20                  | 0.25    | 32203.56 | 2.52    | 0.20    | -99.94  |
| 7              | 7     | 86.88            | 0.94 | 546.00            | 530.73                       | 15.27                   | 0.03    | 32203.56 | 2.52    | 0.20    | -99.94  |

### Reset X Axis

The *Reset X Axis* command item resets the X axis to mm units.

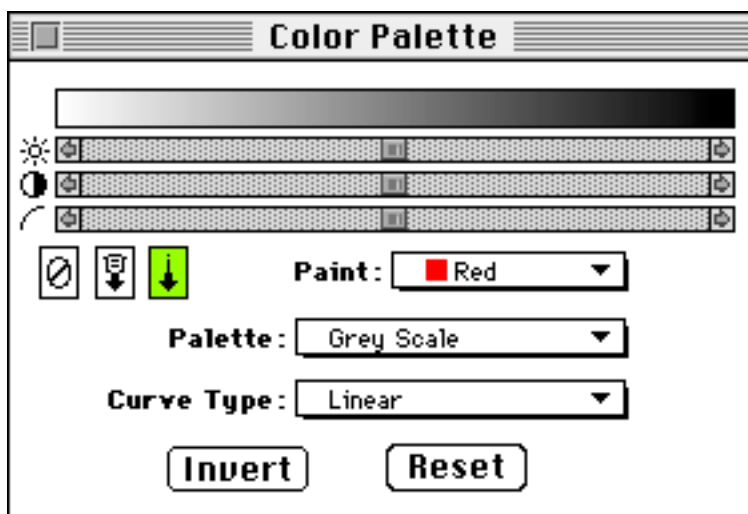
## 4.10 Windows Menu

The *Windows* menu item will contain a list of all currently opened windows. When the user selects one of these menu items, the window associated with that item will be brought forward.



### Palette

Adjusting image brightness and/or contrast using the Color palette is a rapid way to reduce image display noise and enhance display of the important gray level regions of your sample. The Color palette, or look up table (LUT) is dynamically linked to the transformation function, and to the image display window. These features effect only the display of the image and not the underlying image data used in quantitation.



### Palette Highlighting

The palette highlighting allows the user to color parts of the palette which are reflected in the image.

### Clear

This tool allows the user to remove all user defined highlights (colors) within the palette.



### Fill

When the user selects this tool it allows them to fill the image gray scale palette with color. To fill a section of the gray level palette the user selects this tool and a fill color. Then positions the cursor to the start position in the palette and clicks and drags the cursor. As the cursor is dragged through the palette, it is filled with the current fill color. When the user is finished they simply release the mouse button.



### Set Value

When the user selects this tool it allows them to fill a specific section of the image gray level with the current fill color. To fill a position of the palette the user selects this tool and a fill color. Then positions the cursor to the position in the palette to be highlighted and clicks mouse. As the cursor is dragged through the palette, the fill color follows the cursor not filling behind. When finished, simply release the mouse button and the palette is highlighted at that position.



### Fill Color

The user can select one of the pre-defined colors to be used by the fill or set value tools. To select a color, simply click on the desired color within the pull-down menu.



### Curvature



The *Curvature* control allows the user to adjust the curvature (also known as gamma correction) of non-linear curve types. When linear curve types are selected this control is disabled. This control operates like a standard scroll bar.

### Contrast



The *Contrast* control allows the user to adjust the contrast of the palette. This control operates like a standard scroll bar.

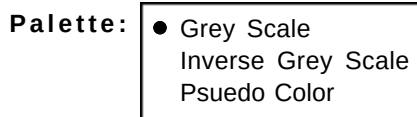
### Brightness



The *Brightness* control allows the user to adjust the brightness of the palette. This control operates like a standard scroll bar.

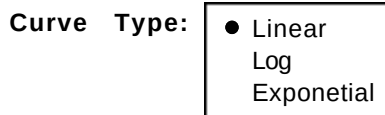
### Palette

The *Palette* pop-up menu contains a list of pre-defined palette types. When a user selects one of these items, the palette will be updated to reflect the selection.



### Curve Type

The *Curve* pop-up menu contains a list of pre-defined curve types. When a user selects one of these items, the palette will be updated to reflect the selection.



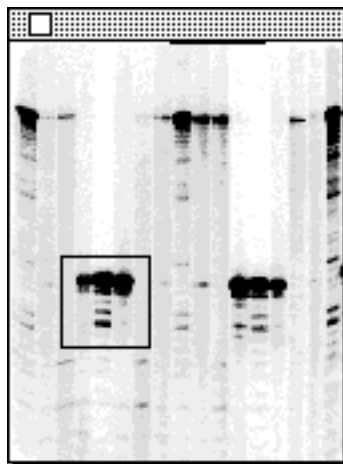
### Invert - Reset

The *Invert* and *Reset* buttons allow the user to invert or reset the palette. The invert button inverts the current palette. The reset button sets the palette to its default values.



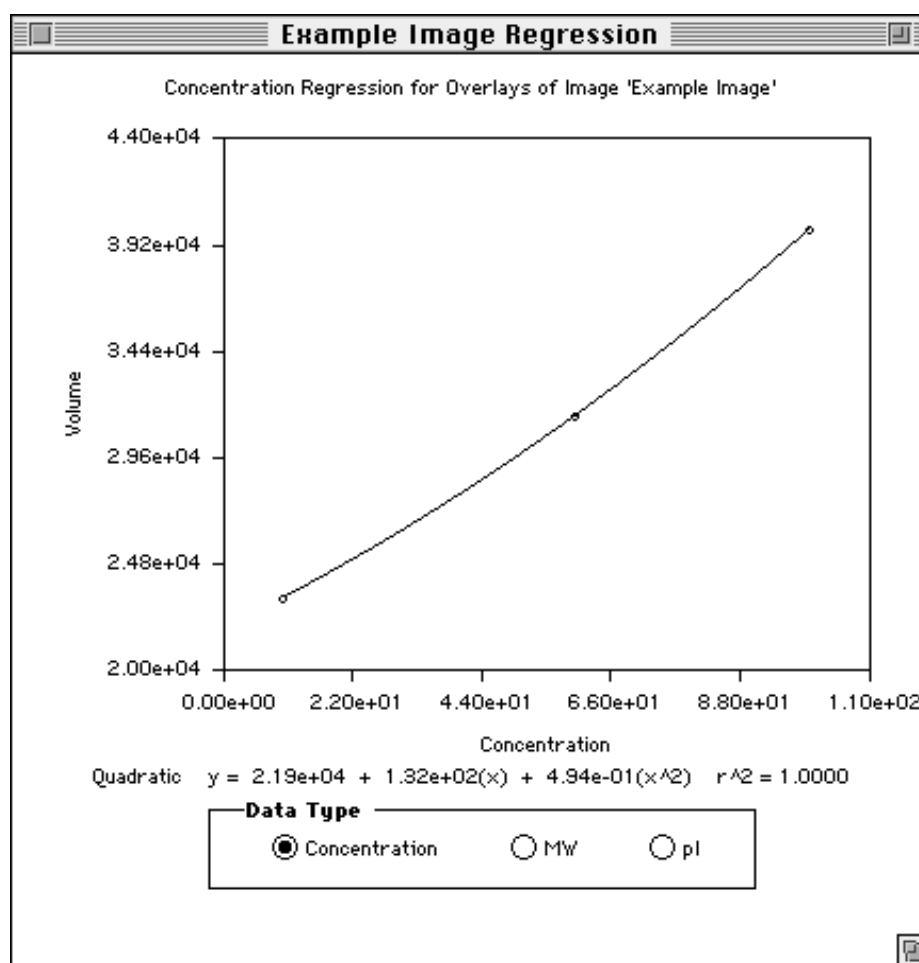
### Preview

The Preview command contains a small footprint of the entire image when the image is zoomed. Within this image footprint is a selection rectangle sized proportional to the amount of image displayed in the main window. The user can drag this rectangle to any position on the small image and upon releasing the mouse, the image in the main window will be positioned accordingly.



## Regression

The *Regression* command displays the regression graph of the standards from profile of volume data. Different regression curves can be generated for best fit determination.



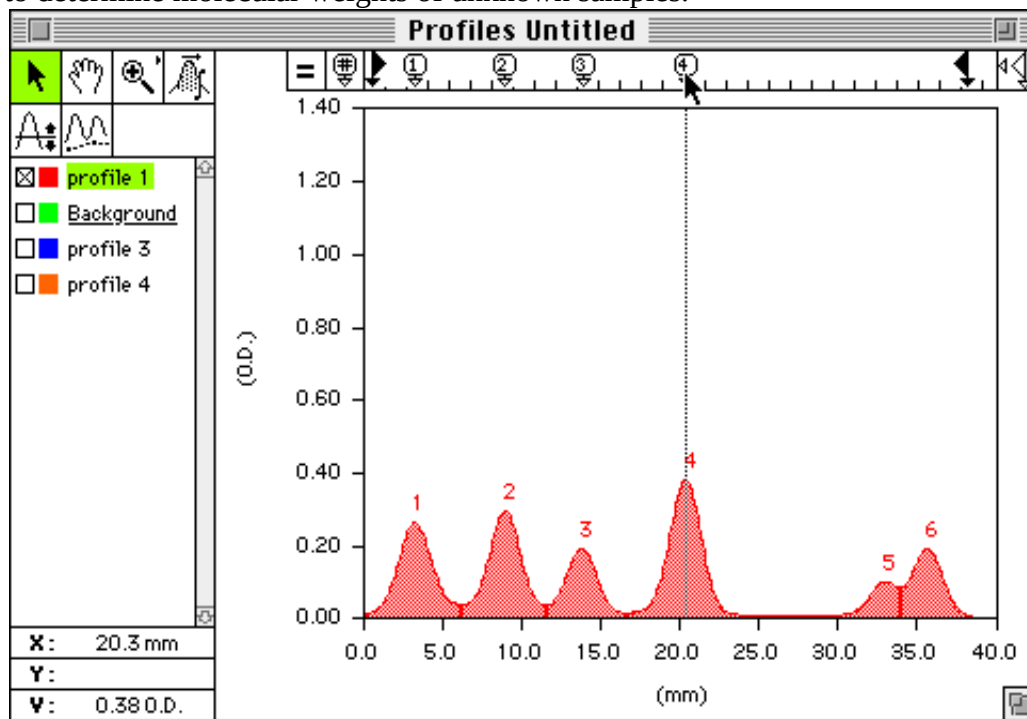
There is one Regression window associated with either a Volume results table or Profile window. The Regression window can only be activated when there is an active molecular weight ruler or volume standard concentrations entered in the results window.

The names and errors next can be displayed by selecting the *Show Names* or the *Show Errors* menu items from the REGRESSION menu.

The standard curve in the Calibration window can be printed or copied to the clipboard.

## Section 5 Profile Window

The Profile window is where profiles from an image are plotted, compared, and analyzed. It makes full use of the Macintosh graphical interface to make previously difficult analyses easy. It allows you to both observe your data in a graphical form and to interact with it. You can edit the profiles as well as determine areas, points, and values. You can overlay multiple profiles to compare them. It also allows you to align profiles for averaging or subsequent subtraction from each other. Finally, the profile window allows you to determine molecular weights of unknown samples.



### Profile Window Components

#### Profile Tool Palette

The tool pallet allows you to easily access the most commonly used functions. These functions allow you to move profiles, expand profiles, or select parts of profiles for expansion or editing. The tools also provide a simple method for creating a baseline to subtract from your profiles. Finally, the tools allow you to read peak areas as well as heights. The tools work only on the front-most profile. Thus, you should select the profile to be manipulated before you select an individual tool. Once a tool is selected, it will remain active until another tool is selected.



**Pointer.** When the pointer is active, multiple profiles could be selected by either shift-click or option-click. The pointer tool also allows single profiles to be moved relative to other profiles on the x axis. This tool is always activated.



**Hand Tool.** The Hand tool allows you to move the position of profiles either relative to each other or to the screen. Use this tool by pressing the mouse button and holding it while moving the mouse. The front profile will move together with the move tool. If the profiles are grouped together, then all the profiles will be moved. The graph coordinates will automatically update to reflect the new profile position. Pressing command key will temporarily switch the cursor to a move tool when the cursor is on the Profile window.



**Zoom.** This tool provides methods to zoom in or out of specific region on the graph. Hence, any section of the profile can be blown up using the variable zoom tool. Zoom factor is always two. The resulting display will always center on the click point of the tool.



These operations take effect from the point of mouse clicking on the profile. Holding the shift or option key will invert the mode of the "zoom in" tool to a "zoom out" tool. The cursor and the zoom tool icon will reflect the changed mode.



When maximum or minimum zoom is reached, the zoom tool will not show + or - sign in it. The maximum and minimum magnification factor is to be determined. Double clicking on the zoom tool will toggle the Profile window to a 100% scale. Double clicking again will toggle back to the original magnification.



**Integrate profile segment.** This tool allows you to manually determine the area under any segment of the curve. This tool is most accurate when the profile is expanded horizontally. Move the tool to the beginning (left-most) position to be included in the area measurement. Press the mouse button and drag the tool to the end of the desired segment. If you drag the tool out of the window, then the profile will be scrolled automatically. The corresponding area under the curve will be highlighted. Releasing the mouse button will send the integral to the Results window.



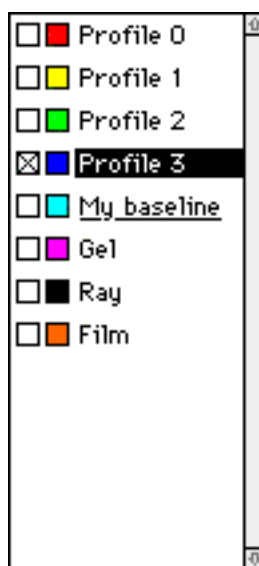
**Flat baseline.** This tool allows you to set a flat baseline. Once the tool is selected, clicking on the Profile window will create a horizontal baseline. Move the tool into position and press the mouse button. The baseline will appear and move with the mouse. Releasing the mouse will fix the baseline into position. Pressing the shift key while moving the baseline will constrain the baseline to remain under the profile. Click again will create another horizontal baseline or delete any existing baseline.



**Point-to-Point baseline.** This tool allows you to draw any baseline to be subtracted from the profile. It is most typically used for creating a sloping baseline to correct for background envelopes or defects in a blot or gel. You must start from the left-most side of the profile and beneath it. Click the mouse where you want the first anchor point to be. As you move the mouse (with the button released), a baseline will be drawn. Note that the line is constrained to remain beneath the profile. Click on the next anchor point and so forth until the baseline is completed. When finished drawing the baseline, double click with mouse button.

### ***Profile Name List***

The corresponding names for displayed profiles are shown in the profile name list. Each profile name has a check box and a color box in front of it. When the box is checked, the profile is shown on the Profile window otherwise, it is hidden. Selecting the profile names in the Name List will select the corresponding profiles as active in the display window. If a profile in the Profile window is selected by using the mouse cursor, the profile name list will scroll the corresponding profile name into view and display it highlighted. Furthermore, any profile name that is underlined is designated as a baseline. However, only one profile could be designated as a baseline at a time.



If a profile name or curve is double clicked, the following dialog will appear with all information on that profile. An alternative way to activate this is to choose the profile name or profile curve and then select the *Get Info* menu item from the *File* menu. If several profiles are selected and one of them is double clicked, there could be several profile information dialogs displayed at a time.

| profile Info    |                    |                                                                                                    |
|-----------------|--------------------|----------------------------------------------------------------------------------------------------|
| <b>Name :</b>   | profile            | <b>Color :</b>  |
| <b>File :</b>   | untitled           | <b>Selected :</b> Yes                                                                              |
| <b>Image :</b>  | Example Image (PA) | <b>Shown :</b> Yes                                                                                 |
| <b>Origin :</b> | 20.8 mm            | <b>Type :</b> Profile                                                                              |
| <b>Front :</b>  | 82.0 mm            | <b>Points :</b> 613                                                                                |

|                    | X       | Y           |
|--------------------|---------|-------------|
| <b>Min :</b>       | 20.8 mm | 17 counts   |
| <b>Max :</b>       | 82.0 mm | 1686 counts |
| <b>Increment :</b> | 0.1 mm  |             |

**Created :** Thu, Jun 30, 1994, 12:51:26 PM  
**Modified :** Thu, Jun 30, 1994, 12:51:26 PM

**Comments :**

### Status (Coordinate Boxes)

|    |          |
|----|----------|
| X: | 21.5 mm  |
| Y: | 18000 OD |
| V: | 5200 OD  |

This status boxes display the instantaneous x-y coordinates (mm or kilobases) of the current **cursor** position on the graph, and currently active **profile value** (V). If a group of profiles are selected, the V value is not displayed. The current profile units is always displayed with the values.

|     |          |             |
|-----|----------|-------------|
| X:  | 21.5 mm  |             |
| Y:  | 18000 OD |             |
| ΔX: | 3.0 mm   | ΔY: 2370 OD |

The values and units of the selected profiles moved will be shown when the move tool is dragging in the profiles.

|    |          |                 |
|----|----------|-----------------|
| X: | 21.5 mm  |                 |
| Y: | 18000 OD |                 |
| V: | 5200 OD  | ∫: 111800 OD*mm |

The integrated result will be shown when the Integrate tool is dragging in the selected profile. Only one profile selected could be integrated at a time.

## Section 6    Appendices

### 6.1    Appendix A    Using Concentration Standards

The following cautions should be kept in mind when using any set of standards: The range of O.D.s of standards should always bracket expected O.D. levels in the sample. The greater the number of points in the standard data, the greater the conversion accuracy will be. The standards should be made in the same type of matrix as the sample.

### 6.2    Appendix B    Technical Support

Bio-Rad provides complete technical support on all features of the Molecular Analyst/Macintosh software. If you have any problems or any questions on the use of the features, contact your local Bio-Rad office, or in the U.S. call Bio-Rad Technical Service at 1-800-4BIORAD (1-800-424-6723). Please report any reproducible software problems to Bio-Rad so we can eliminate them from future versions.

### 6.3    Appendix C    Molecular Analyst / Macintosh Quick Function Keys

|                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| <b>O</b>                           | <b>Open</b>                                                  |
| <b>W</b>                           | <b>Close</b>                                                 |
| <b>S</b>                           | <b>Save</b>                                                  |
| <b>P</b>                           | <b>Print</b>                                                 |
| <b>Q</b>                           | <b>Quit</b>                                                  |
| <b>Z</b>                           | <b>Undo</b>                                                  |
| <b>X</b>                           | <b>Cut</b>                                                   |
| <b>C</b>                           | <b>Copy</b>                                                  |
| <b>V</b>                           | <b>Paste</b>                                                 |
| <b>B</b>                           | <b>Clear</b>                                                 |
| <b>D</b>                           | <b>Duplicate</b>                                             |
| <b>A</b>                           | <b>Select All</b>                                            |
| <b>H</b>                           | <b>Hide All</b>                                              |
| <b>F</b>                           | <b>Find Spots</b>                                            |
| <b>I</b>                           | <b>Get Overlay Or Profile Information</b>                    |
| <b>=</b>                           | <b>Integrate Now</b>                                         |
| <b>T</b>                           | <b>Transformation</b>                                        |
| <b>E</b>                           | <b>Extract Profiles</b>                                      |
| <b>M</b>                           | <b>Smooth Profiles</b>                                       |
| <b>F</b>                           | <b>Find Peaks</b>                                            |
| <b>R</b>                           | <b>New Ruler</b>                                             |
| <b>Option</b> Duplicates an object |                                                              |
| <b>Shift</b>                       | <b>Constrains object to horizontal or vertical direction</b> |
| <b>Delete</b>                      | <b>Deletes selected objects</b>                              |

#### 6.4 Appendix D Adjusting The CCD Camera Focus When Zooming

This procedure should only be done when focus is completely off when zooming from high to low magnification. Please see drawing on following page.

**Tools Required:** Standard small screwdriver, resolution target (provided with camera), and .060" hex tool (provided with camera).

**Purpose of Procedure:** Installation of the lens on camera and adjustment of the flange back so that correct focus is held when zooming from high to low magnification.

1. Loosen lens mount set screw on camera body with hex tool.
2. Install zoom lens by screwing lens into camera body.
3. When lens is completely seated in the camera body, continue to rotate lens clockwise to line up zoom, focus, and iris index line with the face of the camera body. (This will be the face with the two (2) 1/4-20 tapped holes). Resistance will be felt as the lens is rotated to line up the index line. The lens will slip through a maximum of 350 degrees.
4. Tighten the lens mount setscrew with the .060" hex tool
5. Screw the lens diopter and other filters onto the zoom lens.
6. Mount the camera/bracket assembly onto the hood. Place the resolution target into the hood. Connect the camera-lens assembly so that a live image can be viewed on the computer monitor.
7. Remove the tracking adjustment hole cover. Loosen the tracking adjustment locking screw with the screwdriver. **CAUTION:** Do not allow dust or other particulates to accumulate in the camera body during this adjustment procedure.
8. Open up the iris on the lens until maximum illumination (completely open). Source light may need to be reduced in order to prevent saturation of CCD.
9. Rotate the zoom ring to one extreme so that the highest magnification of the resolution target is achieved. Adjust the focus ring for best focus. **Do not adjust the tracking adjustment screw at this zoom.**
10. Rotate the zoom ring to the other extreme so that the lowest magnification of the resolution target is achieved. Adjust the tracking adjustment screw for best focus.
11. Repeat steps 9 and 10 above until rotation of zoom ring requires no adjustment for focus (Generally, this is an iterative procedure and will converge in no more that two steps).
12. Tighten the tracking adjustment screw and replace the tracking adjustment access hole cover.
13. The zoom lens should now be properly installed onto the CCD camera with the focus adjusted properly for high and low magnification.

## 6.5 Appendix E Adjusting The Mitsubishi P68 Thermal Printer for Macintosh

### Rear of printer:

- 1) Mode 1 DSQ pin configuration: All pins are OFF except pin # 2, #7, #8
- 2) Mode 2 DSQ pin configuration: All pins are OFF except pin #5

### Front panel of printer:

- 1) Direction: Normal
- 2) Scan: Frame
- 3) Contrast: Normal

### Recommended function settings

1. Connect the BNC cable (that came with the printer) from "Video" on the camera to "Video-In" on the rear of the printer. Turn on the CCD camera.
2. Connect the Gel Doc integrating cable to the frame grabber card in your Macintosh.
3. Connect the "Camera Video" cable (1 of the 4 cables of the integrating cable) to "Video-Out" on the rear of the printer.
4. Connect the "Printer Remote" cable (1 of the 4 cables of the integrating cable) to "Remote" on the rear of the printer.
5. Turn on the printer (both rear and front power switches).
6. Start Molecular Analyst/Mac and view a live display using the Acquire command from the FILE menu.
7. Open the front panel of the Mitsubishi printer, press the "Function" button and adjust the settings as seen below. Press the "Set" button to accept the settings.

#### RS-170, NTSC Systems

|         | USER |           | USER     |
|---------|------|-----------|----------|
| HWidth: | 794  | B-Lev:- 3 |          |
| HTRC:   | 0    | W-Lev:    | +4 or +5 |
| HRes:   | 640  | Sharp:    | +2       |
| HStart: | 42   | AGC:      | Off      |
| VLines: | 525  | Direct:   | Off      |
| VStart: | 32   | Dither:   | Off      |
| VRes:   | 476  |           |          |
| HSize:  | 100  |           |          |
| VSize:  | 76   |           |          |

#### CCIR, PAL Systems

|         | USER |           | USER     |
|---------|------|-----------|----------|
| HWidth: | 960  | B-Lev:- 3 |          |
| HTRC:   | 0    | W-Lev:    | +4 or +5 |
| HRes:   | 767  | Sharp:    | +2       |
| HStart: | 75   | AGC:      | Off      |
| VLines: | 625  | Direct:   | Off      |
| VStart: | 80   | Dither:   | Off      |
| VRes:   | 508  |           |          |
| HSize:  | 100  |           |          |
| VSize:  | 79   |           |          |

8. Remove the BNC cable (that came with the printer) and reconnect the integrating cable to its appropriate place as noted on each cable on the integrating cable.
9. "Video Print" a saved image to test the printer configuration. If the printout is too light or too dark, repeat the above steps and adjust the "W-Lev" appropriately.

- % Volume 81
- Abs Volume 81
- Acquire 39
- Acquiring an Image from the Gel Doc 1000 44
- Acquiring an Image from the GS-670 Densitometer 40
- Acquiring an Image from the GS-700 Densitometer 41
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- Auto Integrate 65
- Automatic Peak Identification 30
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- Calculating a Standard Curve 31
- Calculating unknown Molecular Weights 36
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- Change Image Size... 46
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 Logistic 80  
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